USER MANUAL

ERP Migration Accelerator

Moving master data from a legacy system into SAP S/4HANA, or into a database or Firestore, under controls that make the result provable.

Document	User Manual
Version	1.0
Date	31 October 2026
Applies to	Application build 8774c9f (main)
Audience	Migration authors, approvers, stewards, administrators and auditors
Practice data	1,980 invented customers; all screenshots are real
Classification	Internal
History	1.0 · 31 October 2026 · First issue as a full manual: the practice migration end to end and every other page (sources of all kinds, Combine, Packs, Assistant, Keys).

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1 Introduction

What this manual is, who it is for, and how to read it.

1.1 Purpose of this manual

This manual describes how to use the **ERP Migration Accelerator**, a web application that moves master data (here: customers) from a legacy source into **SAP S/4HANA** (Business Partner), or into a database or Google Firestore, under controls that make the result provable. It explains the process end to end, the role of each person involved, every screen and button used, and the evidence the application produces.

1.2 Who should read it

READER	READ CHAPTERS	WHY
Migration author (builds the mapping)	2, 3, 4, 5, 6	The whole procedure, from registering a source to a delivered run.
Approver / steward (reviews, approves, signs off, loads)	3, 6.7 to 6.14, 7	Your decisions and the evidence you rely on.
Auditor or project owner	2, 3, 7, 8	How the controls work and where the proof is.
Administrator	4, 9, appendices C and D	Installation, settings and what was and was not tested.

1.3 Scope and what was verified

The manual covers every function of the application as of the build named on the cover. All screenshots are **real**: they were taken by a script that operated the application exactly as a person would, on a practice file of 1,980 invented customers (`mock_data/legacy_ar_customers.csv`). Only the red numbered rings were added, to point at parts of a screen.

CAUTION

SAP itself was not used. Every SAP delivery in this manual went to a **stand-in gateway** that follows SAP's published interface. It proves that the application builds correct messages and handles success and failure; it does not prove that a particular company's SAP system will accept them. Appendix D lists what was tested against what.

1.4 Conventions

ELEMENT	MEANING
Bold	The name of a button, menu or field exactly as it appears on screen.
`Code font`	A value you type, a file or table name, or a technical identifier.
Procedure block	A framed block with Purpose, Performed by, Before you start, Procedure and Expected result. Each task in chapter 6 has one.
NOTE Extra information.	Useful context that does not change what you do.
CAUTION Risk.	Something that can cost time or produce a wrong result if ignored.
CONTROL Safeguard.	A safeguard the application enforces, and what it records as evidence.

2 Concepts and architecture

The ideas the rest of the manual relies on.

2.1 The migration in one paragraph

Data is read from a legacy **source** and frozen into a locked **snapshot**. It is profiled and quality-checked. A **mapping set** (a list of rules) says which legacy column fills which SAP field and how. A second person **approves** the rules, after which they are locked. A **run** applies them to every row, finds duplicates and validates against SAP's constraints. The **gate** blocks while critical defects remain. A named **steward signs off**, the data is **delivered** and read back, and a **reconciliation certificate** proves that every source row is accounted for.

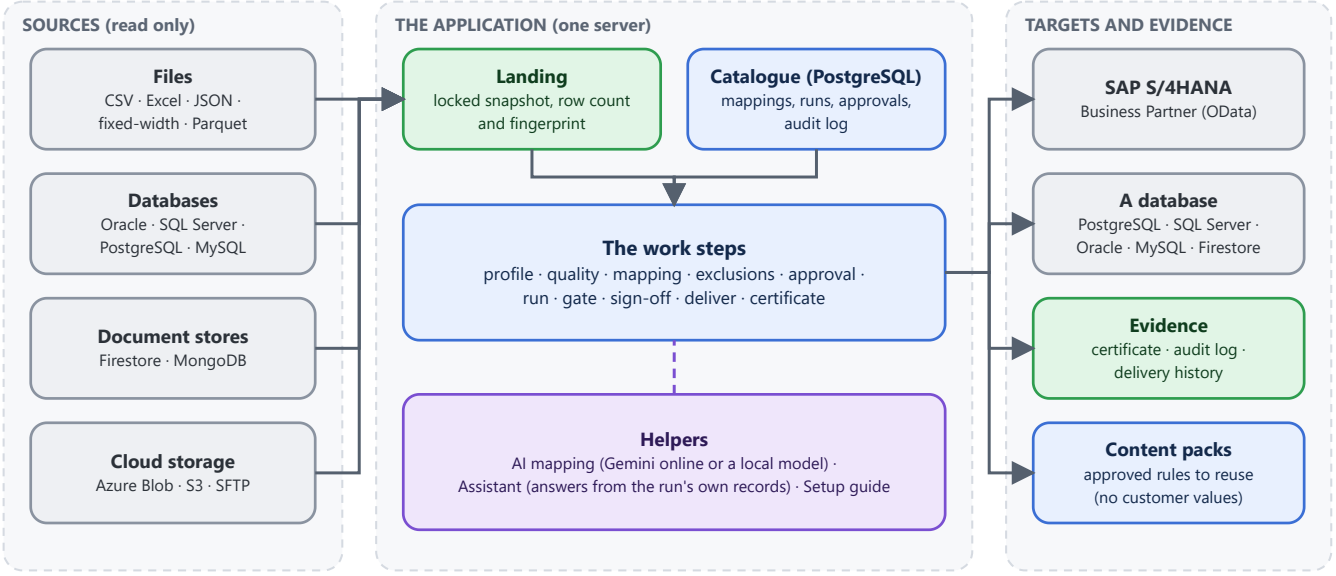


Figure 1. Architecture: sources are only read; the application keeps the snapshot and the catalogue; results go to a target, with evidence.

2.2 The process at a glance

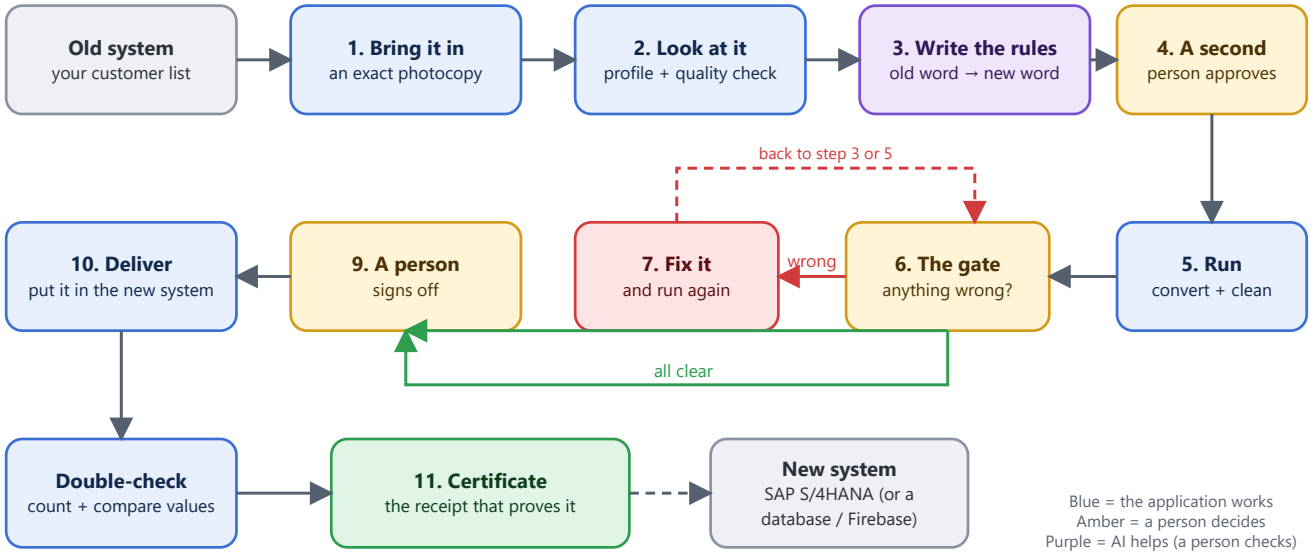


Figure 2. The end-to-end process. Blue steps are performed by the application, amber steps need a person, purple is AI assistance that a person checks.

STEP	PROCEDURE	RESULT	SECTION
1	Register a source	A locked snapshot with a row count and a fingerprint	6.1, 6.2
2	Profile and assess quality	Column statistics; findings ranked by severity	6.3, 6.4
3	Combine tables (optional)	A new source table made from several	6.5
4	Write the mapping	Rules per SAP field, tested on a sample	6.6
5	Approve	Locked rules; a second person's name on record	6.7
6	Run	Converted, de-duplicated, validated rows	6.8
7	Read the gate; fix; run again	READY FOR LOAD	6.9, 6.10, 6.11
8	Sign off	A named person accepts the run	6.12
9	Deliver and verify	Rows in the target, read back and compared	6.13

STEP	PROCEDURE	RESULT	SECTION
10	Certificate; roll back if needed	Proof of completeness; a precise undo	6.14, 6.15

2.3 Glossary

Source / legacy
The old system or file data is read from. Never modified.
Snapshot
A locked copy of the source taken at registration; all later work uses it. It has a row count and a fingerprint.
Fingerprint (hash)
A short code computed from the data; it changes if any value changes. Used to prove two data sets are identical.
Control total
The row count you expect, entered at registration. A different count stops the registration.
Key
The column (or columns) whose value is unique per row, such as `CUST_ID`. It lets every finding name the exact record.
Profile
Statistics per column: type, blank ratio, distinct values, lengths, most frequent values, a masked sample, a health score.
Mapping set / rule
A set is the complete list; a rule fills one SAP field from legacy data. Kinds: `DIRECT_COPY`, `STATIC_VALUE`, `VALUE_MAPPING`, `SQL_EXPRESSION`.
Lookup
A dictionary inside a `VALUE_MAPPING` rule, for example `Deutschland → DE`. Matching ignores letter case.
Status
DRAFT (editable), **SUBMITTED** (awaiting a second person), **APPROVED** (locked).
Run
One complete pass of all rows through an approved mapping set, with validation and de-duplication.

Defect
A group of rows that broke one rule, with examples and a root cause. Severities: CRITICAL, HIGH, MEDIUM, LOW. Only CRITICAL blocks the gate.
Gate
The release check of a run: **BLOCKED** while a critical defect is open, **READY FOR LOAD** when none is.
Exclusion
A rule that leaves rows out on purpose, with a recorded reason. Excluded rows are kept, not deleted.
Sign-off
A steward's recorded acceptance of a run that passed the gate. Required before load.
Target
Where data is delivered: SAP (workbook, sandbox, live OData), a database, or Firestore.
Certificate
The reconciliation report: every source row accounted for, every control listed, who approved and signed off.
Rollback
Removal of exactly the rows one delivery wrote (each row carries its run number).
Content pack
An exported approved mapping (rules, lookups, exclusions; no customer values) reusable in another migration.

3 Roles and responsibilities

Who does what, and what the application prevents.

3.1 Roles

The application records the name in the **Acting as** box (top right of the window) against every action. In an installation with API keys configured, the name comes from the person's key; without keys, anyone may type any name, which is acceptable for practice and not for real data (see 4.4).

ROLE	MAY	MAY NOT
Author	Register sources, profile, write or generate rules, test them on a sample, exclude rows, submit a mapping set, run an approved set.	Approve a set they submitted.
Approver	Read a submitted set and approve it (rules are then locked).	Approve their own submission.
Steward	Sign off a run that passed the gate; execute the load; verify; roll back.	Sign off a run whose mapping they authored (default policy).
Administrator	Install, configure, manage keys, allowed servers, who may approve (`MIGRATION_APPROVERS`).	Bypass the gate.
Auditor	Read the certificate, audit log, compliance page, dashboard.	(read only)

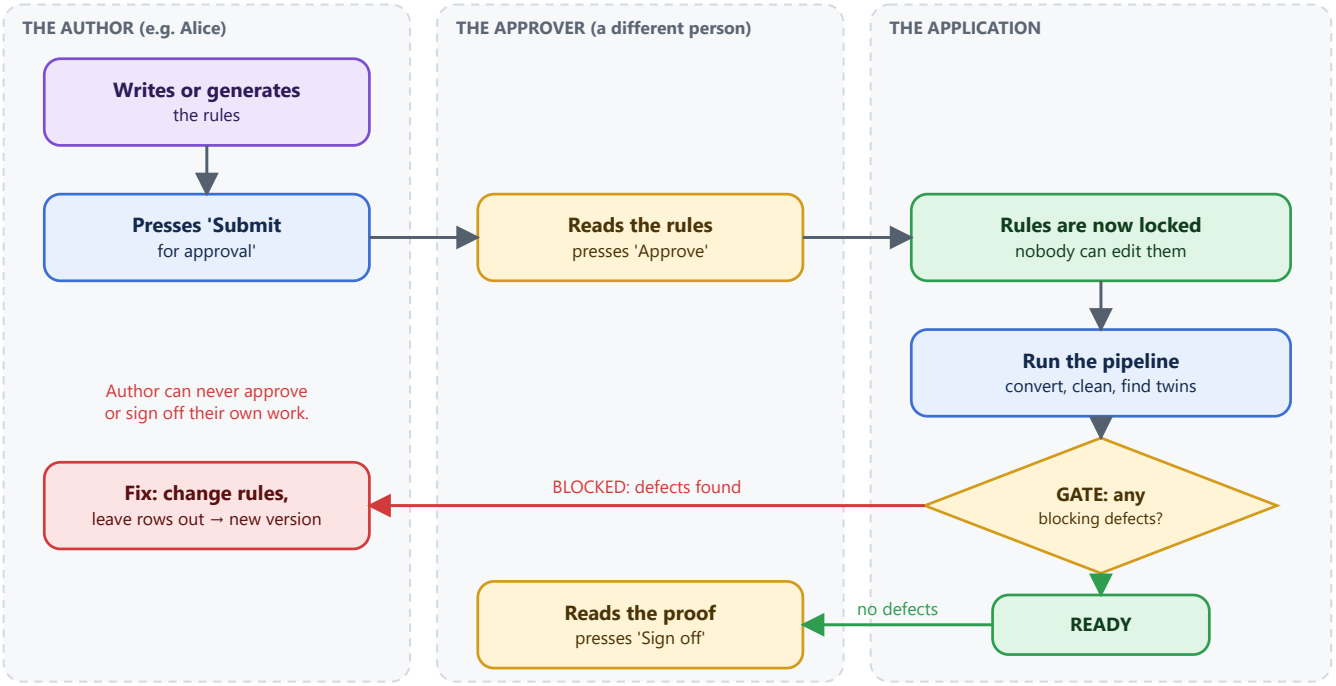


Figure 3. Who may do what. The author prepares and submits; a different person approves; the gate and the sign-off protect the target.

CONTROL

Segregation of duties. The application refuses an approval by the author of the set. The certificate lists the author and the approver and fails its control check if they are the same person.

3.2 Life of a mapping set

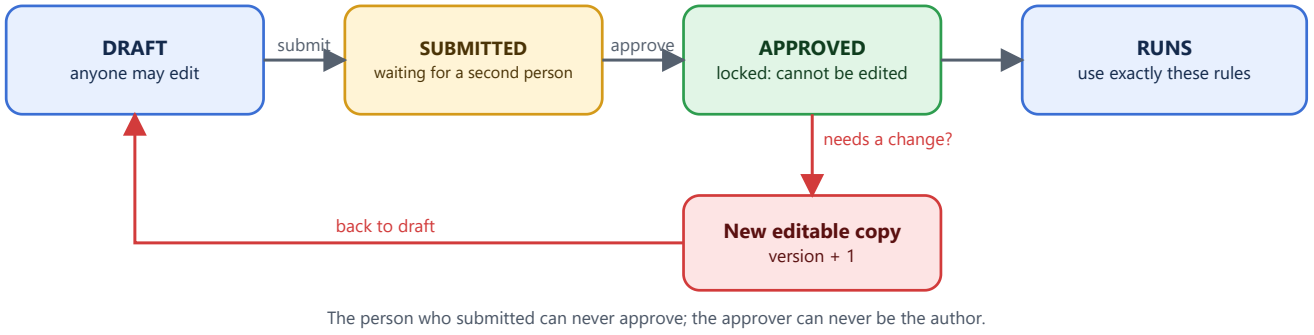


Figure 4. Statuses of a mapping set. An approved set is never edited: a change makes a new version, which goes through the same approval.

4 Getting started

Requirements, installation, the main window and the settings that matter.

4.1 Requirements

ITEM	REQUIREMENT
Server	A computer that runs Python 3 (the practice runs used Python 3.13 on Windows 11); the catalogue needs **PostgreSQL** (Docker is the quickest way)..
Browser	A current Microsoft Edge or Google Chrome. The user interface is a single web page; nothing is installed on user computers.
Disk	A workspace folder for snapshots and run results. It holds copies of customer data: back it up and treat it as confidential.
AI (optional)	Either **Ollama** with a local model (default `qwen2.5-coder:7b`; needs about 6 GB of memory and runs on a CPU, slowly) or an online **Gemini** key. Everything except AI mapping works without either.
SAP (optional)	For live delivery: the address of the SAP OData service and a technical user. Without them the sandbox stand-in and the Migration Cockpit workbook are available.

4.2 Installation

```
# 1. a PostgreSQL database for the catalogue
docker compose up -d postgres

# 2. Python packages
python -m venv .venv
.venv\Scripts\python -m pip install -r requirements.txt

# 3. settings: a file called .env in the project folder
DATABASE_URL=postgresql+psycopg2://migration:<password>@localhost:5432/migration_catalog
MIGRATION_WORKSPACE=workspace

# 4. create the tables and load the SAP field catalogue
.venv\Scripts\python -m alembic upgrade head
.venv\Scripts\python -m src.seeds.seed_sap_catalog
```

```
# 5. start the application, then open http://127.0.0.1:8000
.venv\Scripts\python -m uvicorn --factory src.api.main:create_app --host 127.0.0.1 --port 8000
```

NOTE

After upgrading to a build that changes the SAP catalogue (for example the German column-name aliases), run step 4 again so the new descriptions are loaded.

4.3 The main window

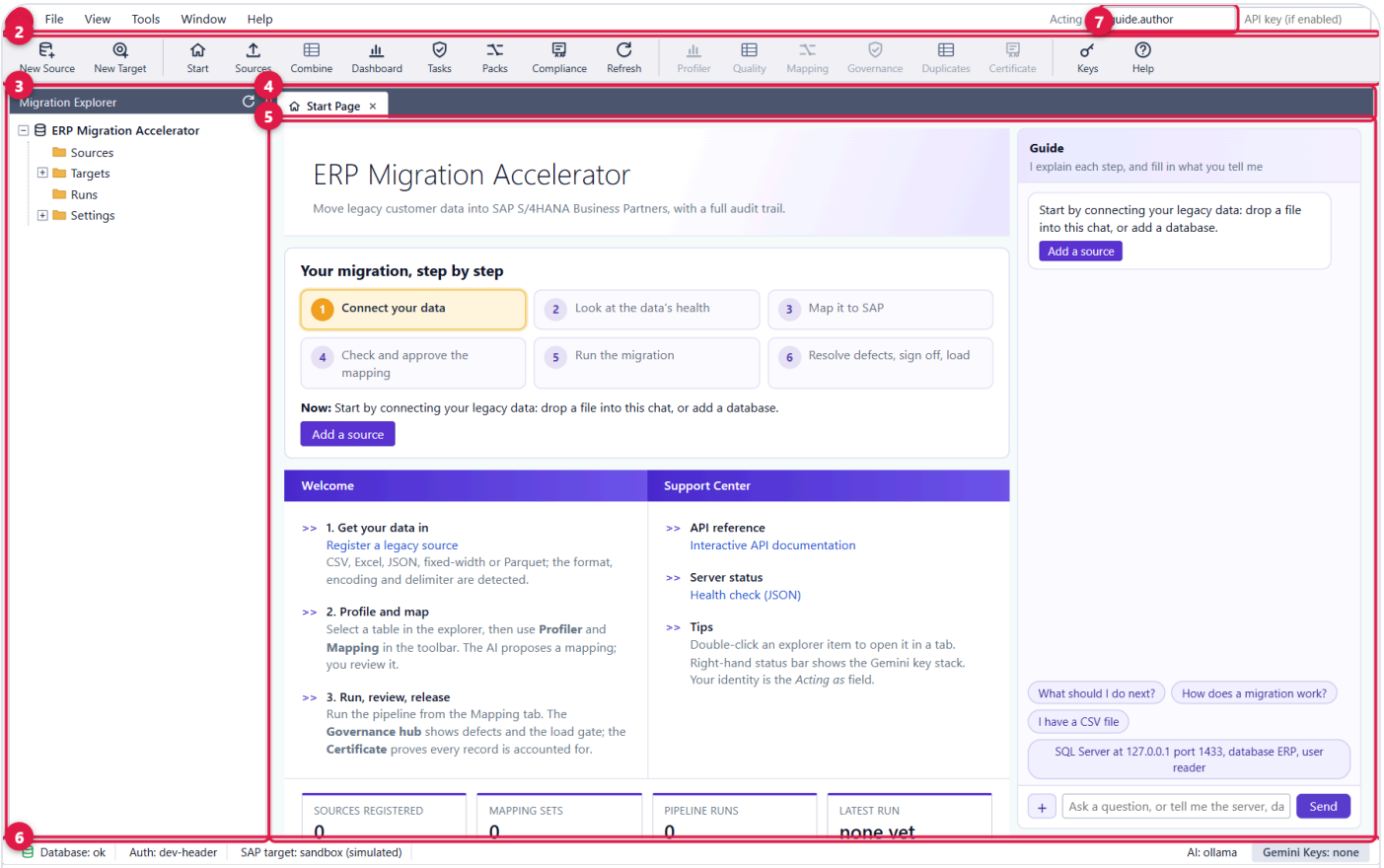


Figure 5. The main window. 1 Menu bar · 2 Toolbar · 3 Explorer · 4 Tabs · 5 Work area · 6 Status bar · 7 Acting as.

#	PART	USE
1	**Menu bar**	File, View, Tools, Window, Help.
2	**Toolbar**	**New Source**, **New Target**, **Start**, **Sources**, **Combine**, **Dashboard**, **Tasks**, **Packs**, **Compliance**, **Refresh**, then **Profiler**, **Quality**, **Mapping**, **Governance**, **Duplicates**, **Certificate** (greyed until a table or run is selected), and **Keys**, **Help**.

#	PART	USE
3	**Explorer**	A tree: **Sources** (systems and tables, with their mapping sets), **Targets**, **Runs**, **Settings**. **Click** to select; **double-click** to open.
4	**Tabs**	Every page opens in a tab. Open tabs are restored when the page is reloaded.
5	**Work area**	The page itself; the first is the Start Page.
6	**Status bar**	Database state, SAP mode (here *sandbox, simulated*), the active AI provider.
7	**Acting as**	Your identity for the audit trail.

CAUTION

Select first, then press. Toolbar buttons act on the table or run selected in the Explorer. A grey button almost always means nothing is selected.

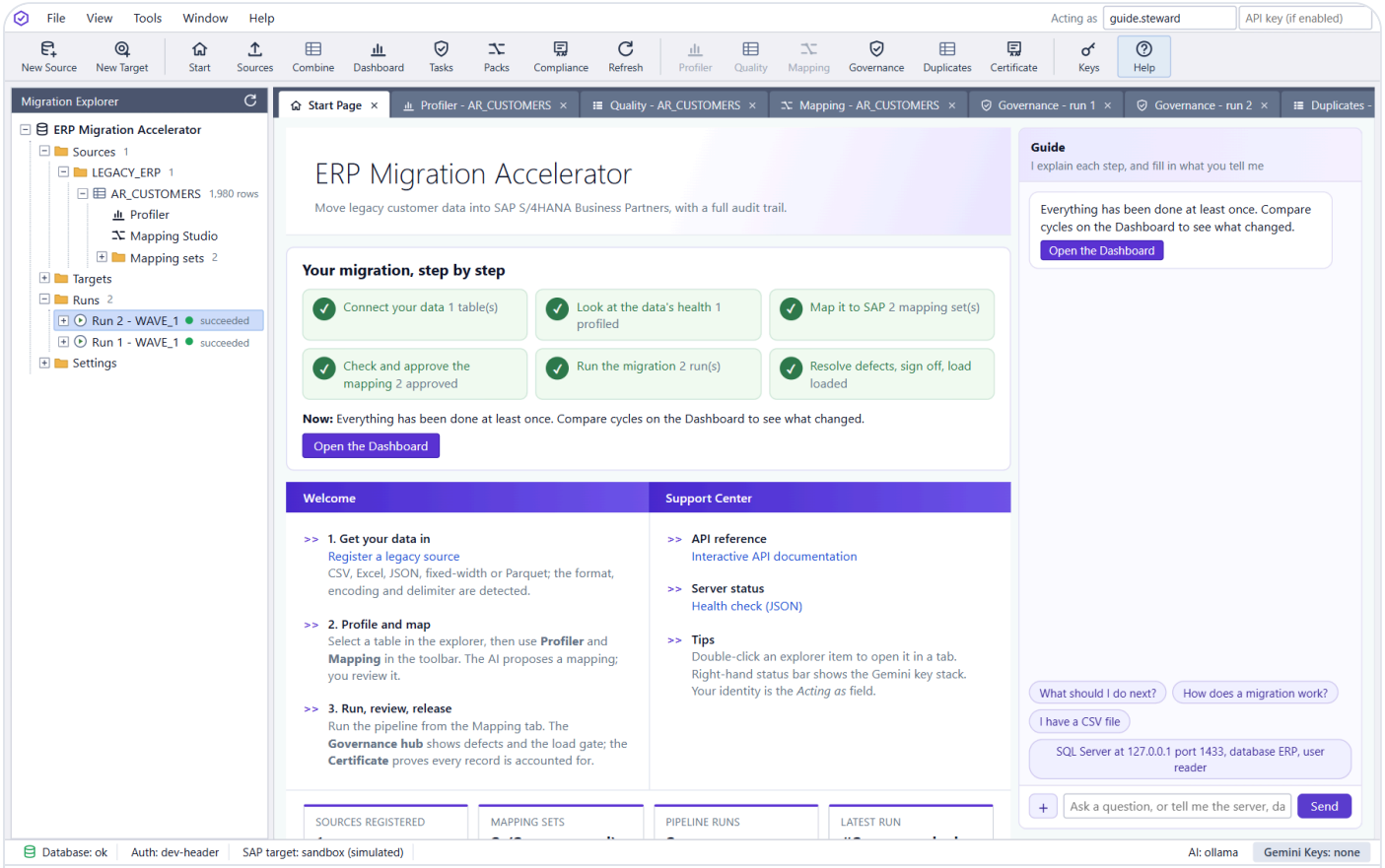


Figure 6. The Start Page: a progress checklist and, on the right, the Guide helper that explains the next step.

4.4 Configuration reference

SETTING	EFFECT
`LLM_PROVIDER`	`ollama` (default), `gemini` (online; keys are added under **Keys** and kept in memory only) or `ollama-only`.
`OLLAMA_MODEL`	The local model, default `qwen2.5-coder:7b`.
`MIGRATION_REQUIRE_AUTH`	`true`: refuse to start unless `MIGRATION_API_KEYS` is set. Recommended for real data.
`MIGRATION_API_KEYS`	Maps keys to user names; the application then ignores a typed name.
`MIGRATION_APPROVERS`	Who may approve and sign off. Default: anybody except the author.
`MIGRATION_WORKSPACE`	Folder for snapshots and results.
`MIGRATION_DB_ALLOWED_HOSTS`	Servers the application may connect to. Default: any, because a user enters the address. **Set this in production.**
`MIGRATION_AUTO_SEED`	Load the SAP field catalogue automatically at start (default on).

5 The practice data set

The file used in the screenshots, and what is wrong with it on purpose.

5.1 The practice file

Procedures in chapter 6 are illustrated with `legacy_ar_customers.csv`: **1,980 rows, 10 columns**, invented names, deliberately untidy in the ways real extracts are.

```
CUST_ID,CUST_NAME,TAX_ID,EMAIL,PHONE,ADDR_STREET,CITY,COUNTRY,POSTAL_CODE,PAYMENT_TERMS
100000,Penene Labs LLC,US974844773,accounts@penene0.example.com,+33-315-7664202,258 Birch Road
0,Miami,usa,24514,NET60
100001,Nomili Labs Ltd,,accounts@nomili1.example.com,+91-996-9238128,408 River Road
1,,India,714169,COD
```

CONDITION IN THE DATA	ROWS	CONSEQUENCE FOR SAP
City blank	62	Mandatory field `ADRC.CITY1` would be empty.
Country blank	42	Mandatory field `ADRC.COUNTRY` would be empty.
Countries that do not exist (Atlantis, Narnia, Wakanda)	44	No lookup entry; rows cannot be mapped.
Country written as USA, usa, US, United States, Deutschland, DE, UK, GB …	most	SAP needs the two-letter code: a lookup is required.
Payment terms written seven ways, including `WHEN-RECEIVED`	~1,700	SAP needs fixed codes; a person decides what `WHEN-RECEIVED` means.
Names longer than 40 characters	14	`BUT000.NAME_ORG1` holds 40.
The same company twice (same tax number)	~100	Should arrive once.

The conditions are the realistic reason a first run is blocked, which is the point of the exercise: the controls are shown working against genuine problems.

6 Procedures

One procedure per task, in the order of a migration.

6.1 Register a file source

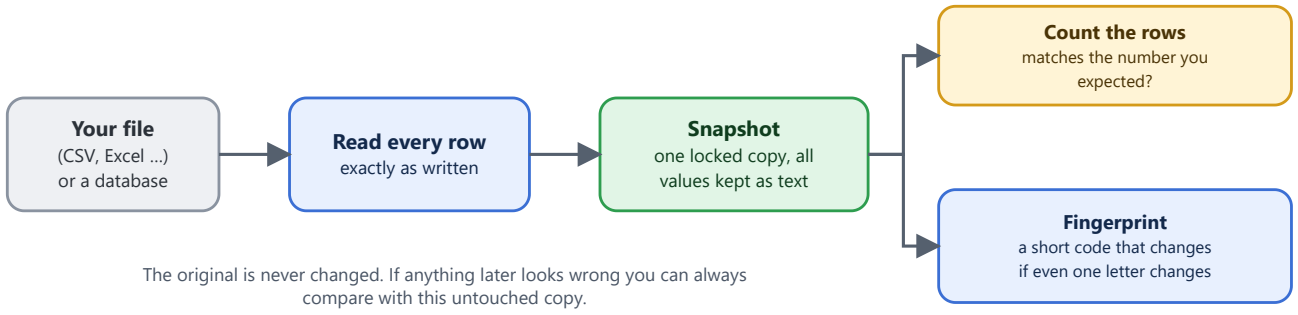


Figure 7. Registration: read every row as written, keep all values as text, count, and fingerprint.

PR-1 Register a source from a file

PURPOSE

Freeze an extract file into a locked snapshot with a verified row count.

PERFORMED BY

Author

BEFORE YOU START

The extract file, the expected number of rows (control total), the name of the column(s) that identify a row.

PROCEDURE

1. In the toolbar press **New Source**. A five-page wizard opens.
2. **Page 1.** Give the source a name, for example `LEGACY_ERP` (the name of the legacy *system*; one system can hold many tables). Leave **Database** on *Extract file* and **Provider** on *Auto-detect*.
3. **Page 2.** Choose the file and press **Test**. The wizard reports the detected format, encoding, delimiter and column count.
4. **Page 3.** Enter the table name (`AR_CUSTOMERS`), tick the key column (`CUST_ID`) and enter the control total (`1980`).
5. **Page 4.** Tick **Profile the table now**. **Page 5** shows a summary; press **Finish**.

EXPECTED RESULT

The wizard closes, a **Profiler** tab opens and a message reads *Source LEGACY_ERP / AR_CUSTOMERS registered: 1,980 rows, 10 columns*.

NOTE

A shorter route exists for a plain file: the **Sources** page (shown in 6.2) has a **Register a legacy source** form with the same fields and a **Register & upload** button.

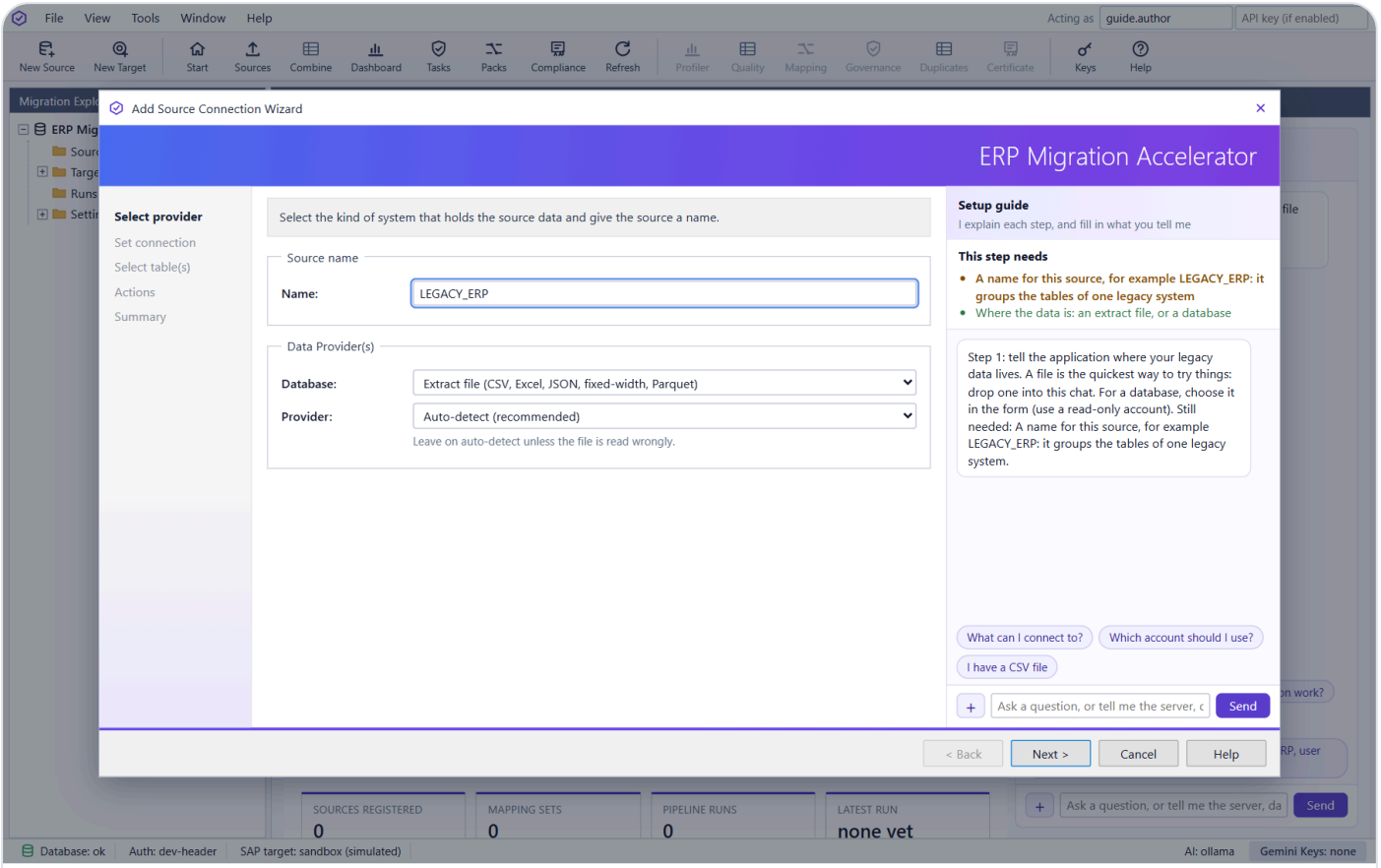


Figure 8. Page 1: name the source. The Setup guide on the right explains each page and can fill in fields.

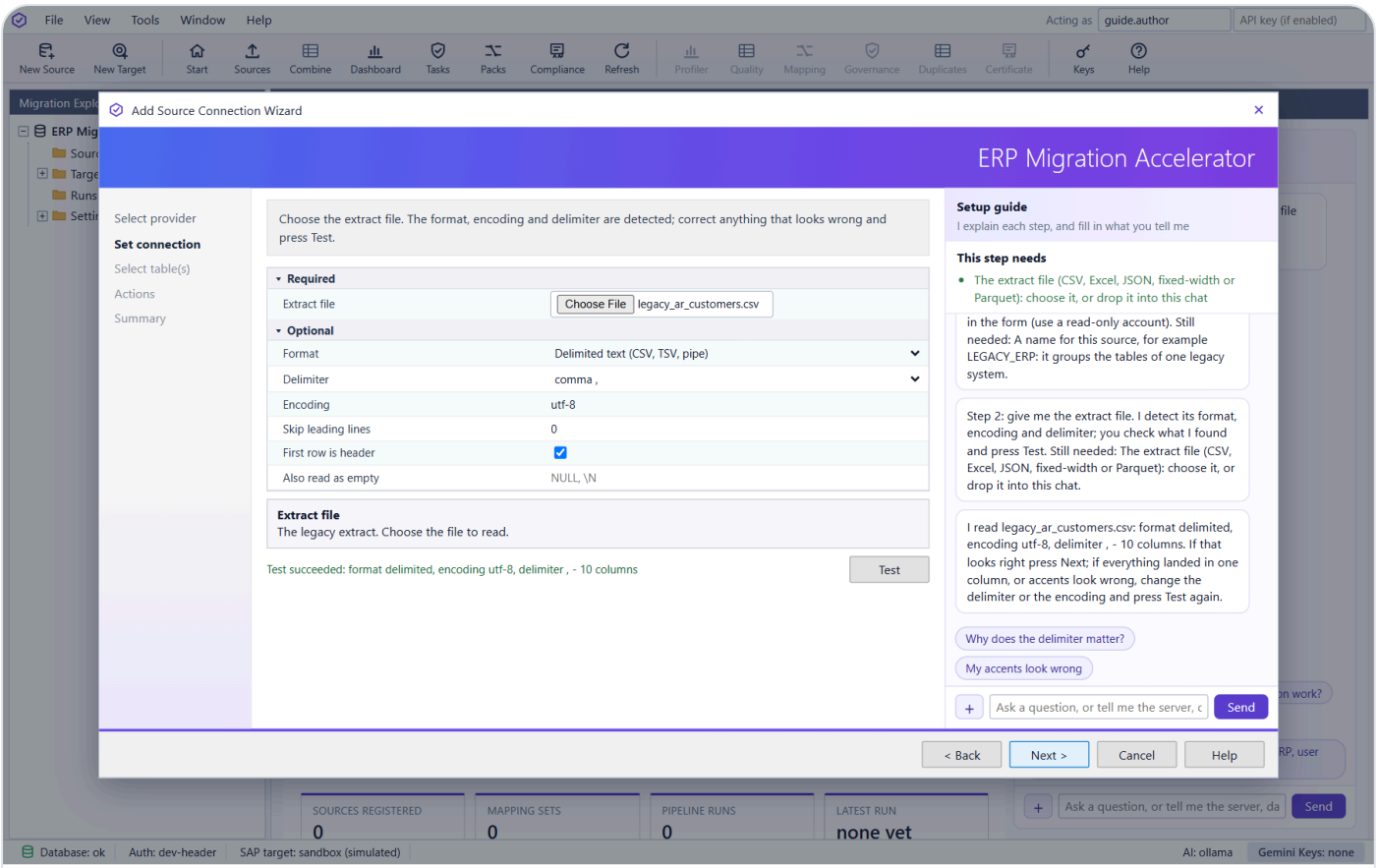


Figure 9. Page 2: the application detected a delimited, utf-8, comma-separated file with 10 columns.

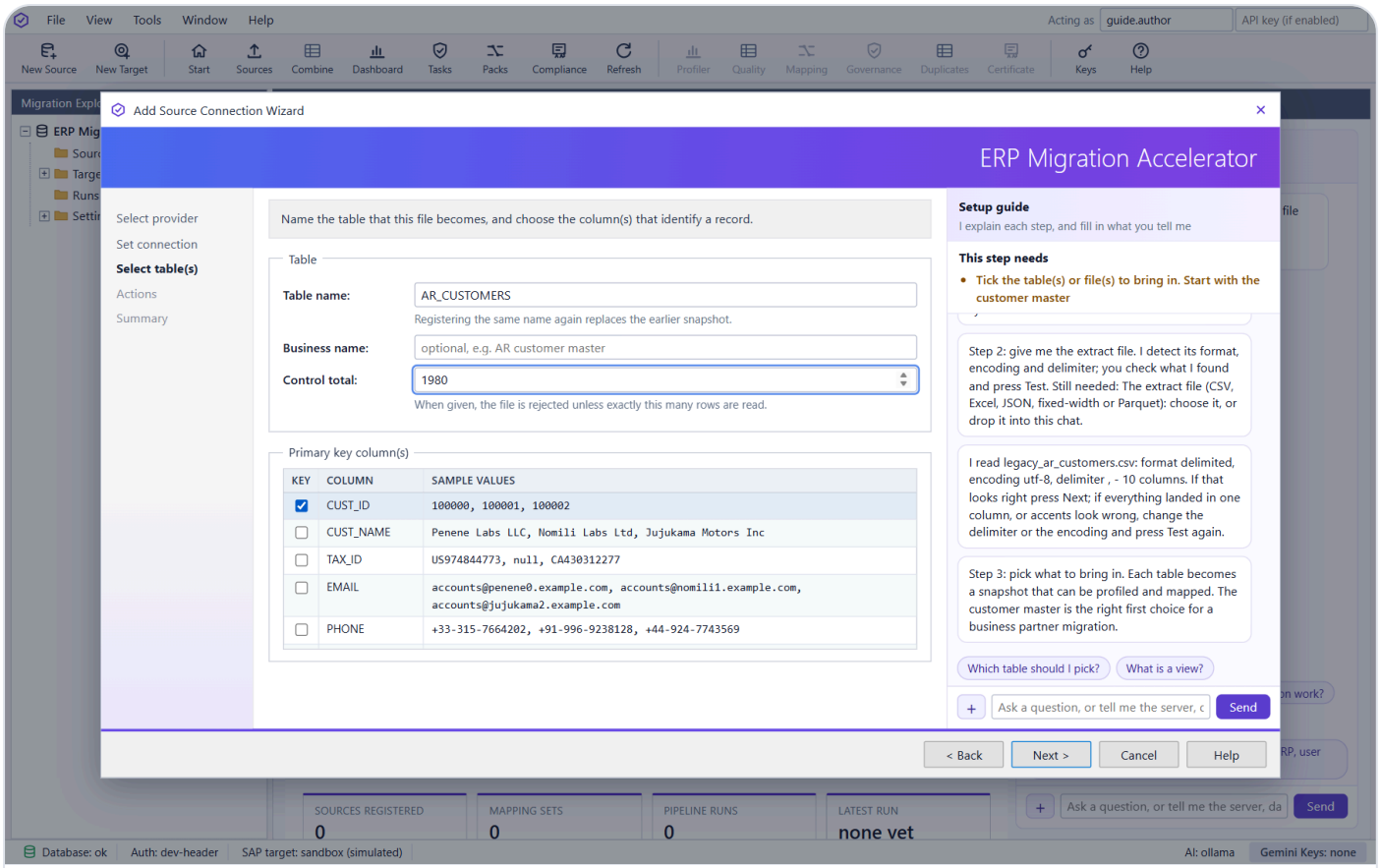


Figure 10. Page 3: table name, key column and control total.

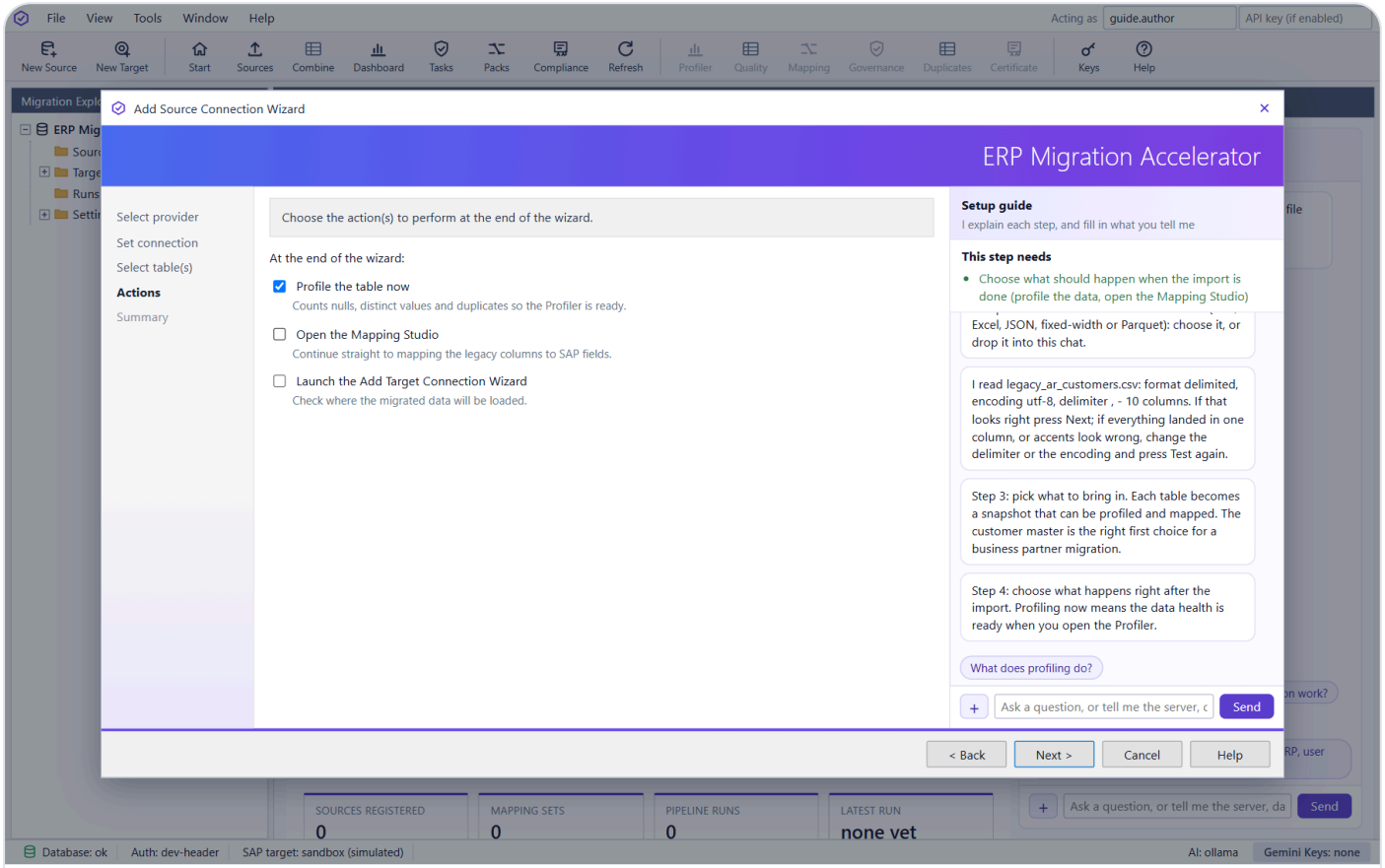


Figure 11. Page 4: actions to run when the table is registered.

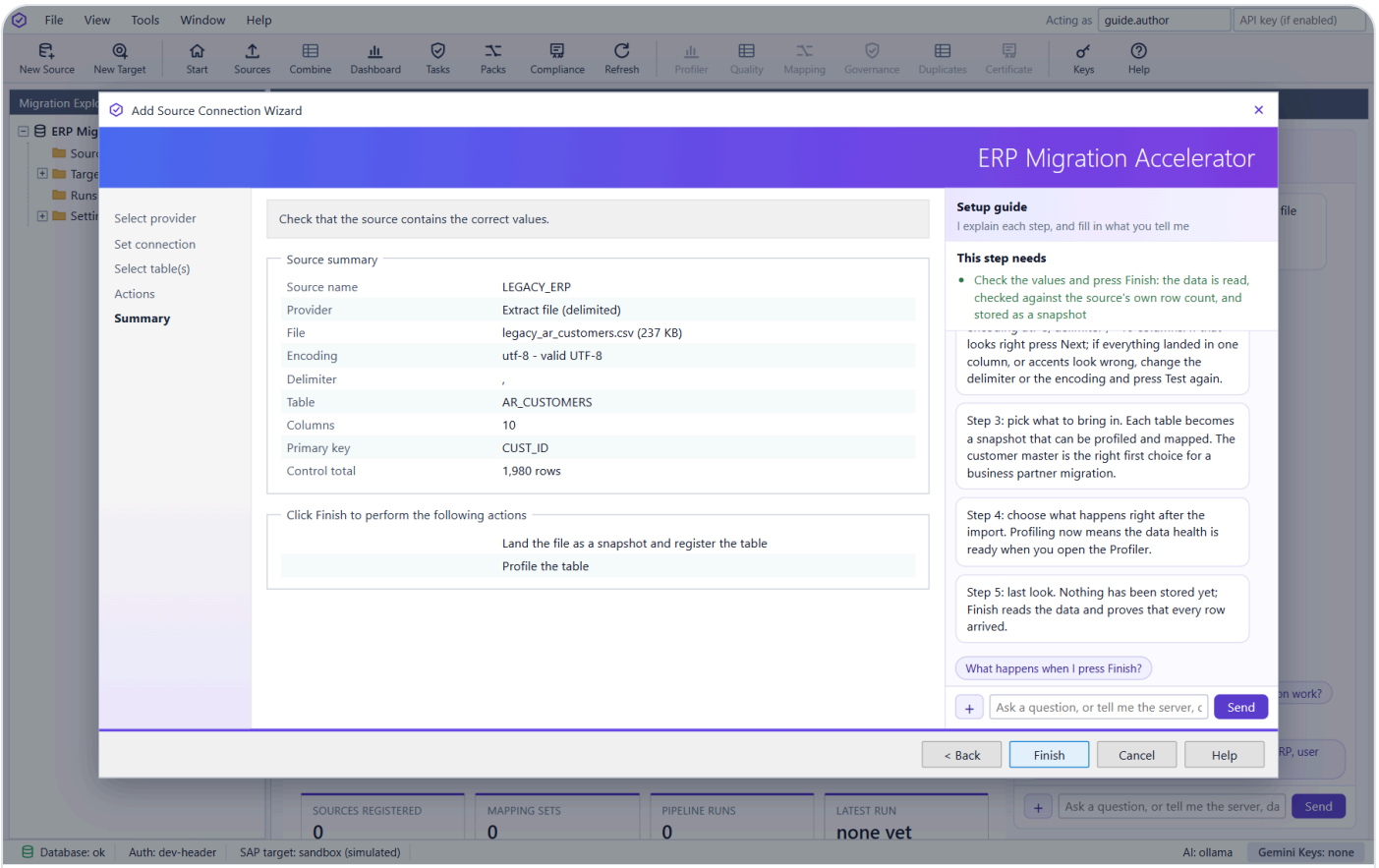


Figure 12. Page 5: summary before anything is written.

CAUTION

If **Test** shows everything in one column or damaged accents, change **Delimiter** or **Encoding** and test again. If the row count differs from the control total, registration stops: obtain a complete extract or correct the number.

CONTROL

The source is only read. All values are stored as text so nothing is rounded or re-typed. The row count and fingerprint of the snapshot are recorded and appear on the certificate.

6.2 Register a database, document store or cloud source

PR-2 Register a database, Firestore, MongoDB or cloud-storage source

PURPOSE

Read a table or collection from a live system into a snapshot, without installing anything on it.

PERFORMED BY

Author

BEFORE YOU START

The server address (or project), a read-only account, and permission to connect. In production, list the permitted servers in `MIGRATION_DB_ALLOWED_HOSTS`.

PROCEDURE

1. Press **New Source** and enter a name (letters, digits, `.` `_` `-`, up to 64).
2. In **Database** choose the kind: PostgreSQL, SQL Server, Oracle, MySQL, **MongoDB**, **Google Firestore**, or cloud file storage (Azure Blob, S3, SFTP). The Provider line shows the driver used.
3. **Connection.** Enter server, port, database, user and password (or, for Firestore, the project and service-account file) and press **Test**. The wizard really connects.
4. **Tables.** Browse the schemas and tables (or collections), choose one, set the key and the control total, and finish as in PR-1.

EXPECTED RESULT

The chosen table or collection is read, counted against the control total and frozen. For MongoDB and Firestore each document becomes one row and nested fields become dotted column names (for example `address.geo.lat`).

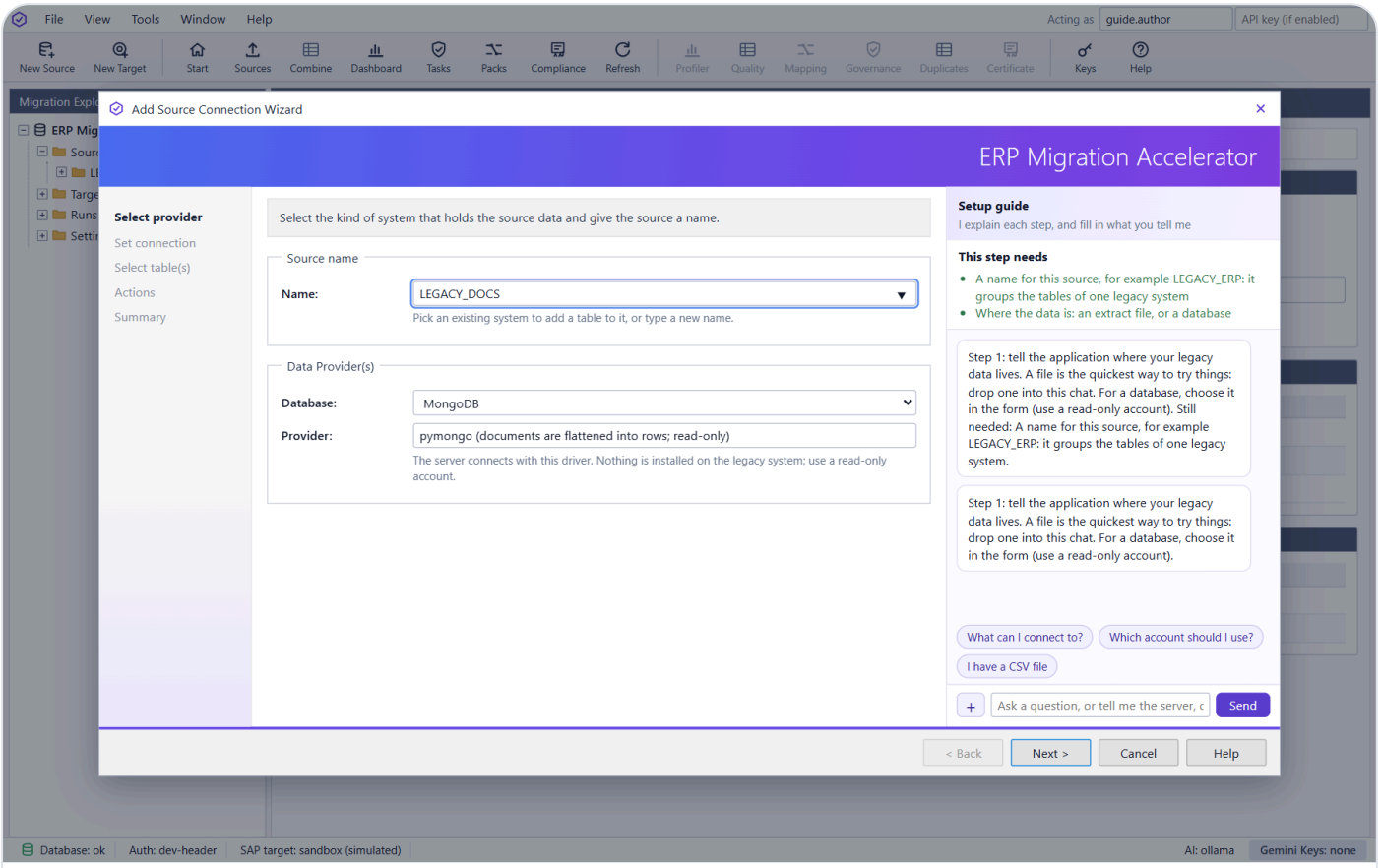


Figure 13. Page 1 for a MongoDB source: the kind list groups Files, Databases (read only) and Cloud file storage.

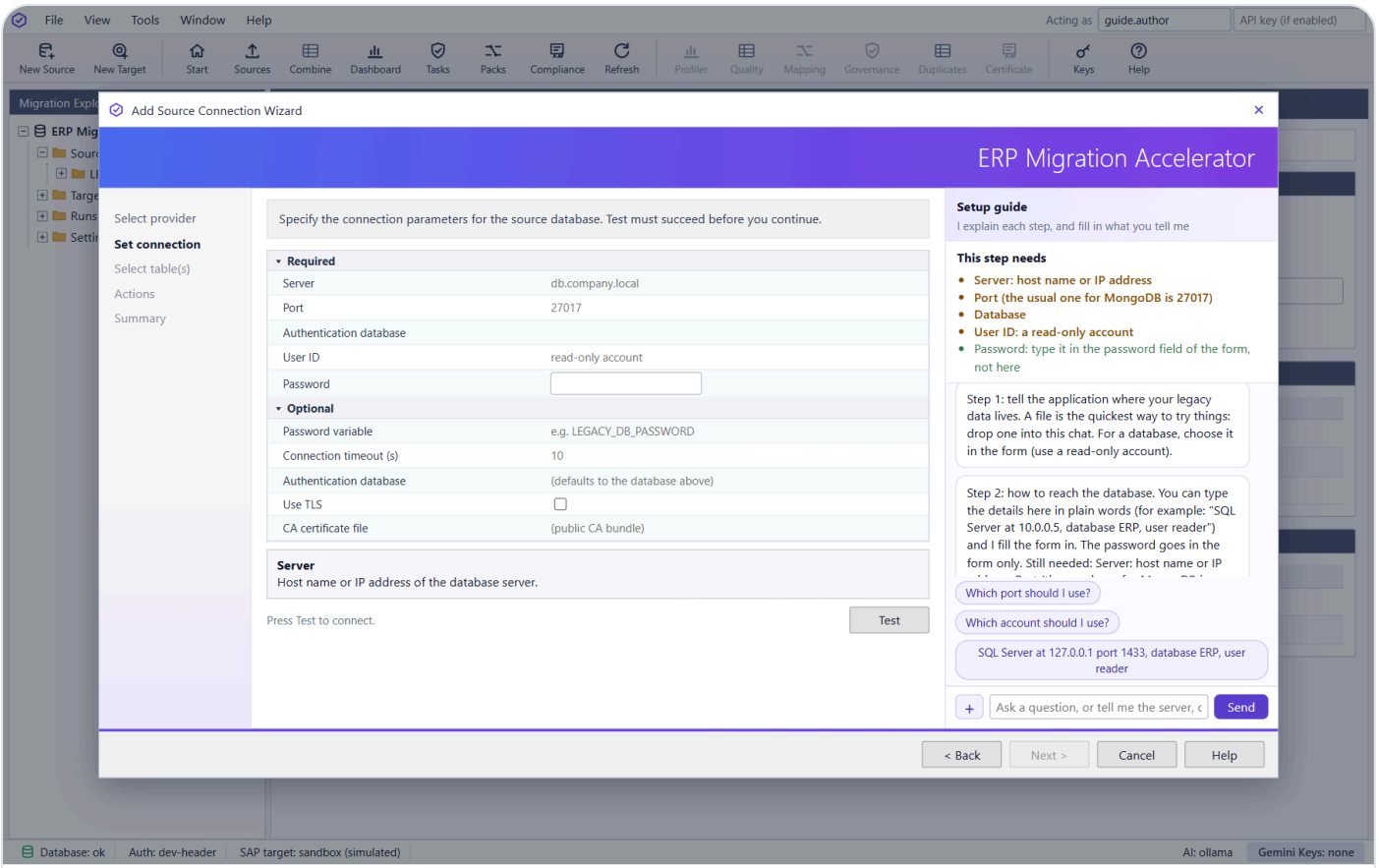


Figure 14. Page 2 for MongoDB: the connection details, with the **Setup guide** on the right. It lists what the step needs, answers questions (*Which port should I use?*) and fills in a field when you tell it the server, database or user. The password goes only into the masked form field, never into the chat.

CONTROL

A password is never written to disk and never echoed back after a failed attempt. MongoDB is read-only by design. Firestore bills one read per document; the wizard says so next to the driver.

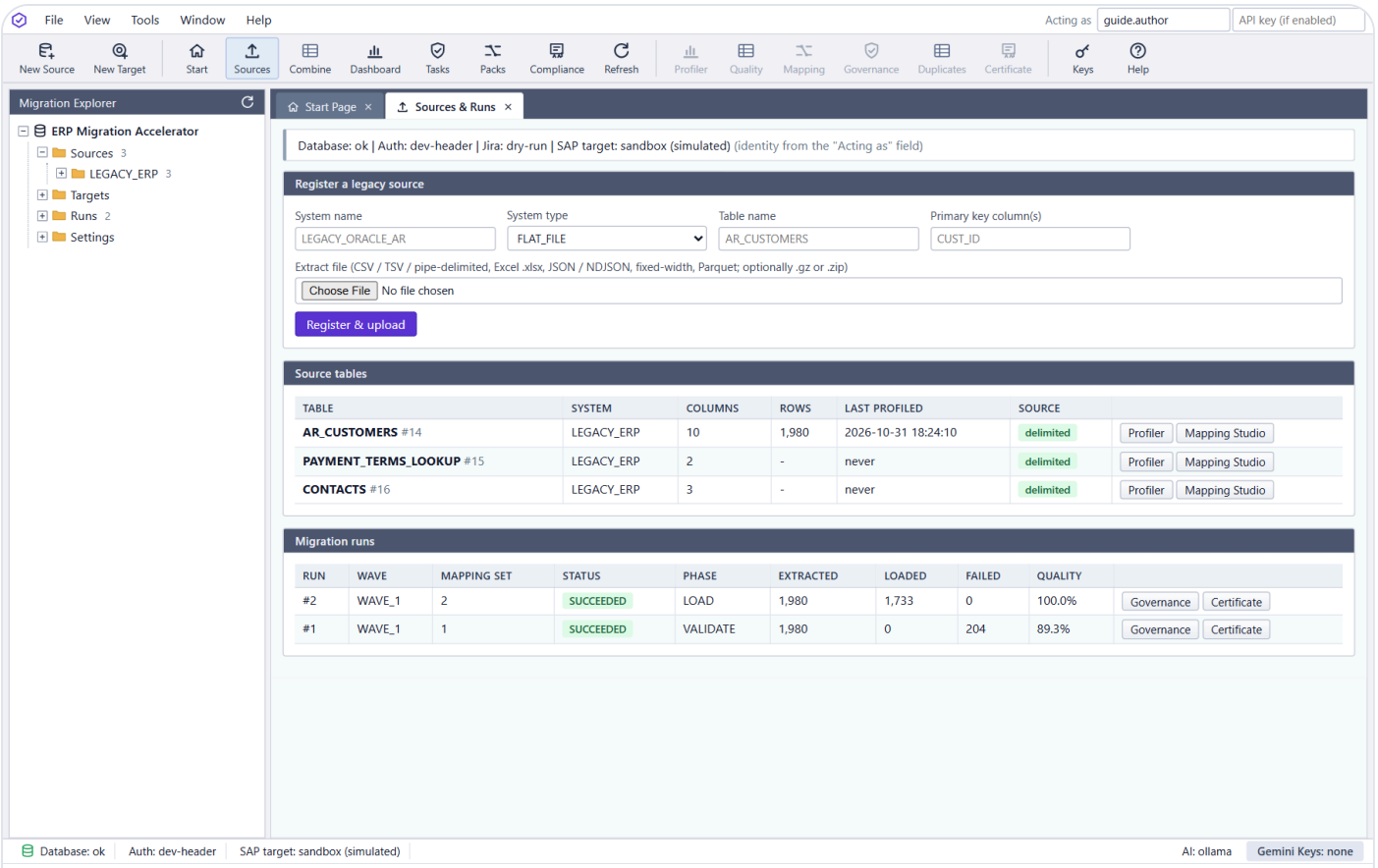


Figure 15. The **Sources & Runs** page: a quick form to register an extract file, the list of source tables (with buttons to open the Profiler or Mapping Studio) and the migration runs (with buttons for Governance and Certificate).

6.3 Profile a table

PR-3 Profile a table

PURPOSE

Measure what the data contains before any rule is written.

PERFORMED BY

Author

BEFORE YOU START

A registered table.

PROCEDURE

1. Select the table in the Explorer and press **Profiler** (it also opens automatically after registration).
2. Read the summary cards, then the column table, then the duplicates panel.

EXPECTED RESULT

Cards show rows, columns, average health, fully identical rows and candidate keys; one line per column follows.

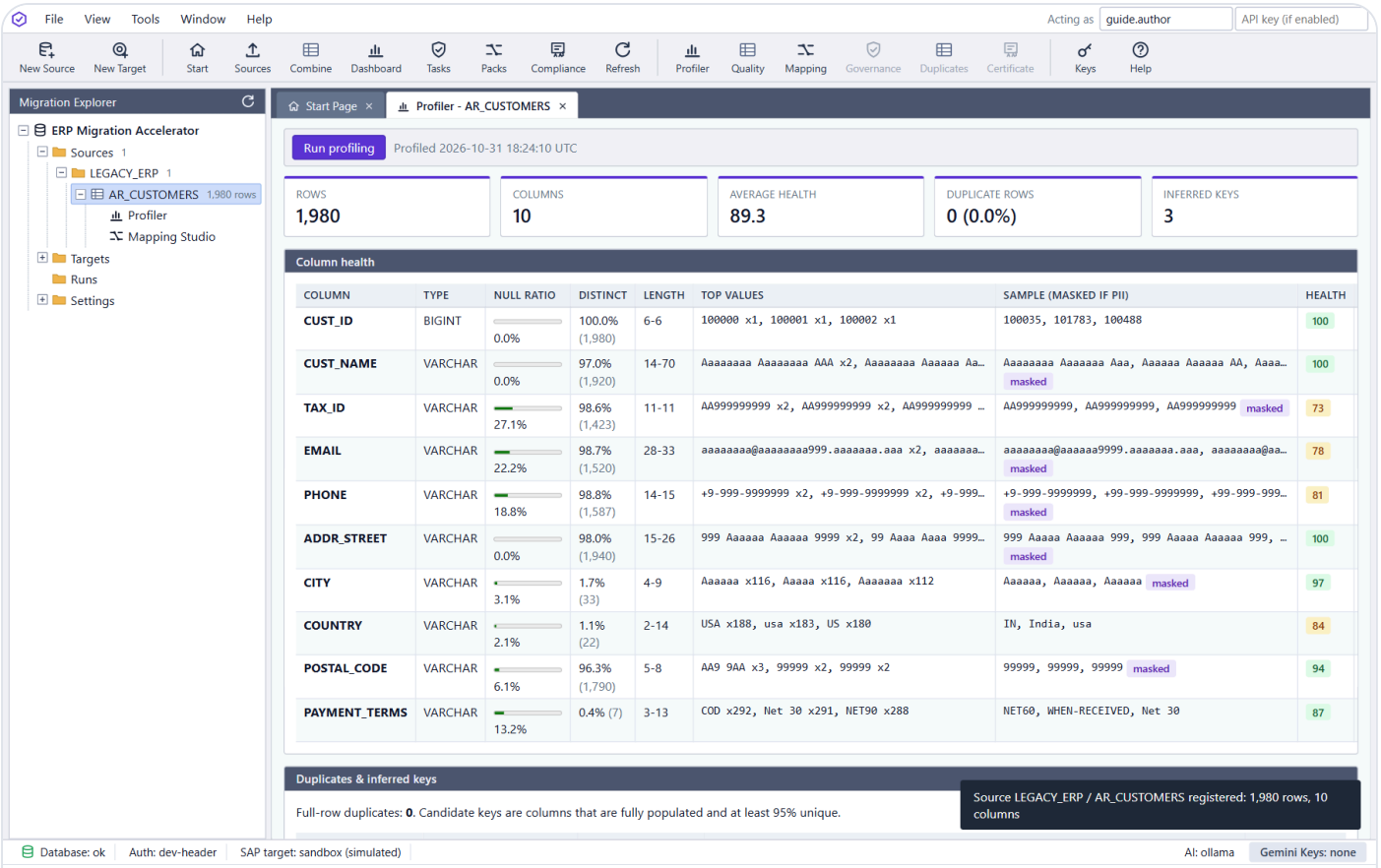


Figure 16. The Profiler: 1,980 rows, 10 columns, average health 89.3, 3 candidate keys.

COLUMN	MEANING
Type	The detected type: BIGINT, VARCHAR …
Null ratio	Share of blank cells (`TAX_ID` 27.1%, `CITY` 3.1%).
Distinct	Different values: `CUST_ID` 100% (a key), `COUNTRY` 22 spellings.
Length	Shortest and longest value: `CUST_NAME` 14 to 70 against SAP's 40.
Top values	Most frequent values: `USA` 188, `usa` 183, `US` 180: one country, three spellings.
Sample (masked)	Examples with letters shown as `A` / `a` and digits as `9`, so personal data is not exposed.
Health	0 to 100; green is good, amber deserves a look.

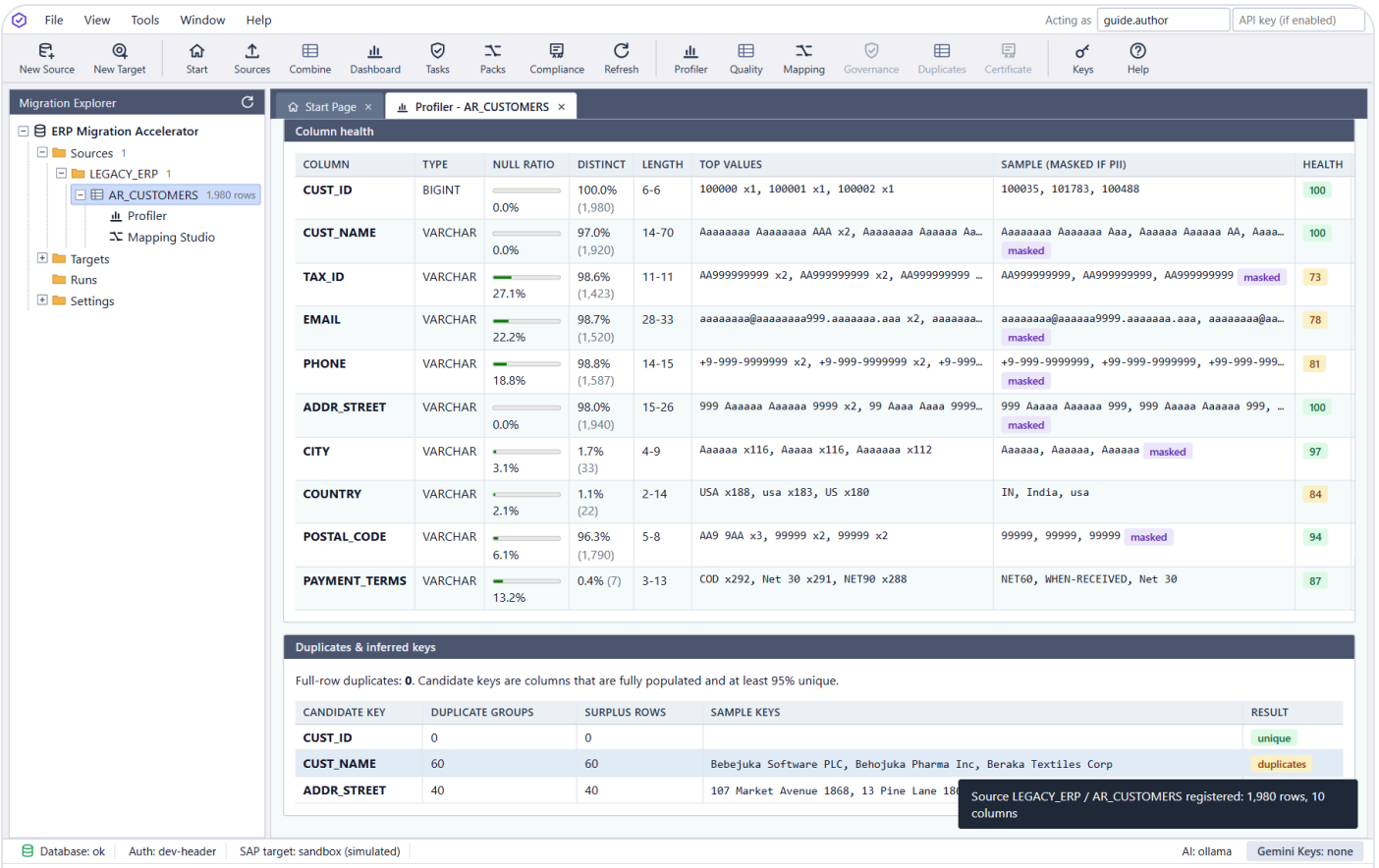


Figure 17. The column table.

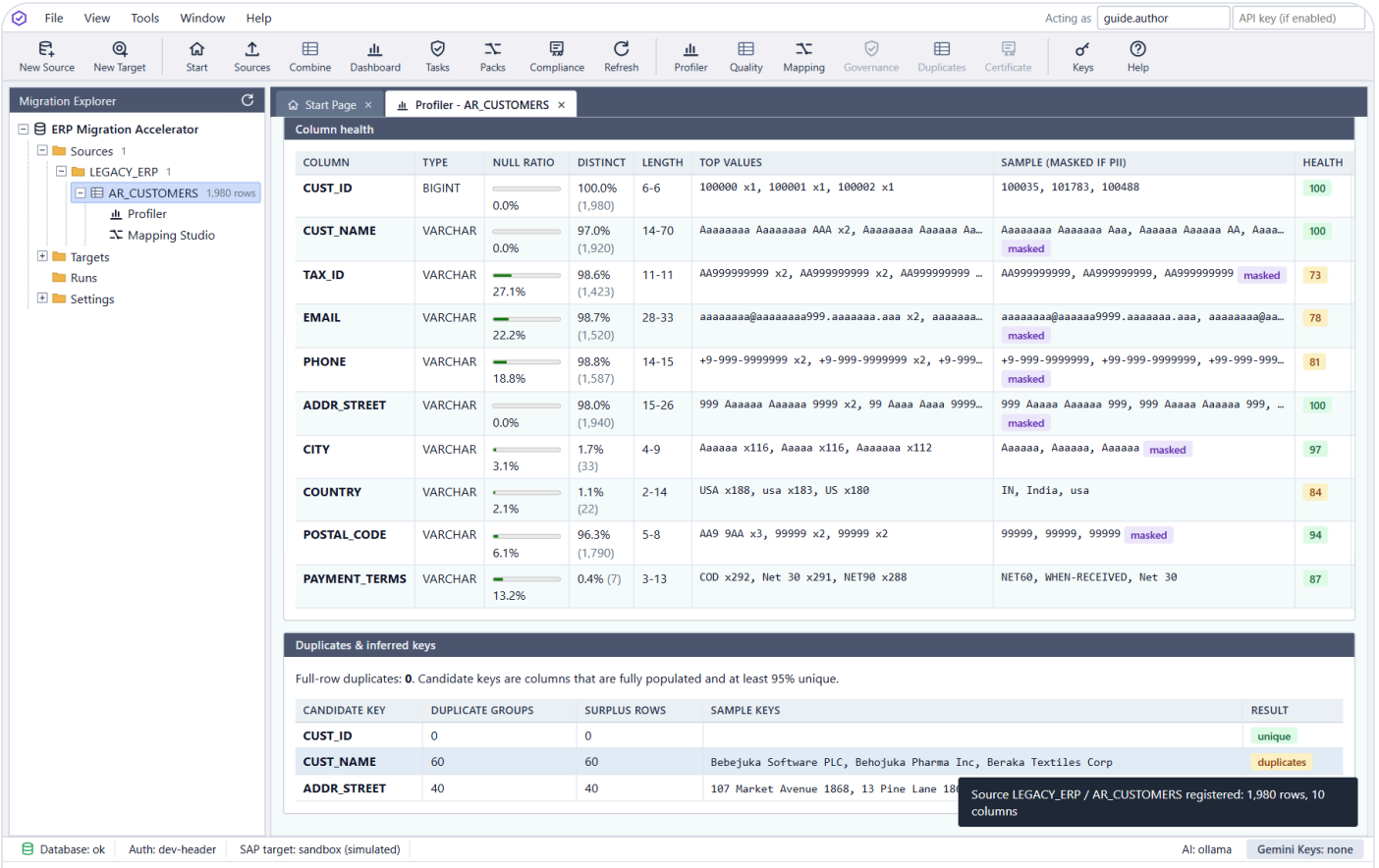


Figure 18. Duplicates and keys: `CUST_ID` is unique; 60 surplus rows share a name, 40 a street.

6.4 Assess data quality

PR-4 Assess data quality

PURPOSE

Run a library of tests and receive ranked findings with a recommended action each.

PERFORMED BY

Author

BEFORE YOU START

A profiled table.

PROCEDURE

1. Select the table and press **Quality**.
2. Press **Assess data quality now**.
3. Sort the findings by severity and read the **what to do** column.

EXPECTED RESULT

A score, the number of tests run and a findings table ranked worst first.

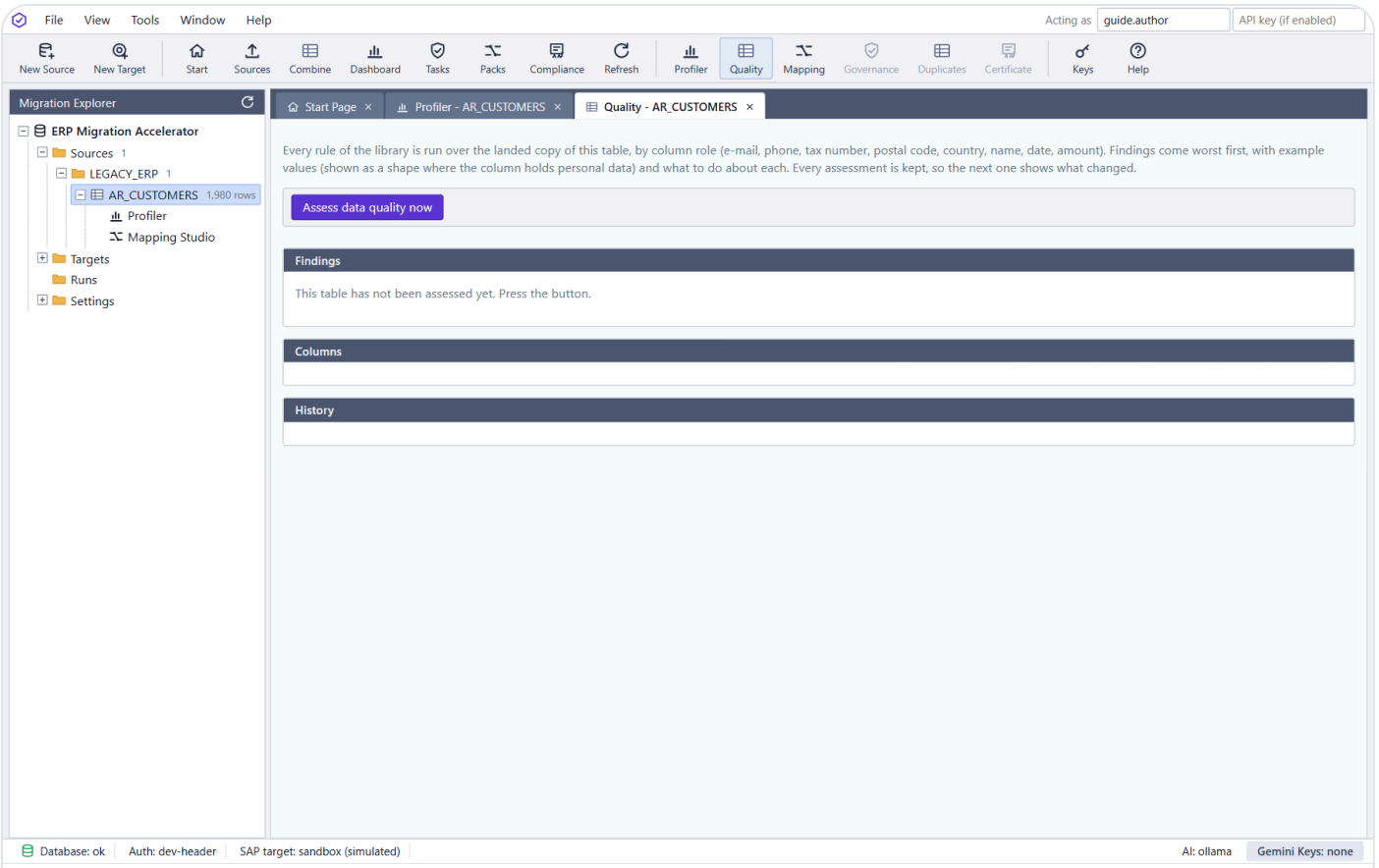


Figure 19. Quality page before the assessment.

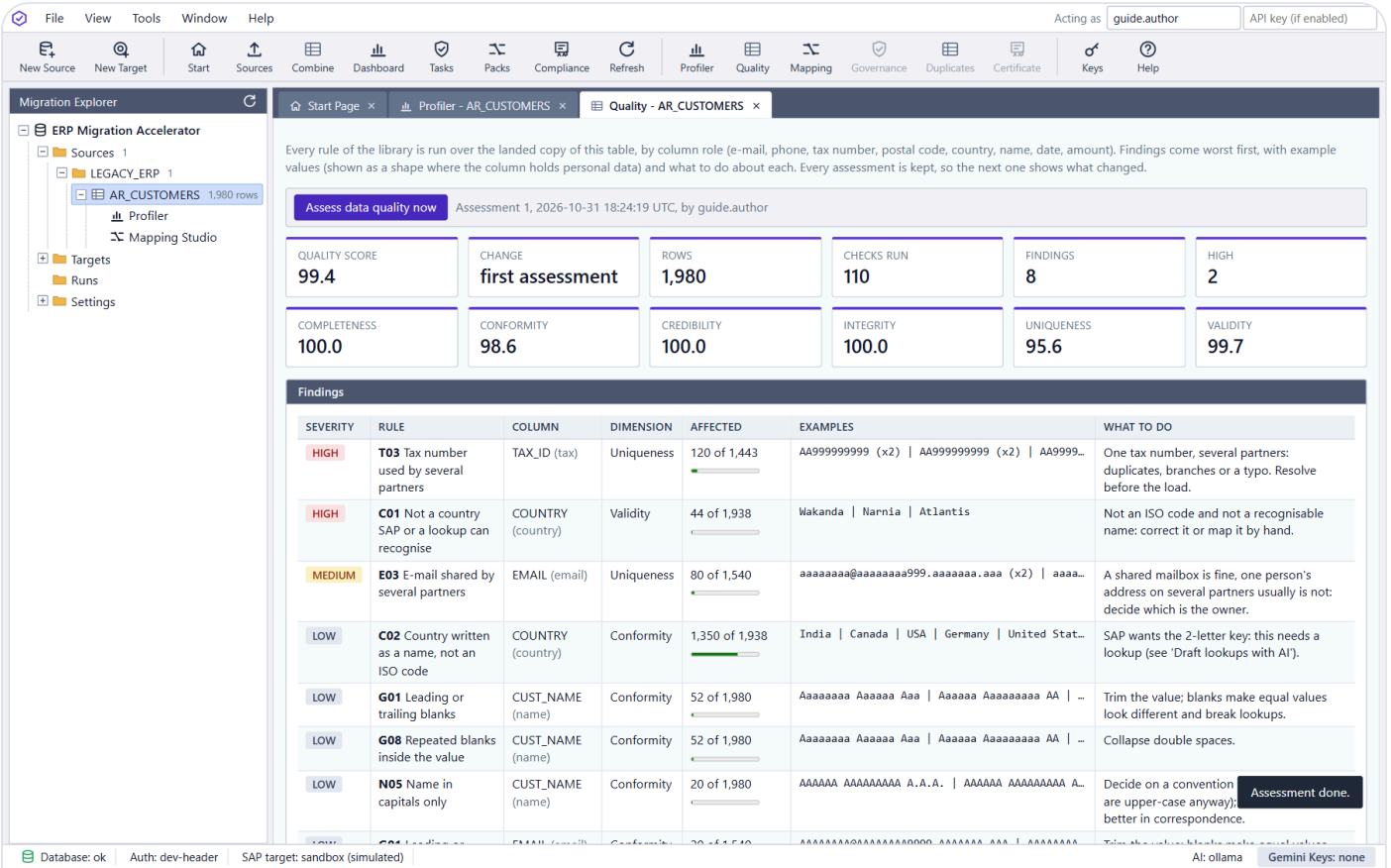


Figure 20. Result: 110 tests, score 99.4, 8 findings (2 HIGH).

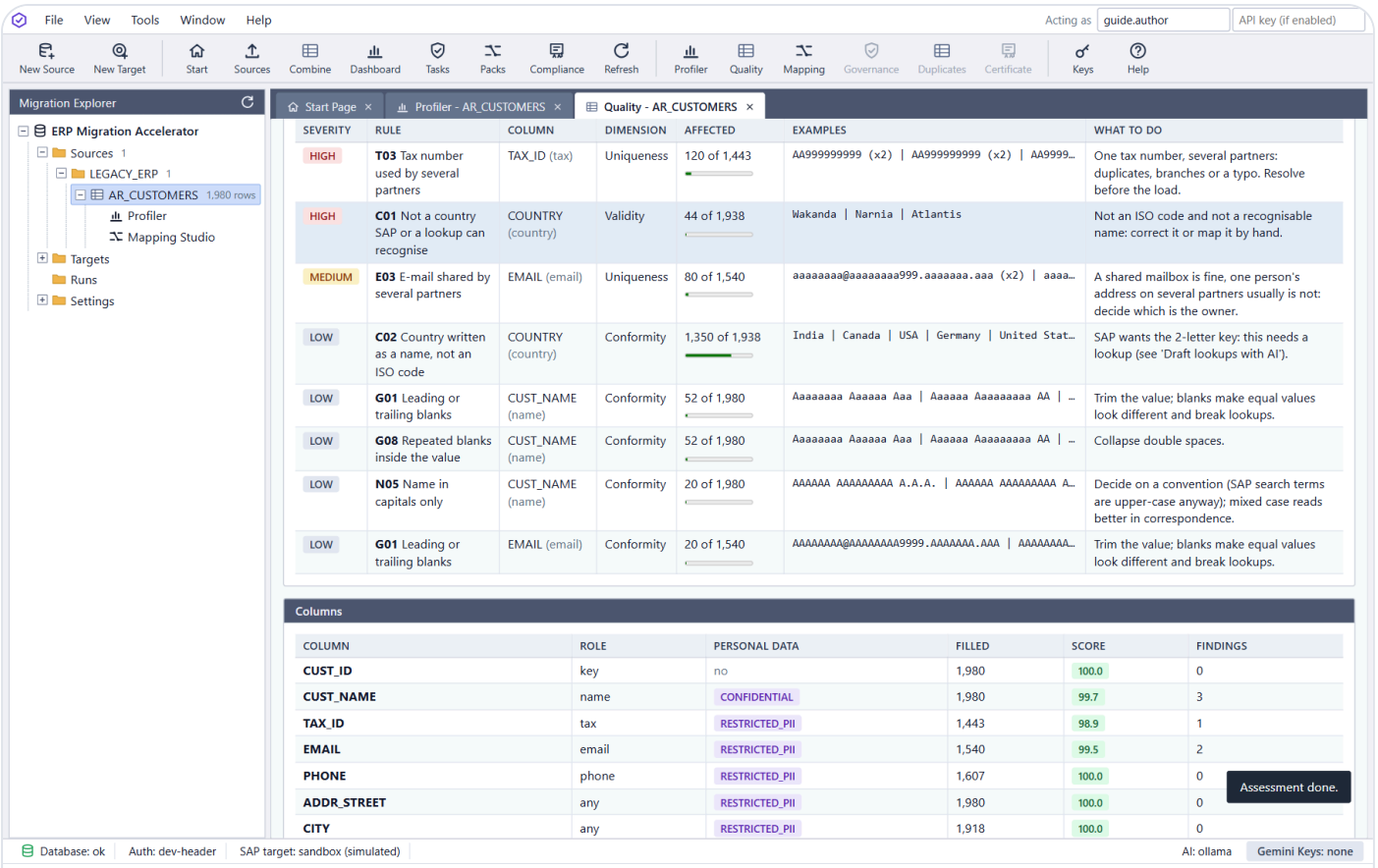


Figure 21. Each finding names the test, column, row count, an example and the recommended action.

FINDING	MEANING
HIGH T03 Tax number used by several partners	120 customers share a tax number: duplicates, branches or a typing error. A person decides.
HIGH C01 Not a country	44 rows name Wakanda, Narnia or Atlantis.
MEDIUM E03 E-mail shared	A shared mailbox is fine; one person's address on several customers usually is not.
LOW C02 Country written as a name	SAP needs `US`, not `United States`: a lookup is required.
LOW G01, G08, N05 …	Extra spaces, double spaces, names written in capitals.

CAUTION

A score of 99.4 is an average over many tests; a few rows with a grave problem hardly move it. Read the findings, not only the score.

6.5 Combine tables

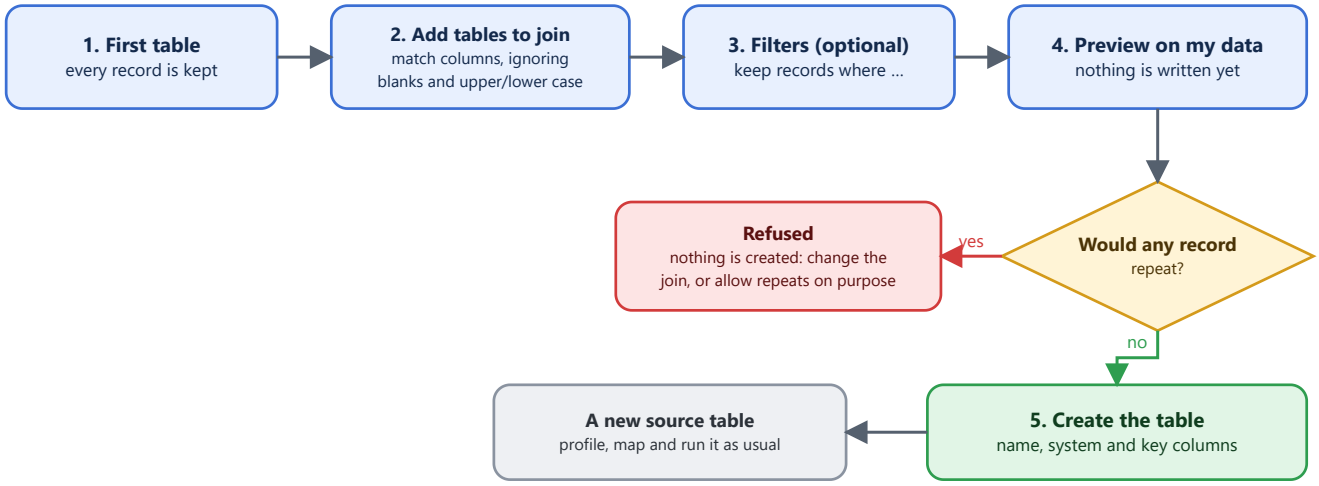


Figure 22. Combining tables. The preview refuses a join that would silently repeat records.

PR-5 Combine registered tables into one source table

PURPOSE

Join and filter tables you already registered (for example customers and payment terms) into one new source table, before mapping.

PERFORMED BY

Author

BEFORE YOU START

Two or more registered tables with a column that matches.

PROCEDURE

1. Press **Combine** in the toolbar.
2. In **Start from** choose the first table; every record of it is kept unless a filter or an inner join removes it.
3. Press **+ Add a table to join**. Choose the table, the column on each side and how to compare (ignoring blanks and letter case, or exact).
4. Optionally press **+ Add a filter** (*keep records where ...*).
5. Press **Preview on my data**. Read the counts: records in the first table, rows after joining, and for each join how many records find **no partner**.
6. If the preview is accepted, enter the **System name**, the **Table name** and the **Primary key column(s)** and press **Create table**.

EXPECTED RESULT

A banner reads *Created table #N with R rows* and links to the Profiler. The new table appears under Sources and is used like any other source; the snapshot records which tables it was built from, so it can be rebuilt.

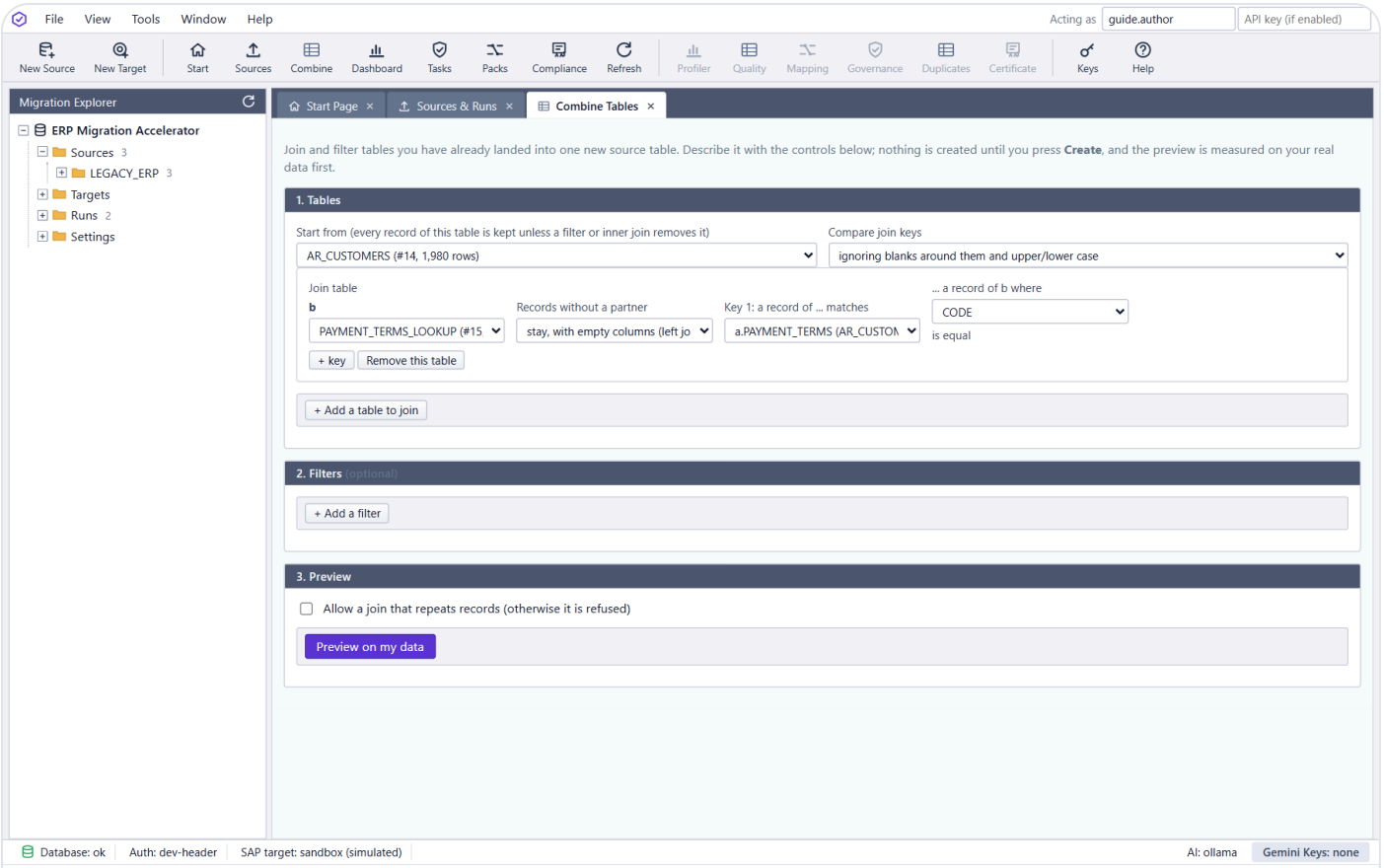


Figure 23. Join set-up: customers joined to a payment-terms table on `PAYMENT_TERMS = CODE`.

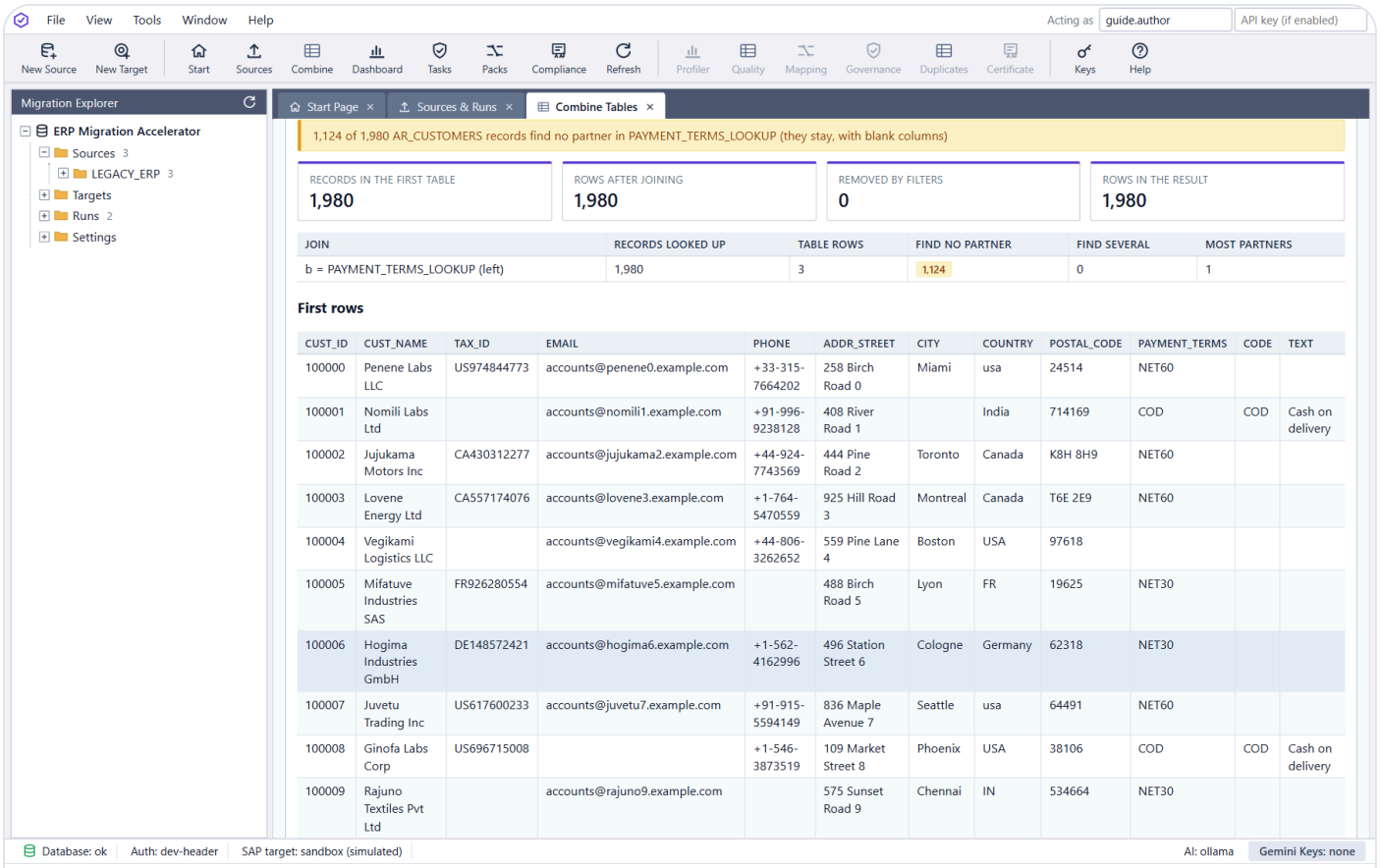


Figure 24. The preview before anything is created: 1,980 records stay 1,980 rows, and a warning says 1,124 customers find no partner in the lookup table (the data spells *NET30* where the table says *Net 30*), so the problem is seen before the table exists.

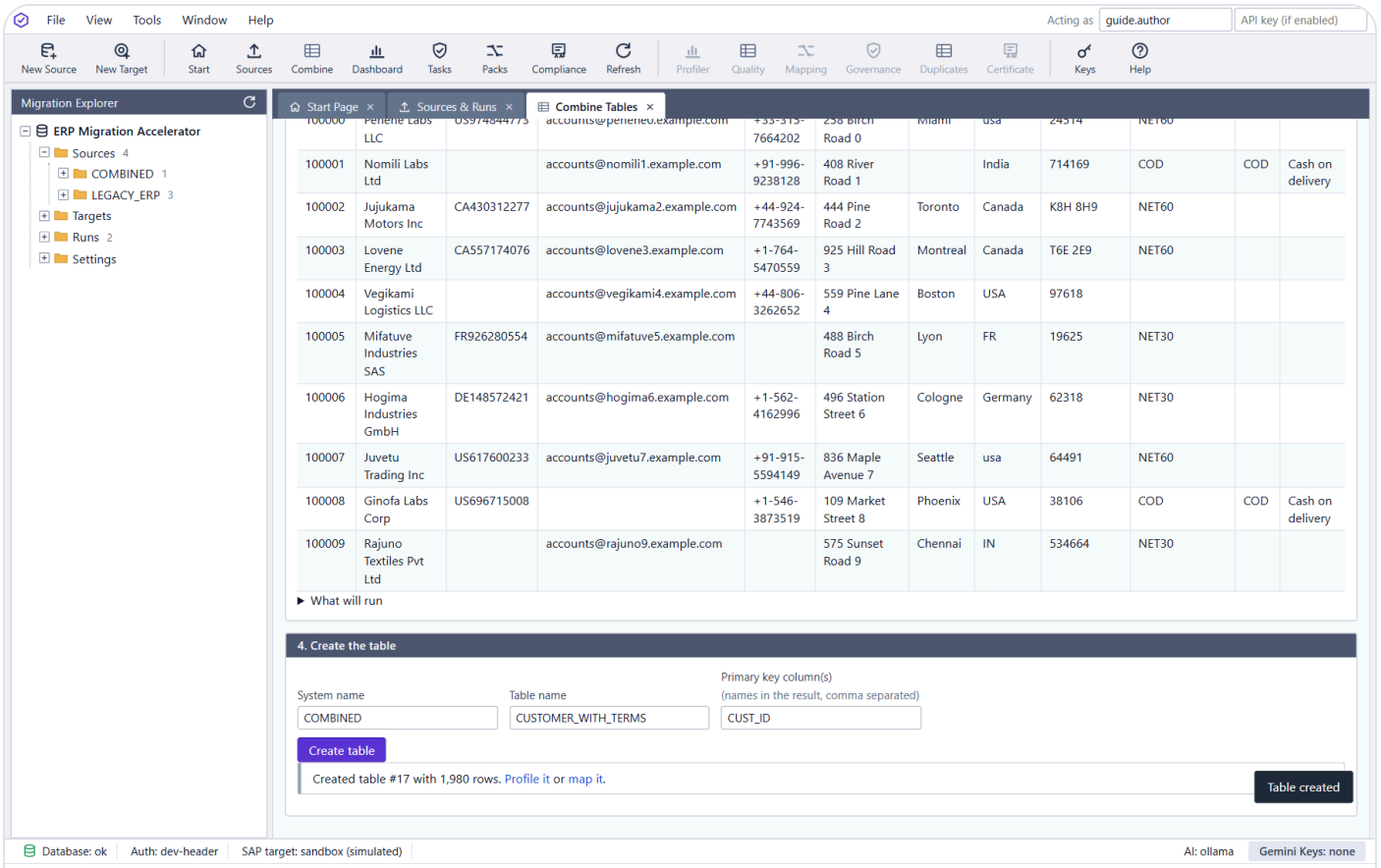


Figure 25. Creating the table: name, system and key columns.

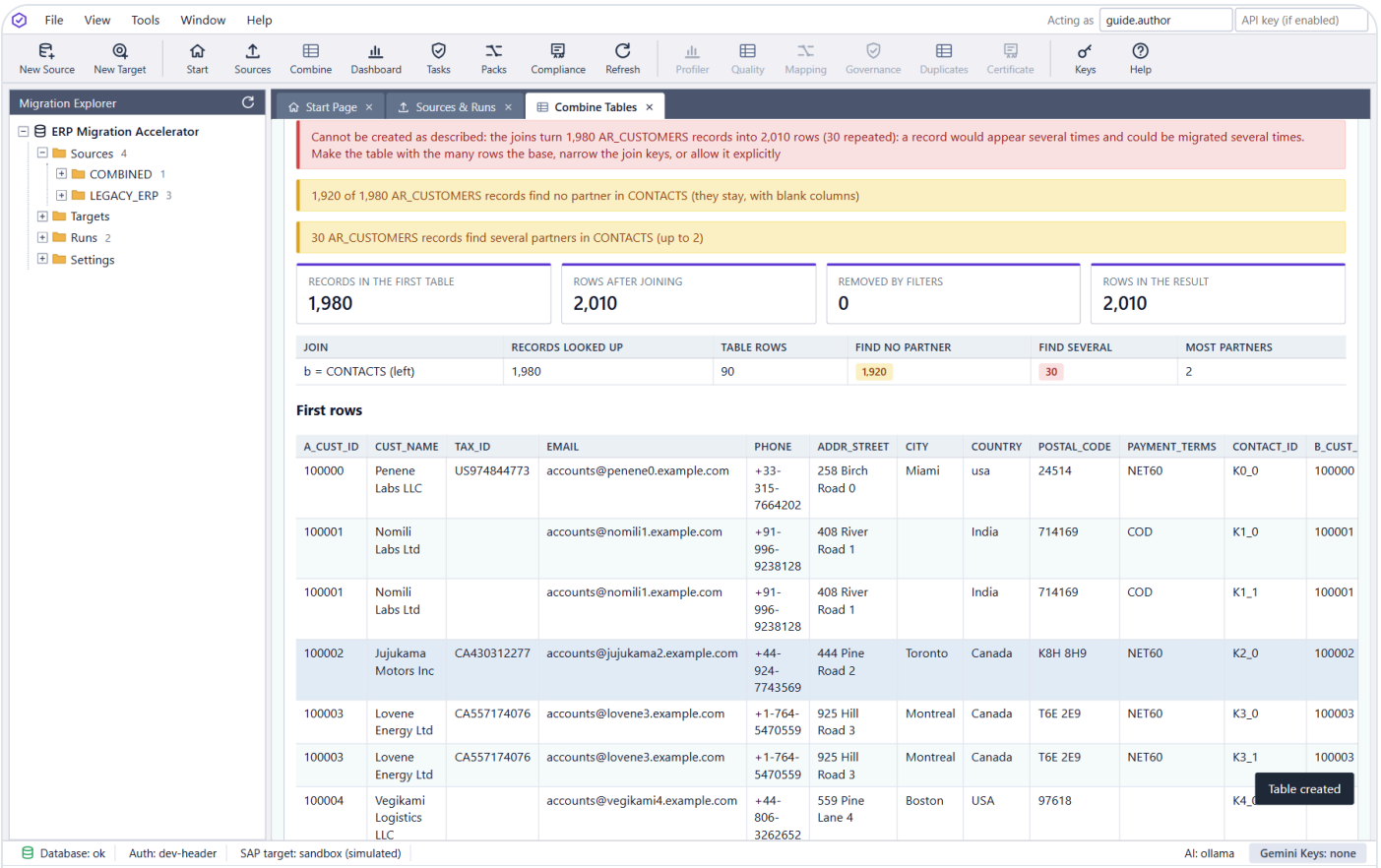


Figure 26. A join to a contacts table would turn 1,980 customers into 2,010 rows (30 repeated). The preview is refused in red and **Create table** is not offered.

CONTROL

A join that would repeat records is **refused** unless you tick **Allow a join that repeats records**, in which case the preview says so in plain words. Nothing is written until you press **Create table**.

6.6 Write the mapping

PR-6 Create a mapping set

PURPOSE

State, per SAP field, which legacy column fills it and how.

PERFORMED BY

Author

BEFORE YOU START

A registered, profiled table.

PROCEDURE

1. Select the table and press **Mapping** (or double-click **Mapping Studio** under the table).
2. Create rules in one of two ways, or both: **Generate AI mapping** (6.6.1) or **New empty mapping set** followed by **Add manual rule** (6.6.2).
3. Complete lookup dictionaries (6.6.3).
4. Press **Check rules on sample** (6.6.4) until it reports that the set can be submitted.

EXPECTED RESULT

A DRAFT mapping set with a card per rule, each showing the legacy column, the SAP field and its length, the kind of rule and a sample pass rate.

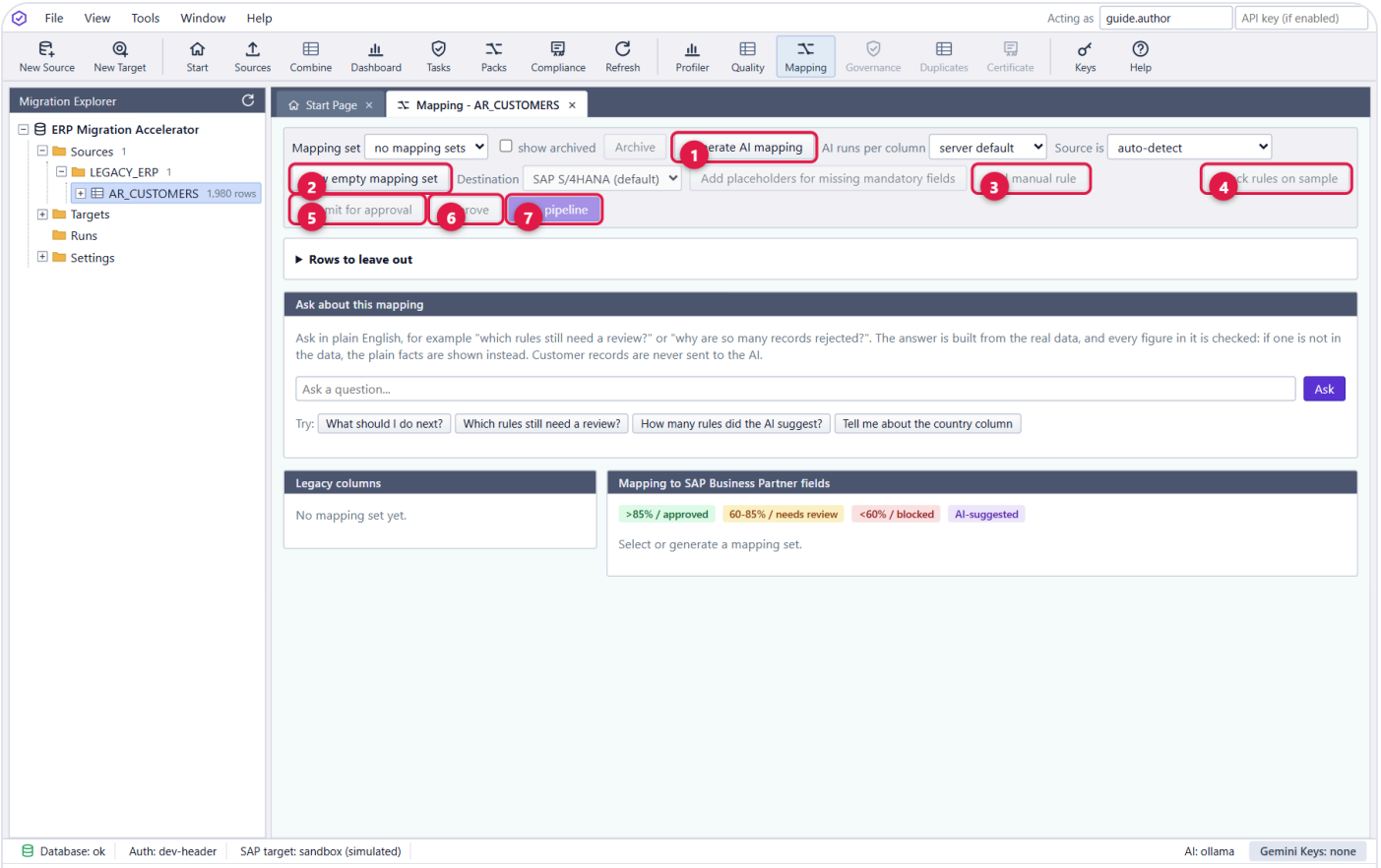


Figure 27. The Mapping Studio. 1 Generate AI mapping · 2 New empty mapping set · 3 Add manual rule · 4 Check rules on sample · 5 Submit · 6 Approve · 7 Run pipeline.

6.6.1 Let the AI propose

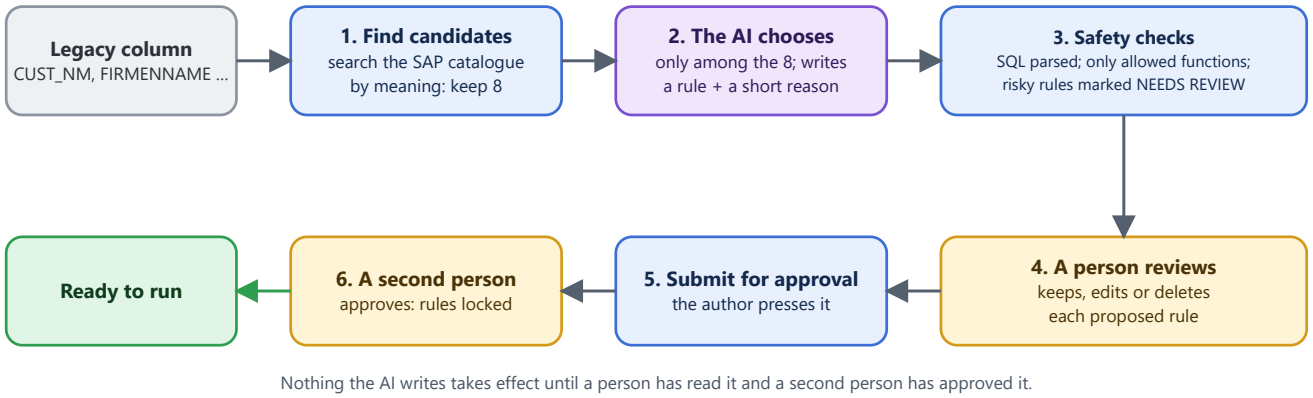


Figure 28. AI mapping: the model chooses only among SAP fields found by a semantic search, and every proposal is checked by rules and by people.

1. Press **Generate AI mapping**. Choose a consistency level if offered.
2. The job runs on the server; a progress bar shows the column being examined. You may leave the page.
3. Review every proposal: confidence, reason and the tag *AI-suggested*.

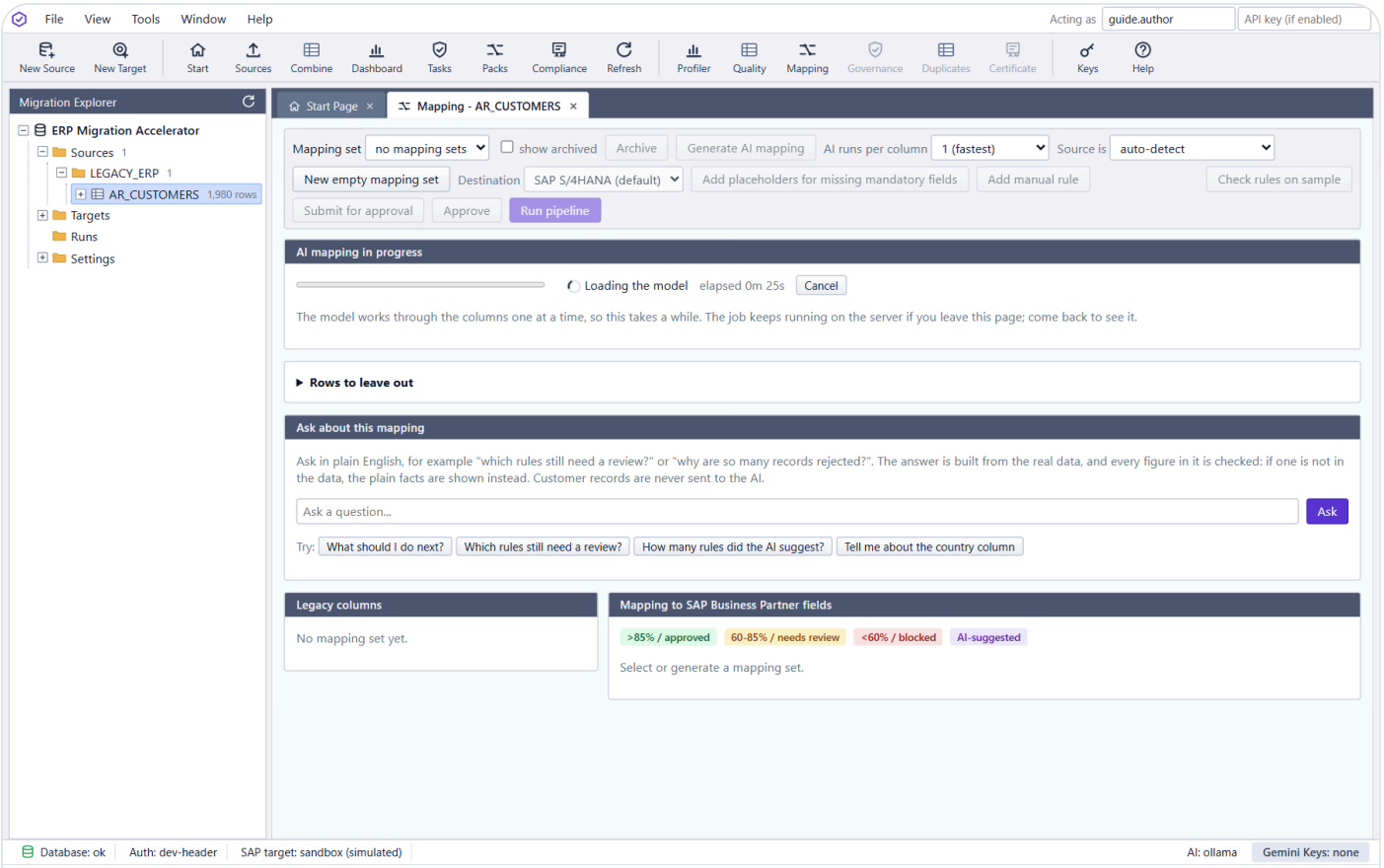


Figure 29. The AI examines one column at a time.

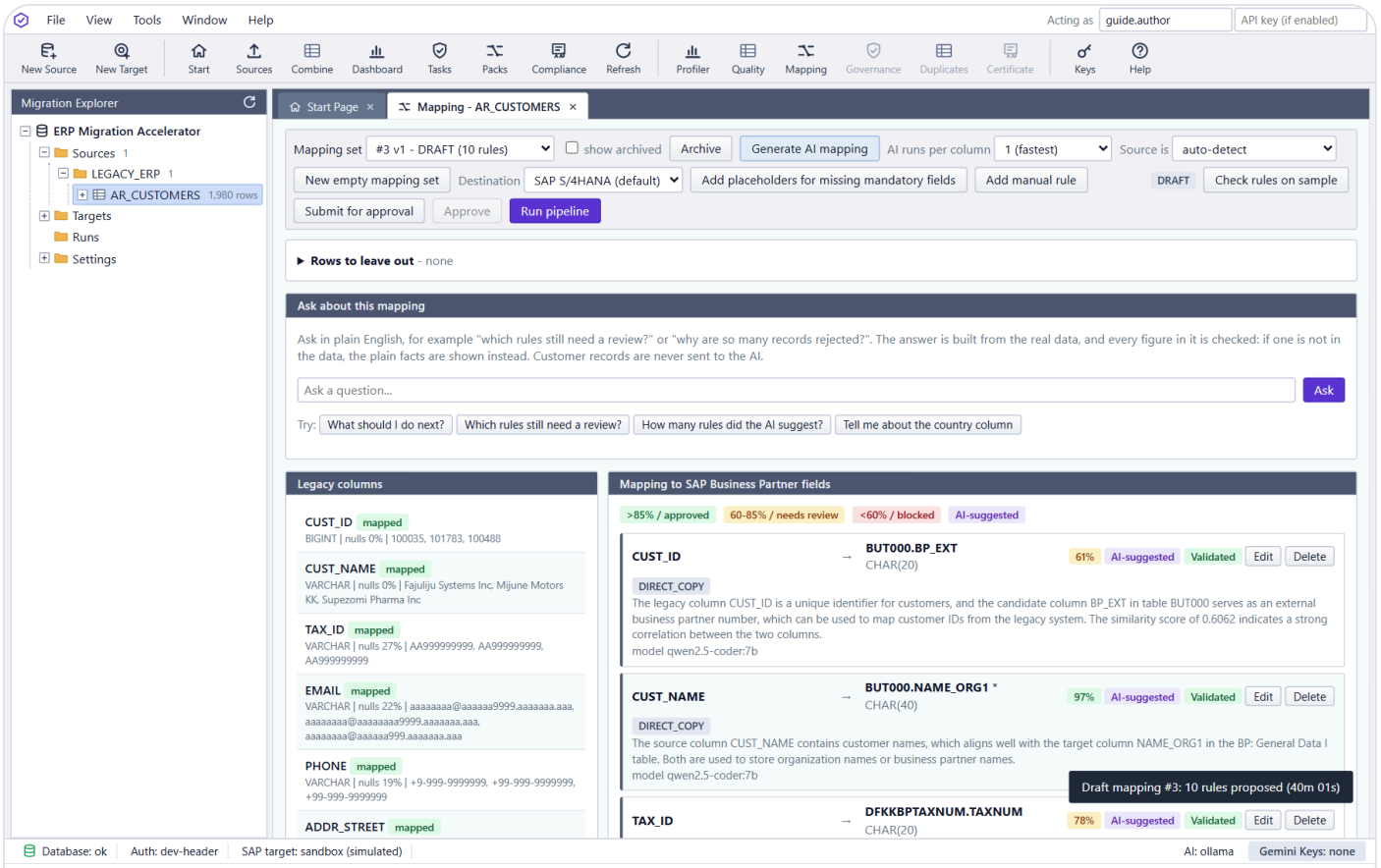


Figure 30. Each proposed rule carries a confidence and the model's name.

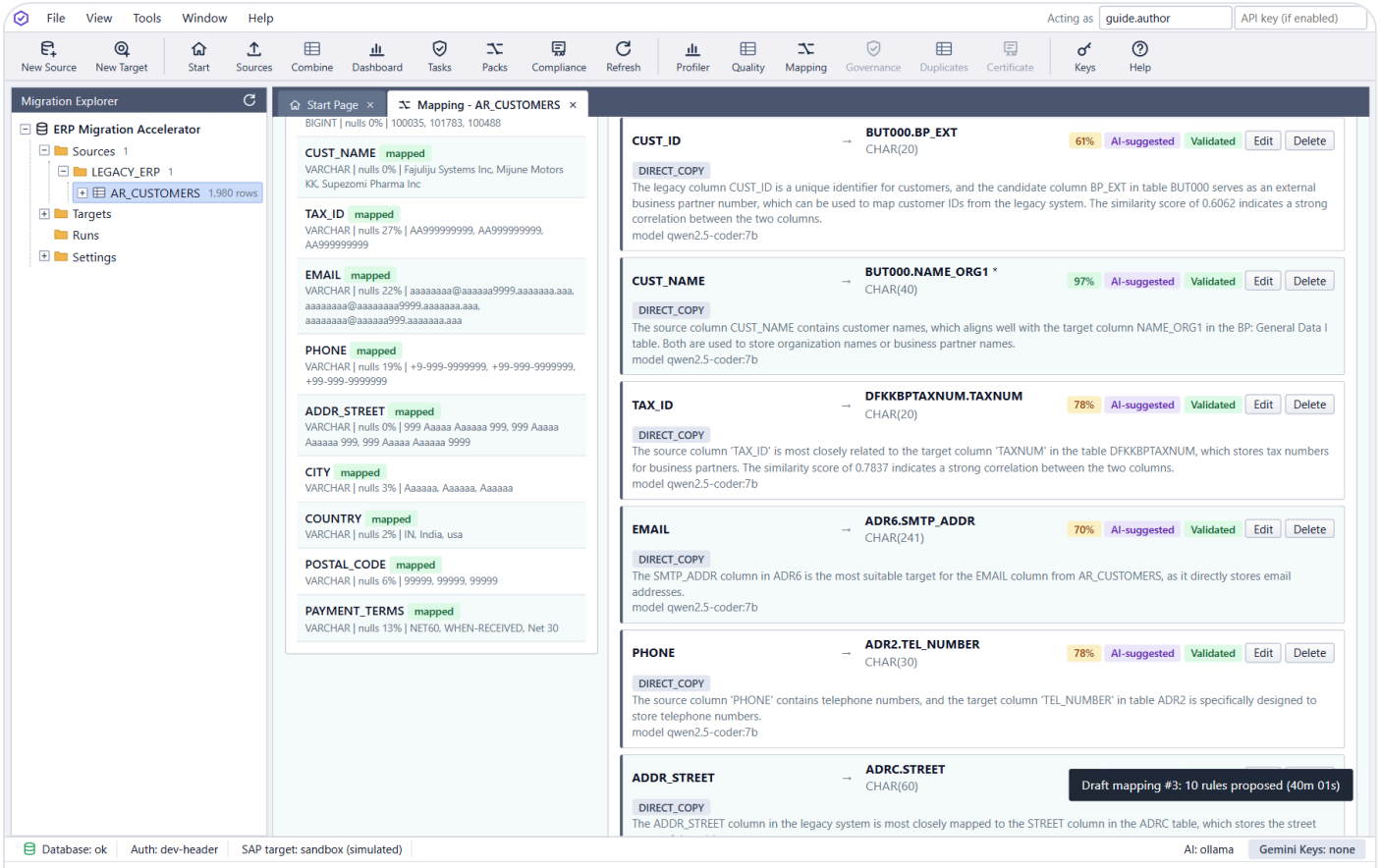


Figure 31. Each proposal states its reason in words, so a person can judge it.

SAFEGUARD	WHAT IT DOES
Candidate retrieval	The model may choose only among the closest SAP fields found by meaning (including German names such as *Firmenname*).
Rule validator	Parses SQL, allows only listed functions, rejects absurd lengths, and marks rules that rewrite values (padding, replacing, translating) **NEEDS_REVIEW**.
Value check	Samples real values: if most are not valid country codes, a country rule is marked for review.
Colours	Green: confident (over 85%). Amber: review (60 to 85%). Red: blocked (under 60%). Never auto-approved.
Human approval	Nothing takes effect until a second person approves the set.

CAUTION

What the AI does not do. In the evaluation recorded in `docs/ai-evaluation.md` it proposed plain copies and chose the right SAP field for 34 of 34 test columns after the German-alias fix. It did not invent the country dictionary, the payment-term codes or the fixed values your business must supply. Confidence percentages did not by themselves separate right from wrong answers; the value check did. On a CPU-only laptop a local 7B model needs minutes per column; an online model is faster if policy allows it.

6.6.2 Write the rules yourself

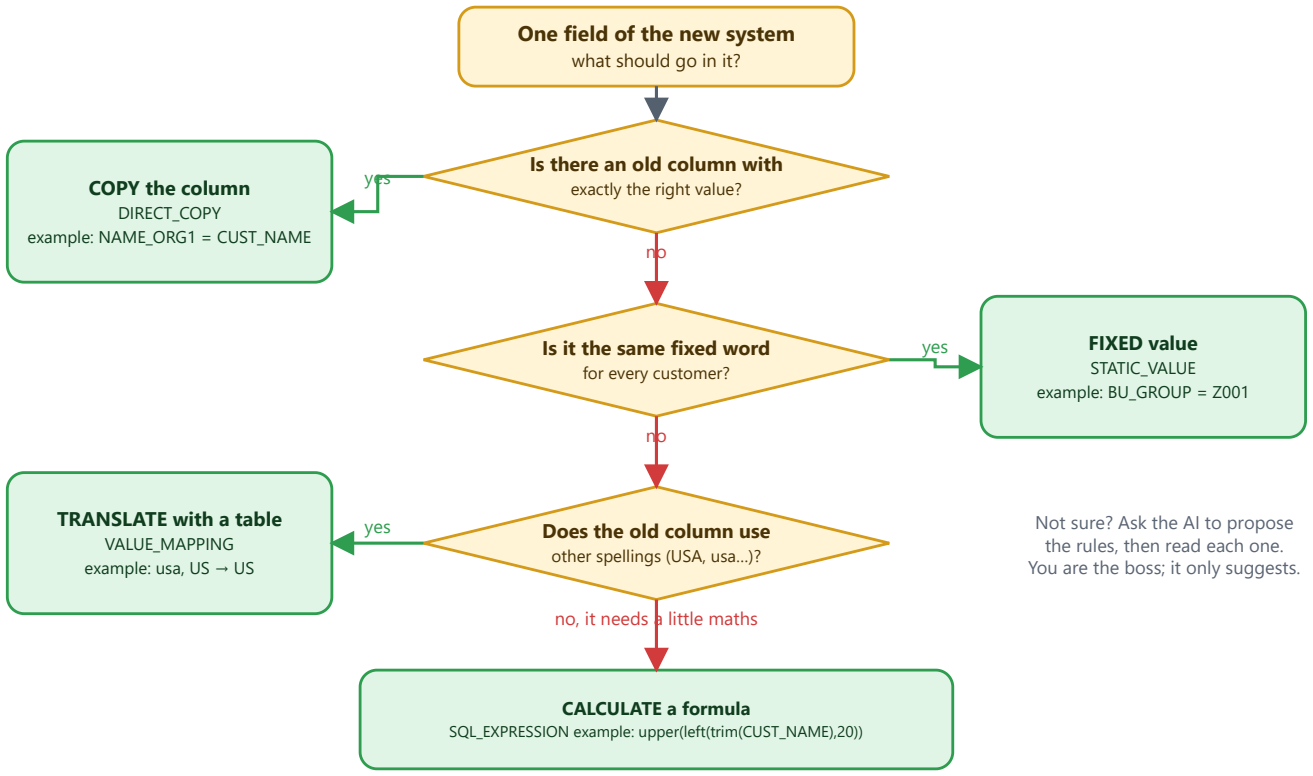


Figure 32. Choosing the kind of rule.

KIND	MEANING	EXAMPLE
`DIRECT_COPY`	Copy the column unchanged	`CUST_NAME` → `BUT000.NAME_ORG1`
`STATIC_VALUE`	The same fixed value for every row	`'Z001'` → `BUT000.BU_GROUP`
`VALUE_MAPPING`	Translate through a lookup	`usa, United States → US` → `ADRC.COUNTRY`
`SQL_EXPRESSION`	A small formula	`upper(left(trim(CUST_NAME), 20))` → `BU_SORT1`

1. Press **New empty mapping set**. Its status is DRAFT.
2. Press **Add manual rule**; choose the SAP table and field, the kind, the legacy column and the formula or fixed value (text in quotes). Press **Add rule**.
3. Repeat for every field.

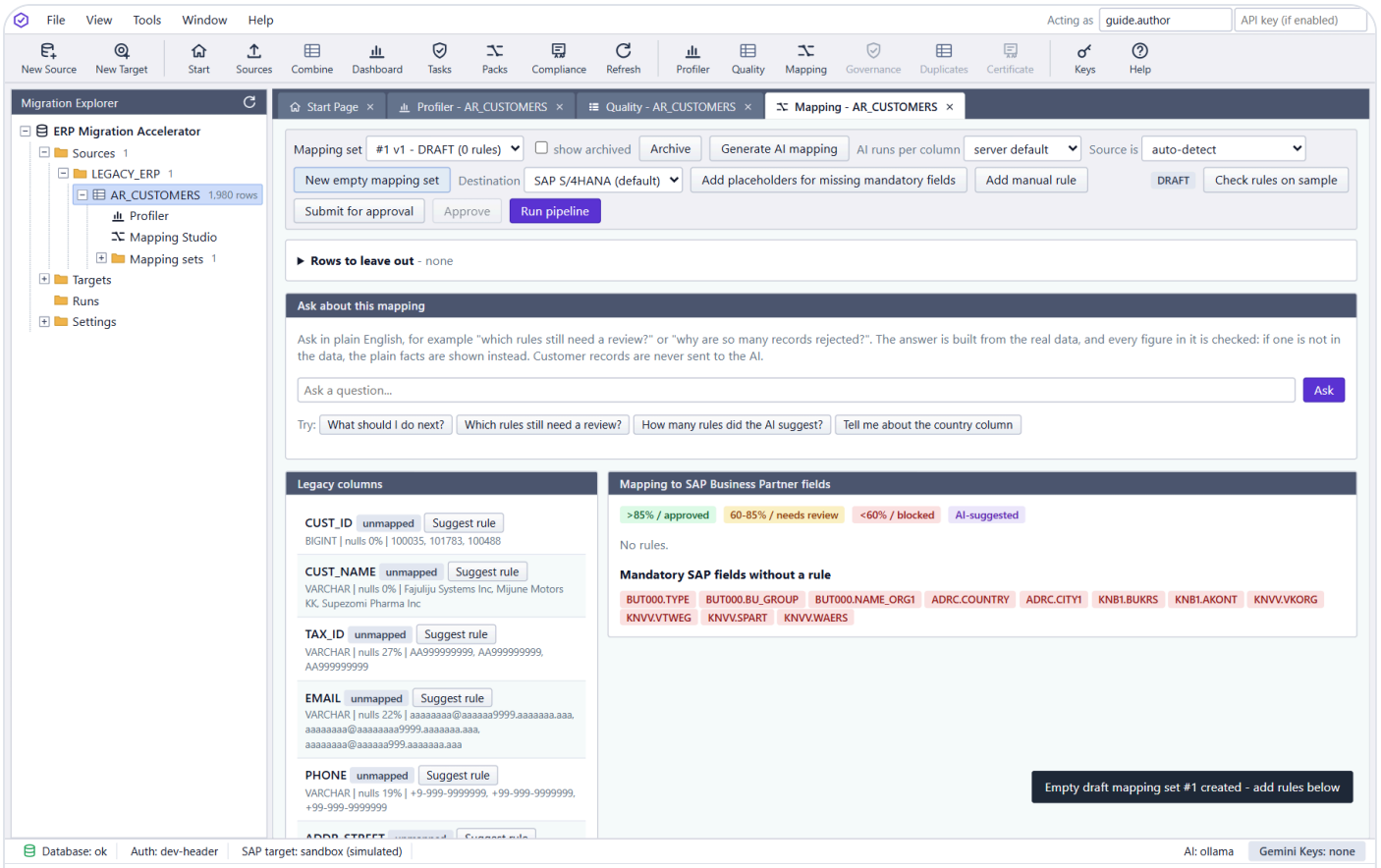


Figure 33. A new, empty mapping set in status DRAFT.

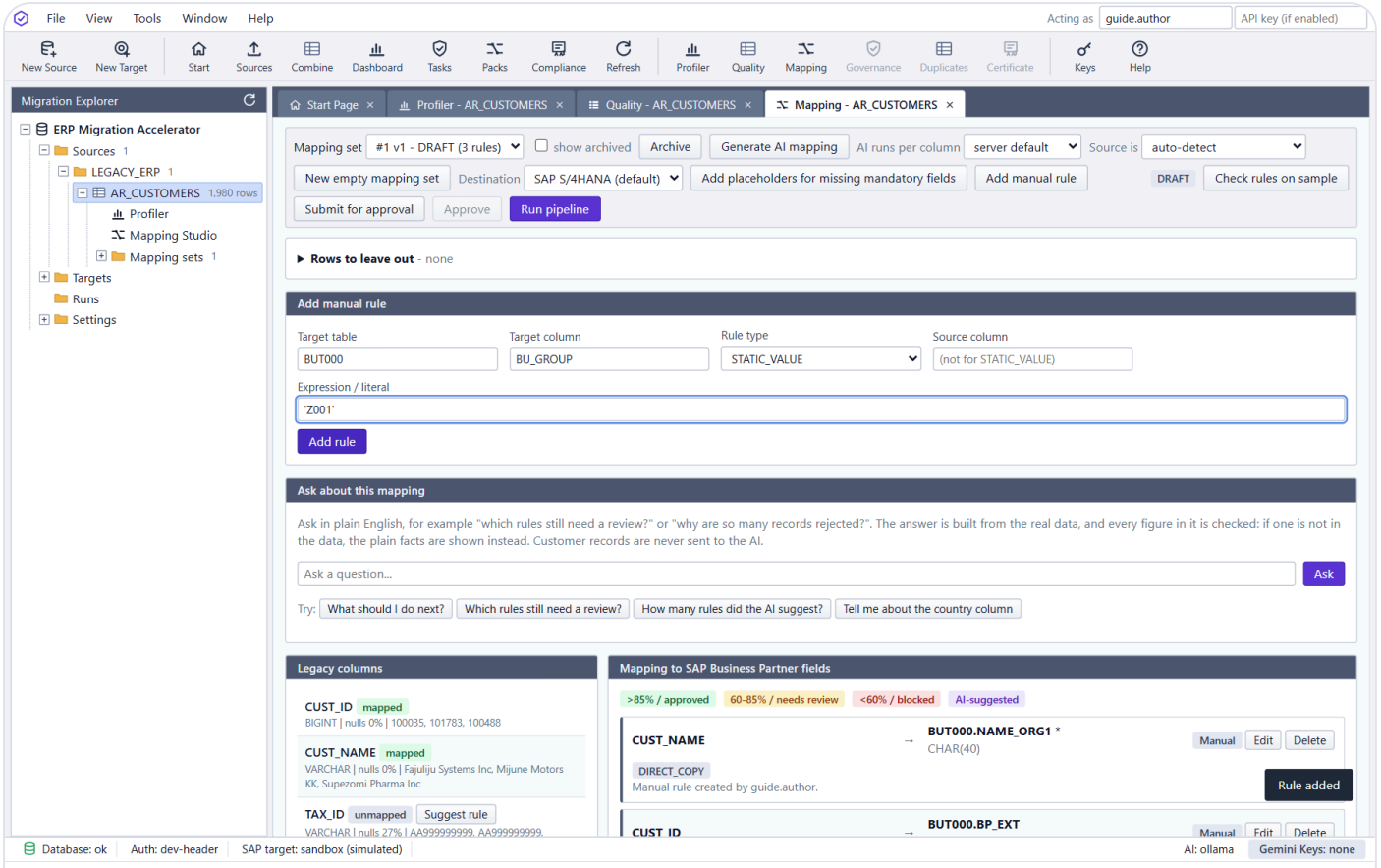


Figure 34. Adding a rule: the fixed value `'Z001'` for `BUT000.BU_GROUP`.

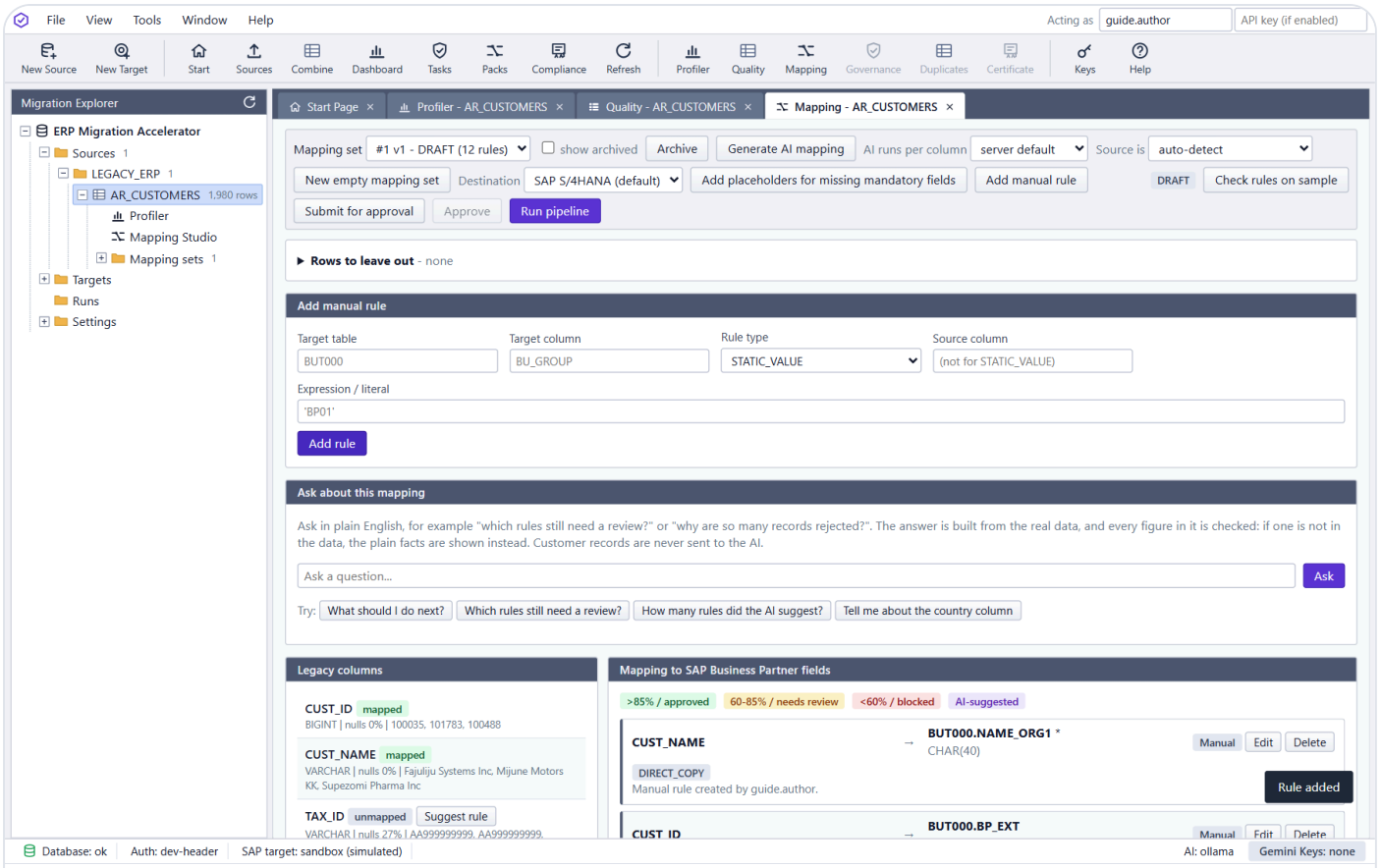


Figure 35. The rules with their SAP field lengths and sample pass rates.

SAP TABLE	HOLDS	FIELDS USED
`BUT000`	The business partner	`NAME_ORG1` (name), `BP_EXT`, `TYPE`, `BU_GROUP`, `BU_SORT1`
`ADRC`	Address	`STREET`, `CITY1`, `POST_CODE1`, `COUNTRY`
`ADR6` / `ADR2`	E-mail / telephone	`SMTP_ADDR` / `TEL_NUMBER`
`KNB1`	Accounting data	`ZTERM` (payment terms), `BUKRS` (company code)

6.6.3 Complete the lookup dictionaries

1. On a `VALUE_MAPPING` rule press **Edit**.
2. In **Lookups** enter pairs as `{"old value": "new value"}`.
3. Tick **Confirm as reviewed** (you state that you read them) and press **Save rule**.

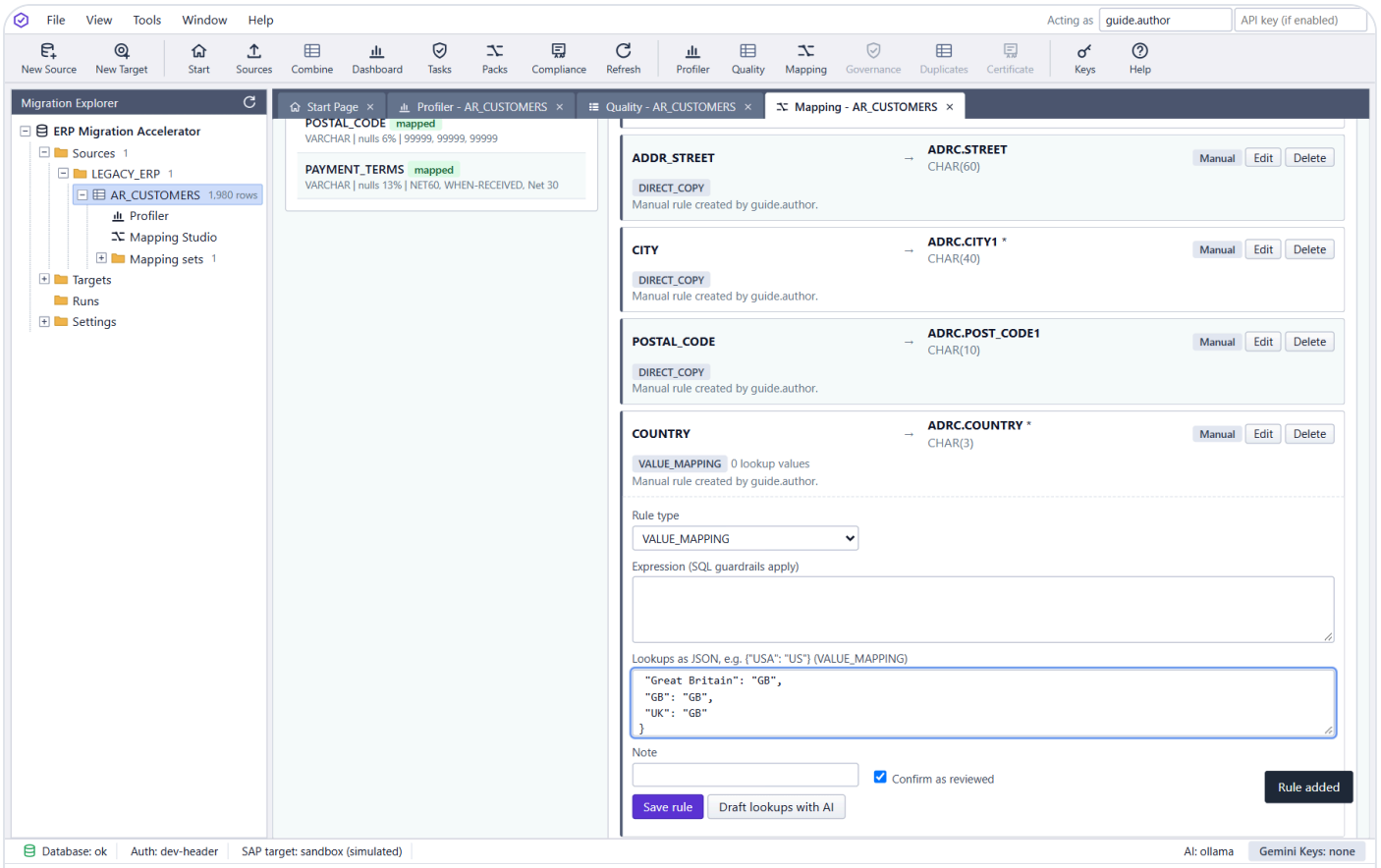


Figure 36. A country dictionary being edited: `Great Britain → GB`, `UK → GB`.

> **NOTE**
>
> Matching ignores letter case: `USA → US` also finds `usa` and `Usa`.

6.6.4 Check the rules on a sample

1. Press **Check rules on sample**. The rules are applied to the real rows as a dry run; nothing is saved.
2. Read the result: whether the set can be submitted, warnings, the pass rate and **What to do next**.

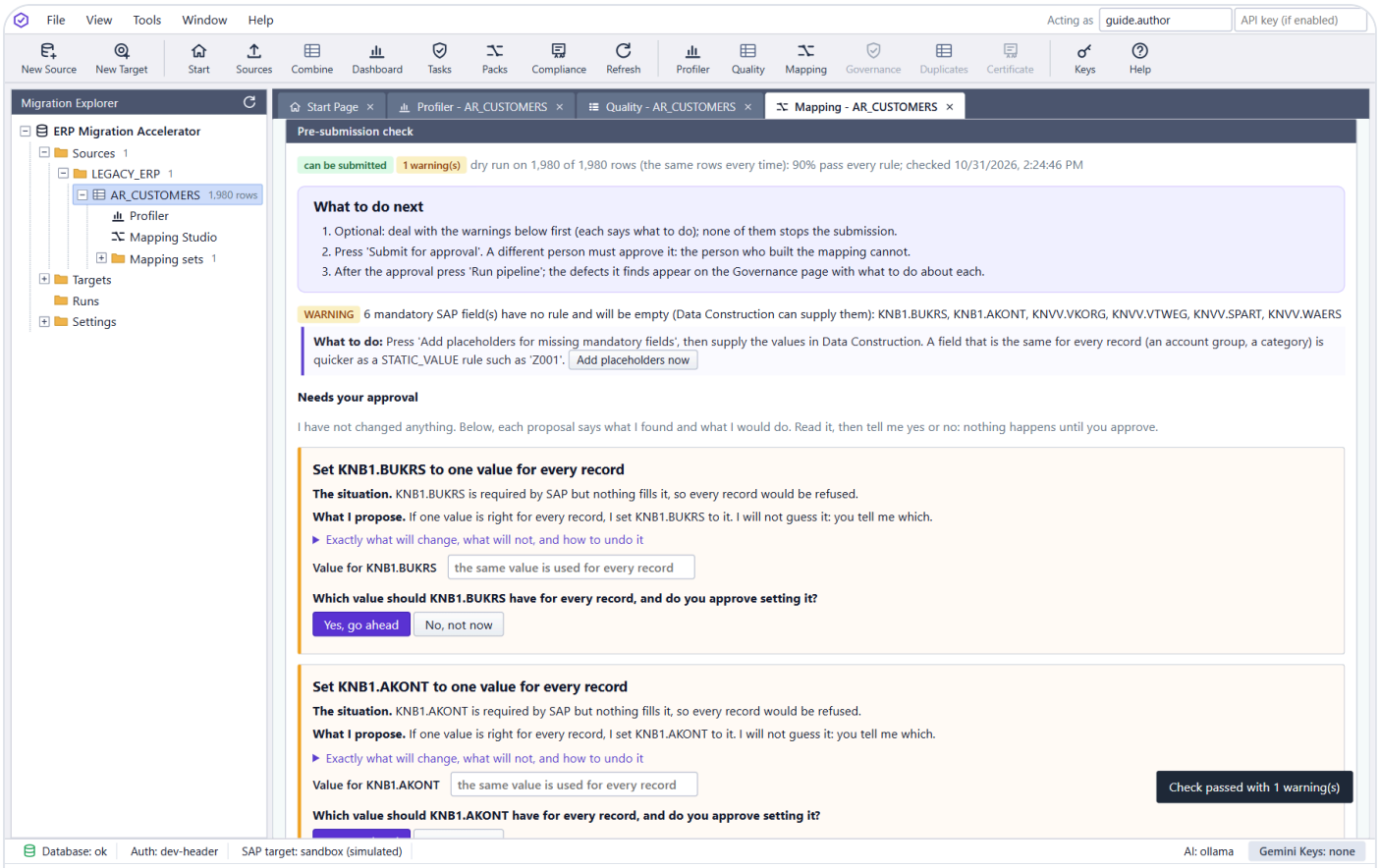


Figure 37. The pre-submission check: can be submitted; one warning (6 mandatory SAP fields have no rule); 90% of 1,980 rows pass every rule.

> **NOTE**
>
> A warning does not prevent submission. The 10% that fail are exactly what the gate will list after the first run (6.9).

6.7 Submit and approve

PR-7 Submit a mapping set and approve it

PURPOSE

Obtain an independent review; lock the rules.

PERFORMED BY

Author submits; a different person approves

BEFORE YOU START

A DRAFT set whose check passed.

PROCEDURE

1. As the author press **Submit for approval**. The status becomes **SUBMITTED**.
2. Change **Acting as** to the approver's name (in a keyed installation the approver signs in with their own key).
3. Read the rules and press **Approve**.

EXPECTED RESULT

Status **APPROVED**, with the approver's name and time. The rules can no longer be edited.

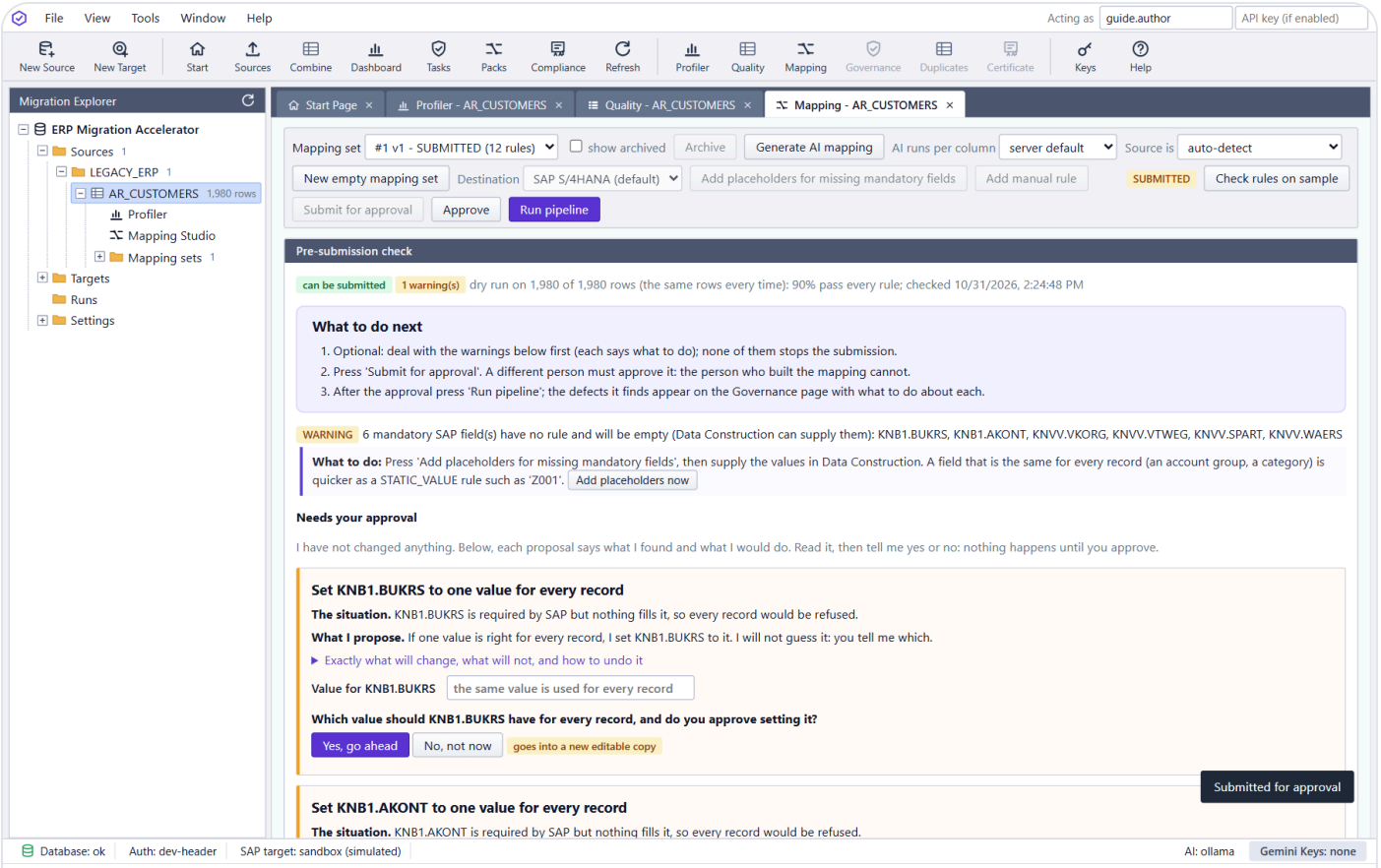


Figure 38. Submitted: waiting for somebody else.

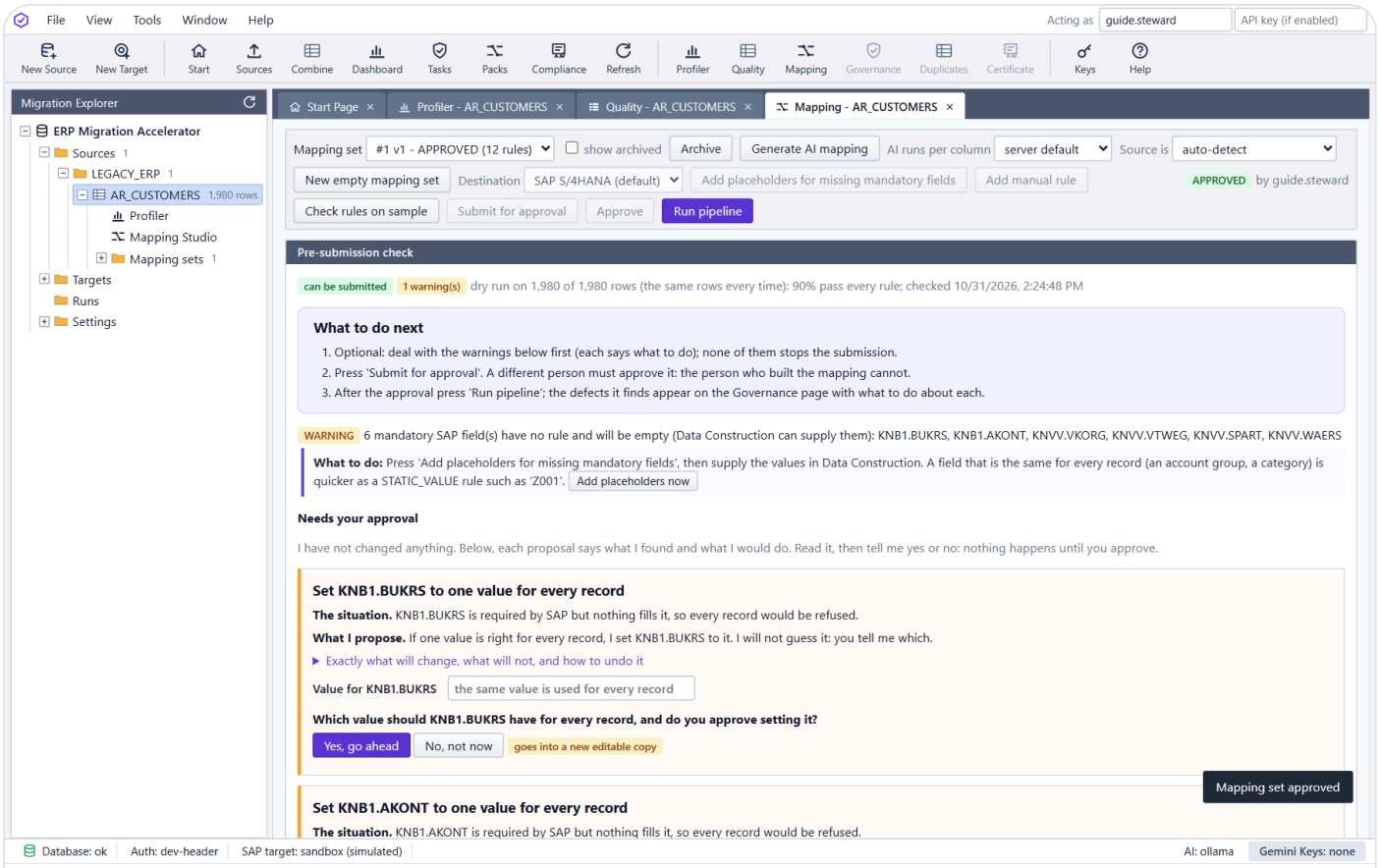


Figure 39. Approved by a different user; the rules are locked.

CONTROL

An approval by the author is refused (segregation of duties). The approval is part of the audit trail and the certificate.

6.8 Run the pipeline

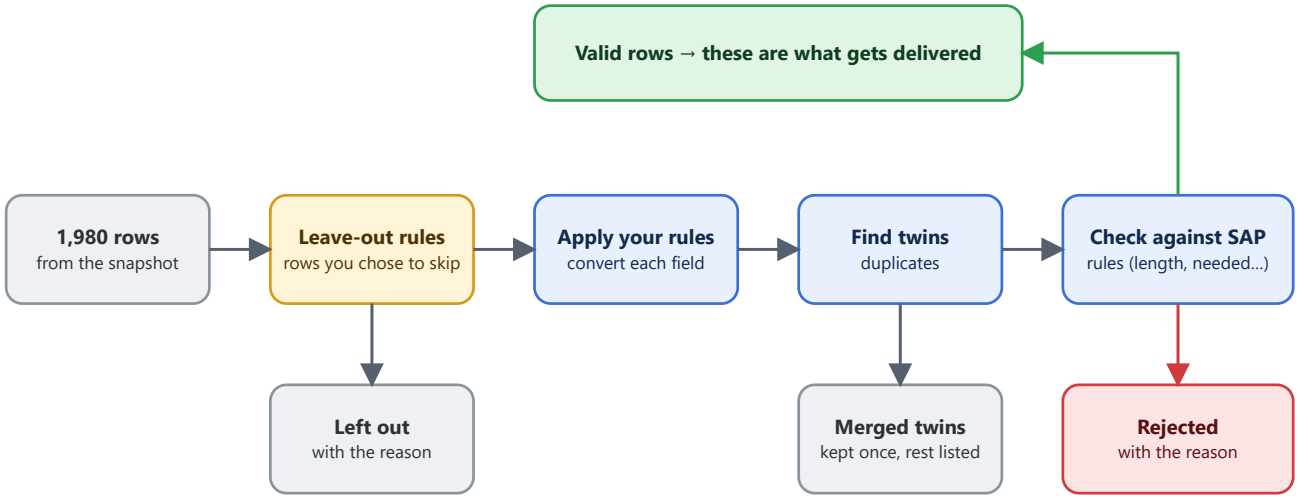


Figure 40. The run. Every row ends in exactly one place.

PR-8 Run an approved mapping set

PURPOSE

Convert, de-duplicate and validate every row.

PERFORMED BY

Author or steward

BEFORE YOU START

An APPROVED mapping set.

PROCEDURE

1. In the Mapping Studio press **Run pipeline**.
2. Wait; a **Governance** tab opens when the run ends (seconds for thousands of rows; the scale test processed 1,000,000 rows, see `docs/scale-test.md`).

EXPECTED RESULT

The Governance page of the new run.

CONTROL

The accounting identity always holds: *left out + merged + rejected + valid = rows extracted*. A row cannot disappear without a reason.

6.9 Read the gate and the defects

PR-9 Interpret the gate

PURPOSE

Decide whether the run may be signed off, and why not if it may not.

PERFORMED BY

Author, approver, steward

BEFORE YOU START

A finished run.

PROCEDURE

1. On the Governance page read the banner: **BLOCKED** or **READY FOR LOAD**.
2. Scroll to **Rule-aggregated defects**. Each group gives the rule, severity, row count, examples and the root cause.
3. Optionally use **Ask about this run** (6.17).

EXPECTED RESULT

A clear statement of what blocks the gate and how many rows are affected.

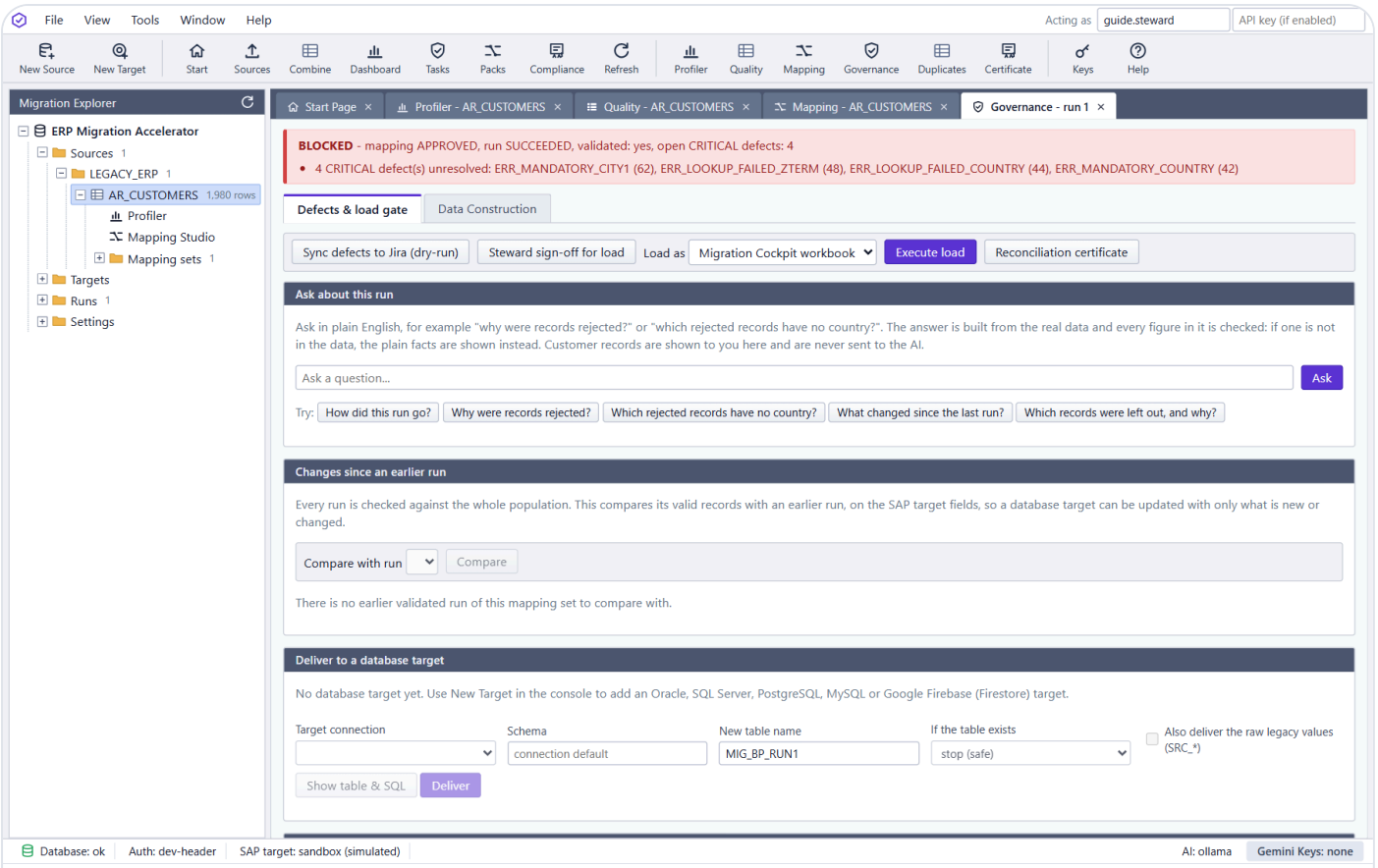


Figure 41. First run: BLOCKED, four critical defects unresolved.

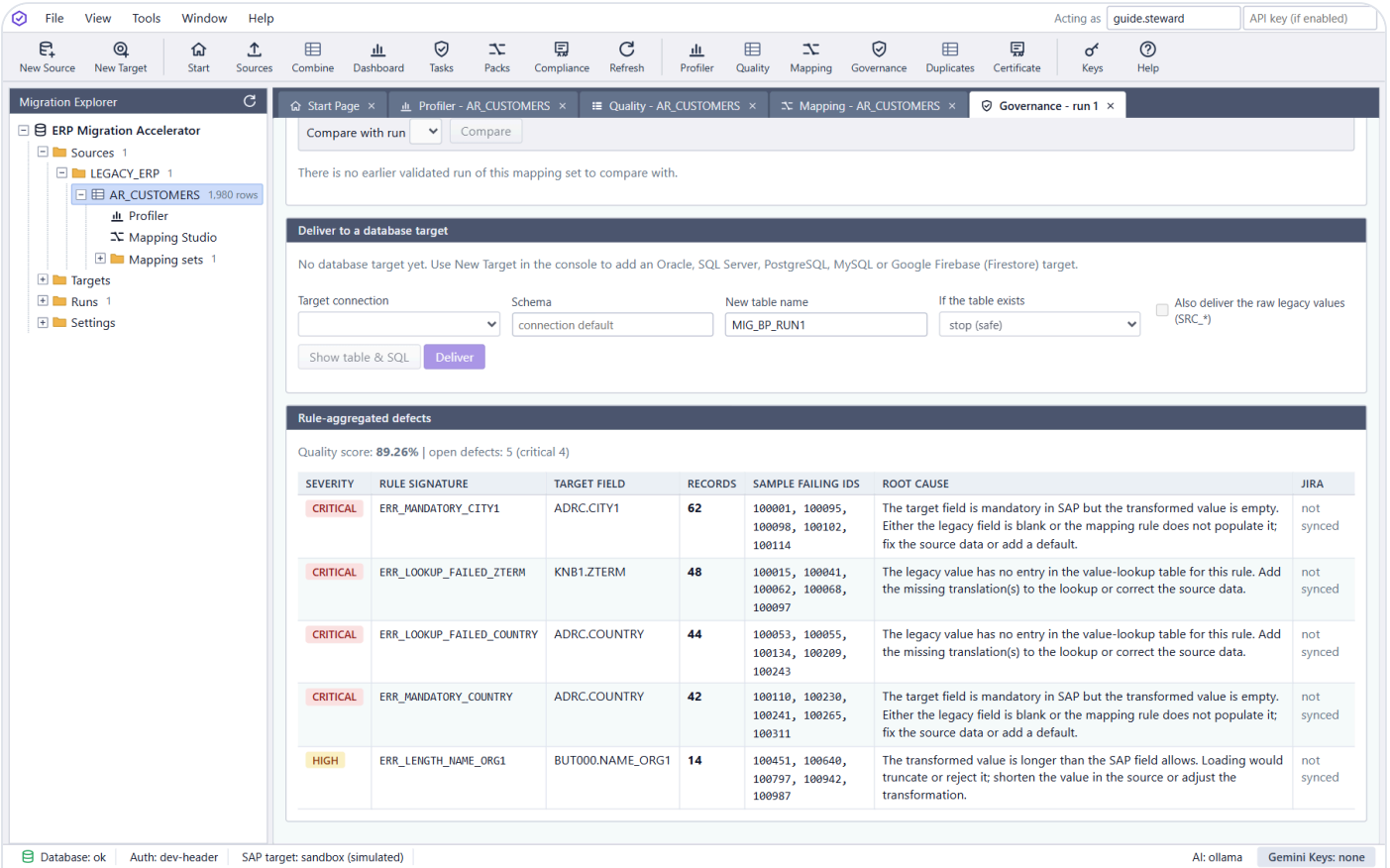


Figure 42. The defects of the first run; quality score 89.26%.

SEVERITY	DEFECT	ROWS	CAUSE	REMEDY
CRITICAL	`ERR_MANDATORY_CITY1`	62	SAP requires a city; none supplied.	Exclude the rows, or source the city.
CRITICAL	`ERR_LOOKUP_FAILED_ZTERM`	48	`WHEN-RECEIVED` is not in the dictionary.	Decide its meaning; add it to the lookup.
CRITICAL	`ERR_LOOKUP_FAILED_COUNTRY`	44	Atlantis, Narnia, Wakanda.	Not countries: exclude.
CRITICAL	`ERR_MANDATORY_COUNTRY`	42	SAP requires a country; none supplied.	Exclude the rows.
HIGH	`ERR_LENGTH_NAME_ORG1`	14	Name longer than SAP's 40.	Shorten in the rule.

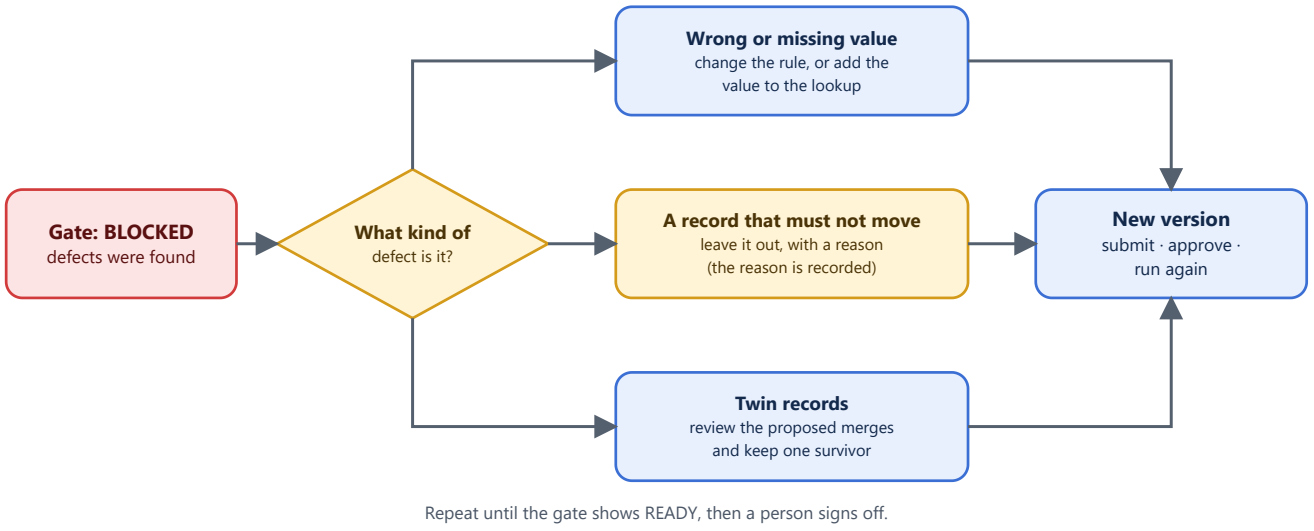


Figure 43. Choosing the remedy for a blocked gate.

6.10 Correct the rules and exclude rows

PR-10 Create version 2 and run again

PURPOSE

Resolve the defects in a new, editable copy; the approved set stays as evidence of run 1.

PERFORMED BY

Author, then approver

BEFORE YOU START

A run whose gate is BLOCKED.

PROCEDURE

1. In the Mapping Studio press **Check rules on sample** again. Under **Needs your approval** read each **Fix-it** card: the situation, the proposed change and how to undo it. Enter any value the card asks for and press **Yes, go ahead**. A new draft (a copy) is created.
2. Open **Rows to leave out**. Choose a column, a condition and a reason, press **Check how many rows**, then **Leave these rows out**.
3. Edit rules as needed: add `WHEN-RECEIVED → CASH` to the payment-term lookup; change the name rule to `left(trim(CUST_NAME), 40)`.
4. Press **Check rules on sample** until 100% pass; then submit, approve (a different person) and run (PR-7, PR-8).

EXPECTED RESULT

The new run's gate reads **READY FOR LOAD** with no open critical defects.

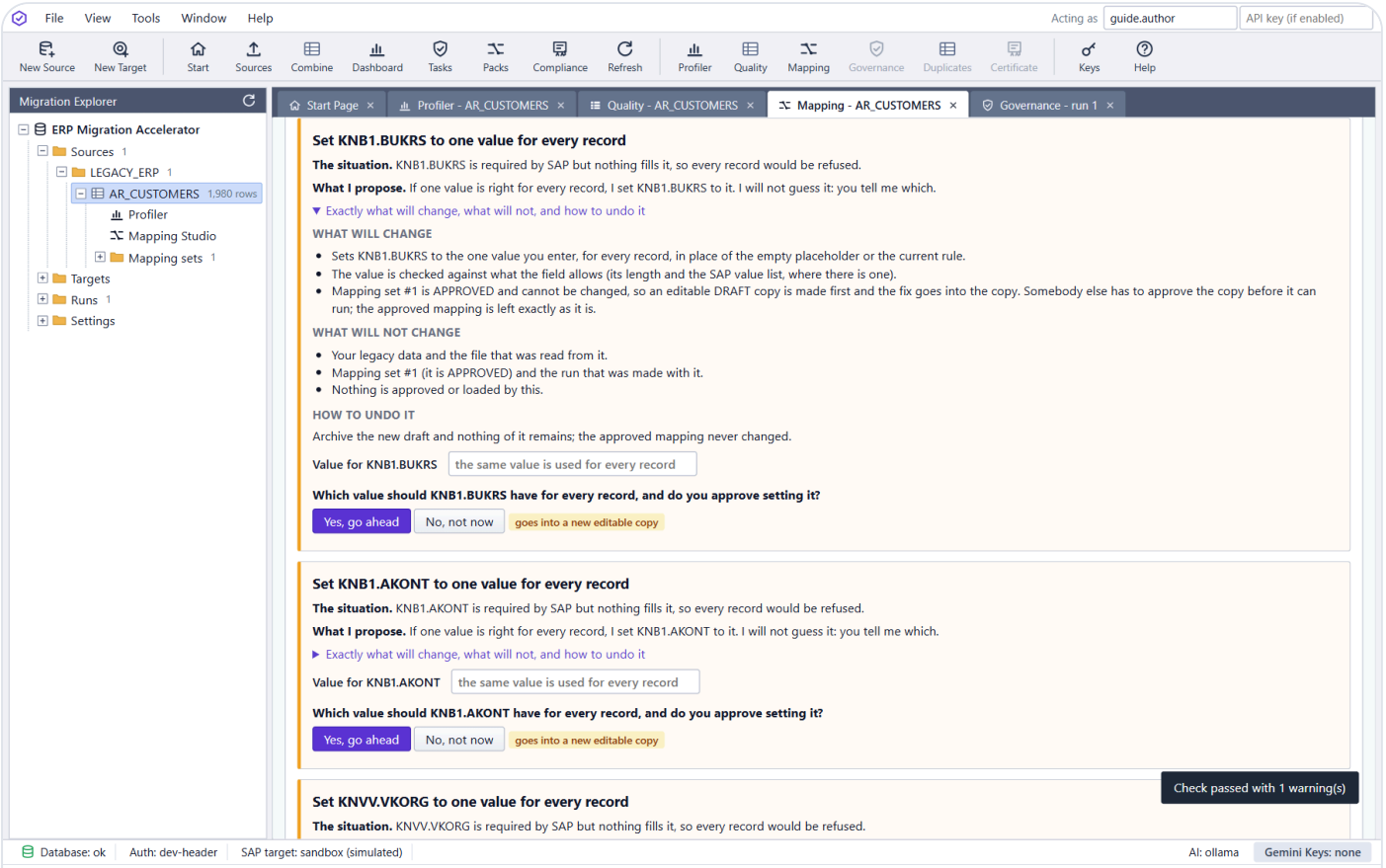


Figure 44. A Fix-it card: SAP requires the company code `KNB1.BUKRS`; the application asks for the value and will not guess.

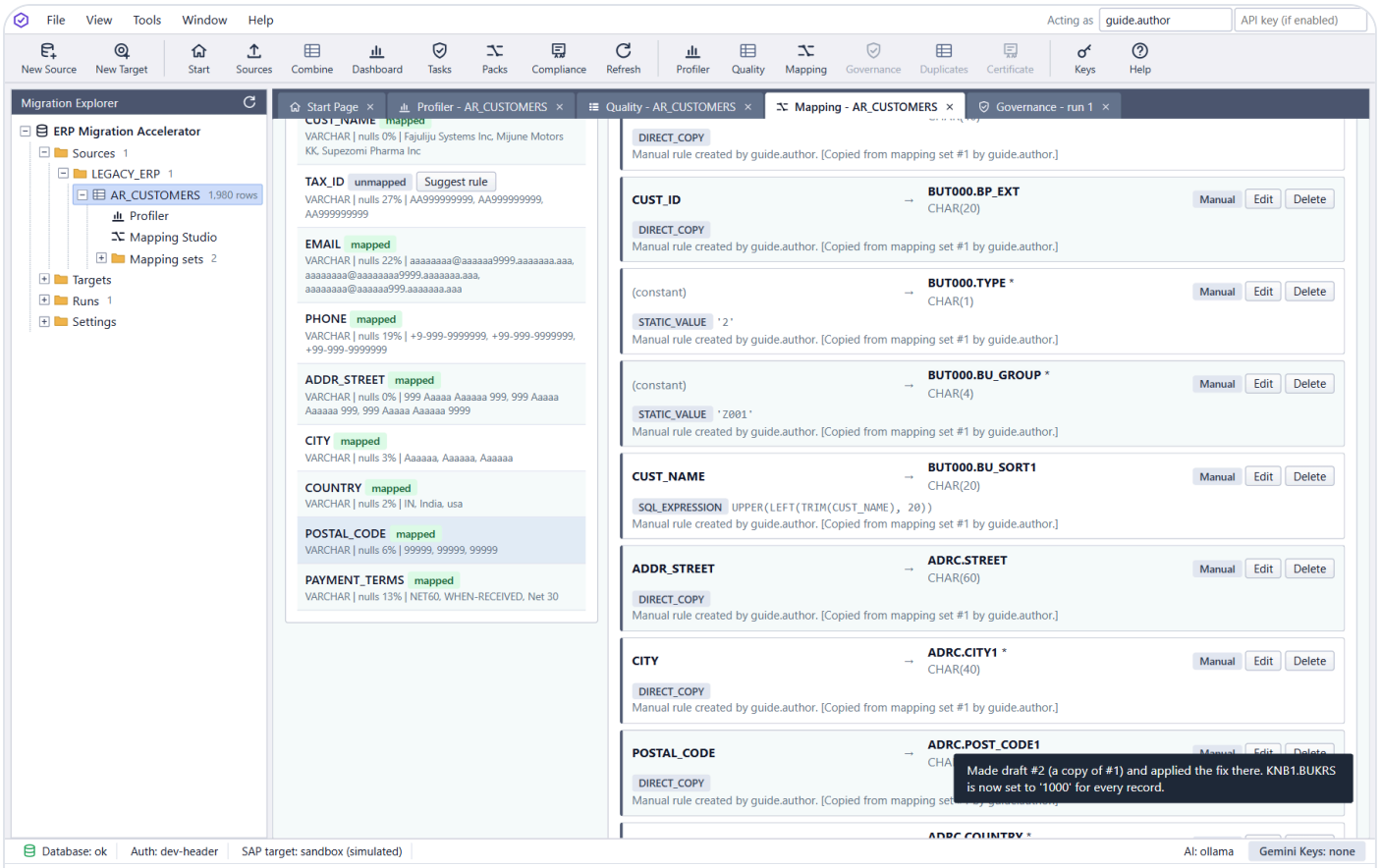


Figure 45. A new draft was made: *draft #2 (a copy of #1)*. The approved set is untouched.

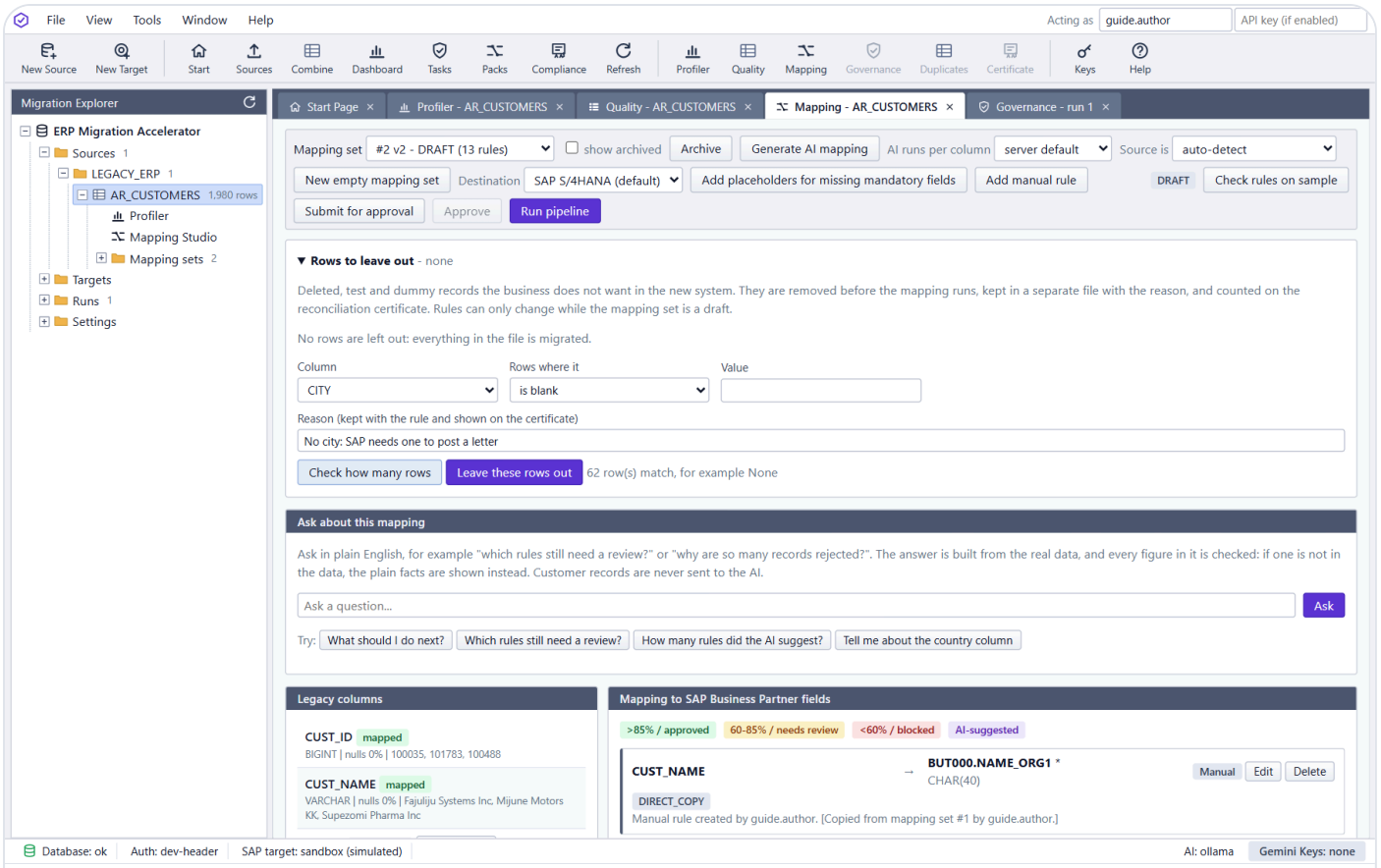


Figure 46. One exclusion rule: *CITY is blank*, with a reason. The application first says how many rows match (62).

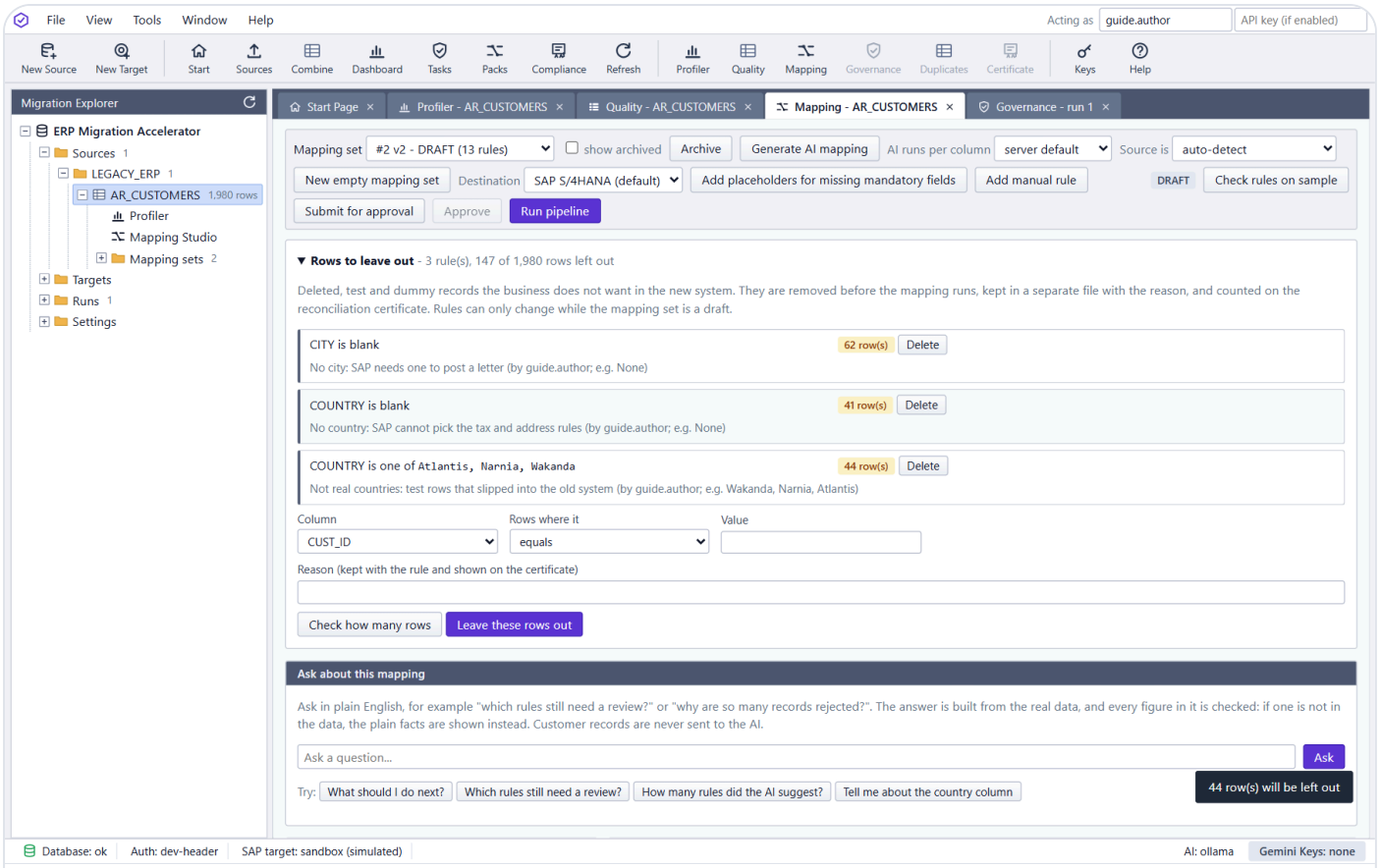


Figure 47. Three exclusion rules: 147 of 1,980 rows left out.

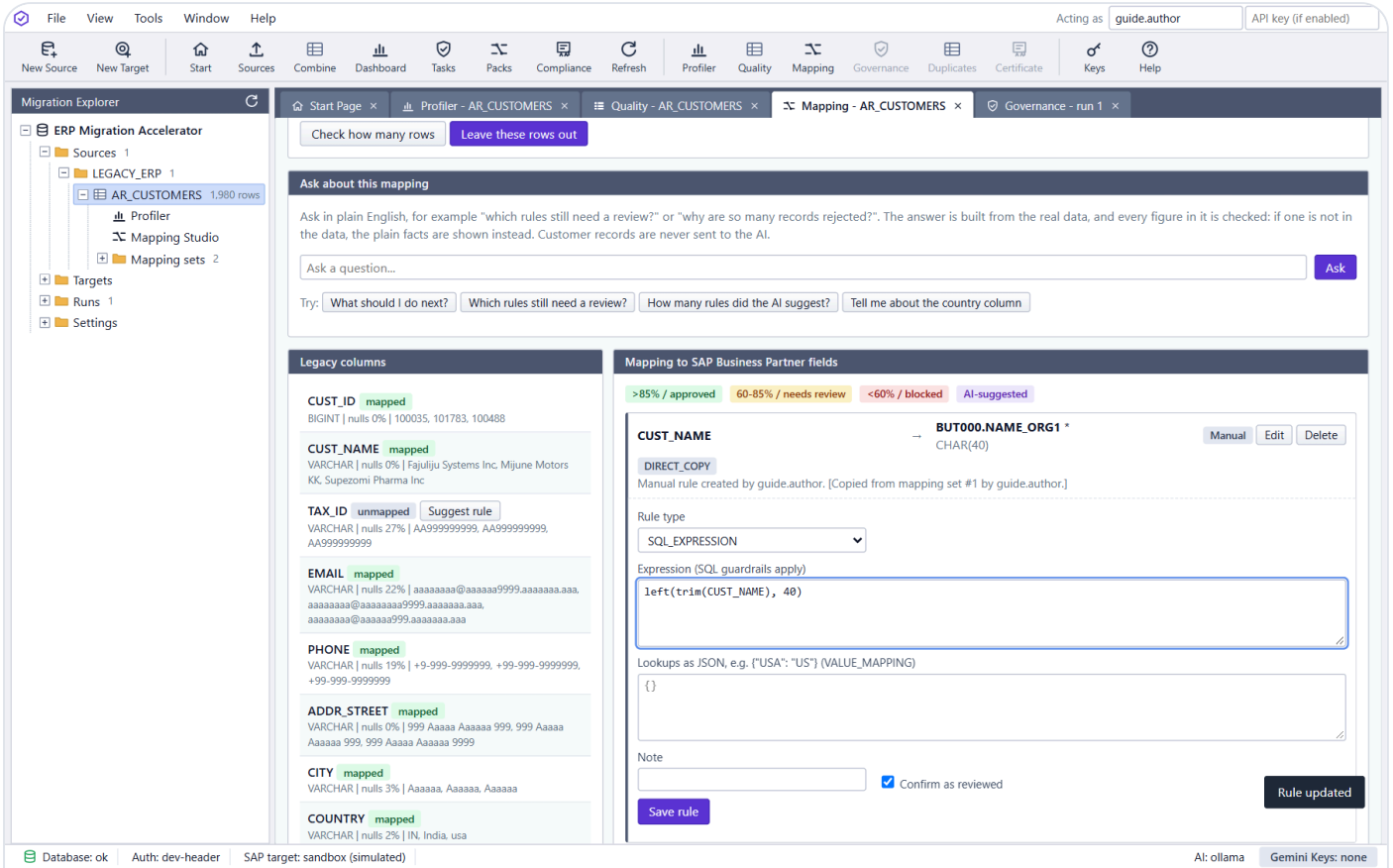


Figure 48. Changing a rule from a plain copy to `left(trim(CUST_NAME), 40)`.

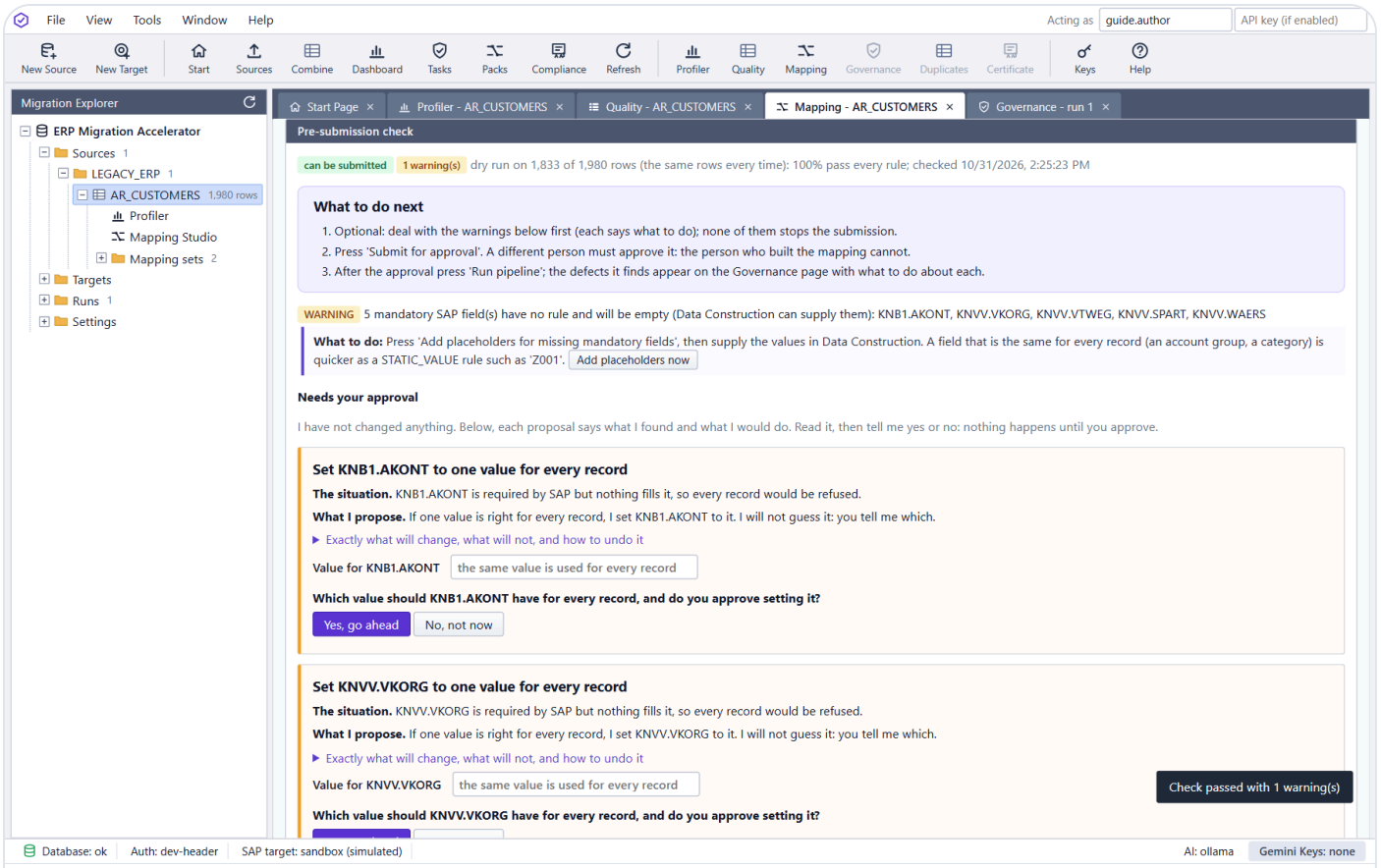


Figure 49. The check on version 2: 1,833 rows in scope, 100% pass.

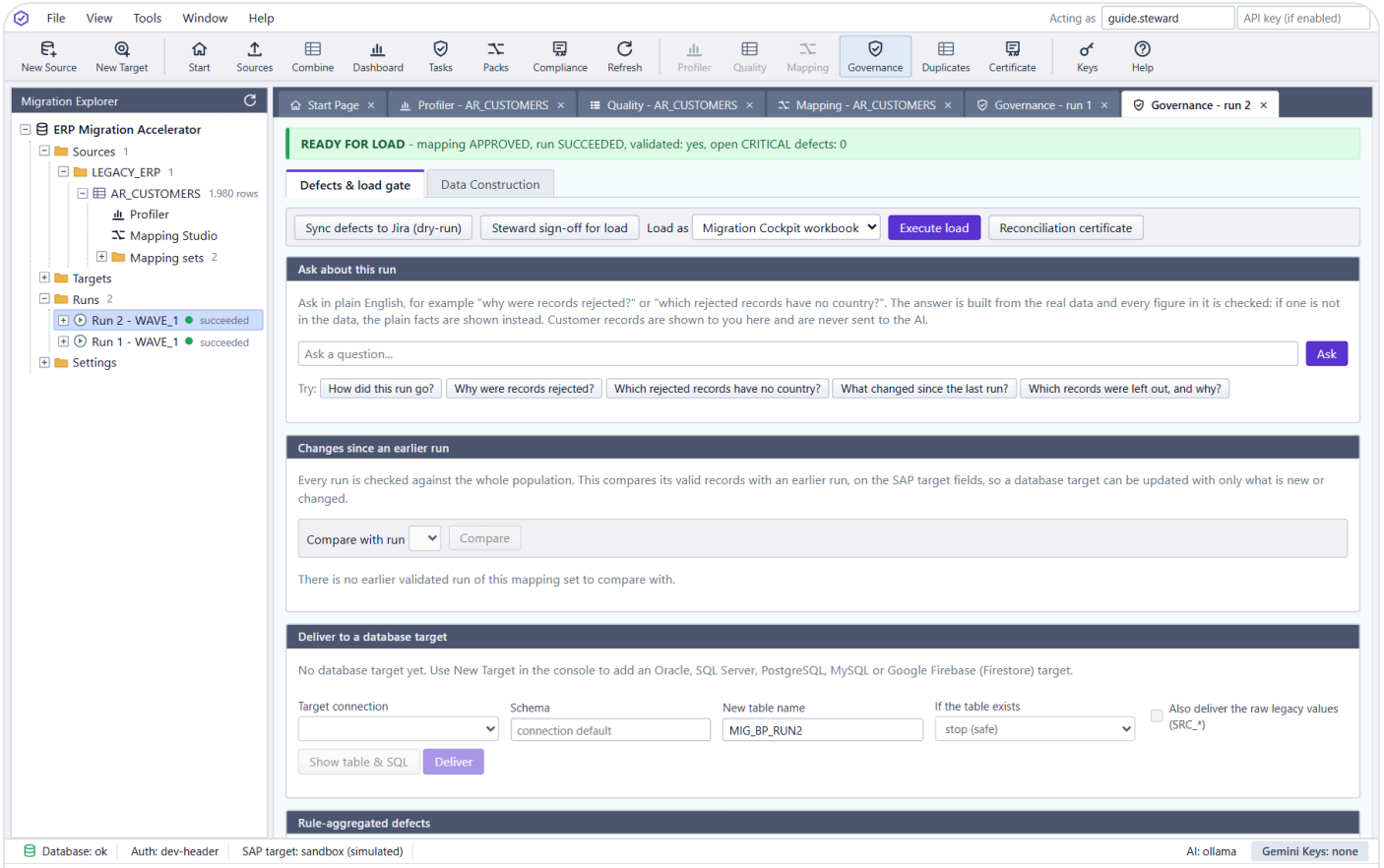


Figure 50. Version 2: READY FOR LOAD; 0 open critical defects.

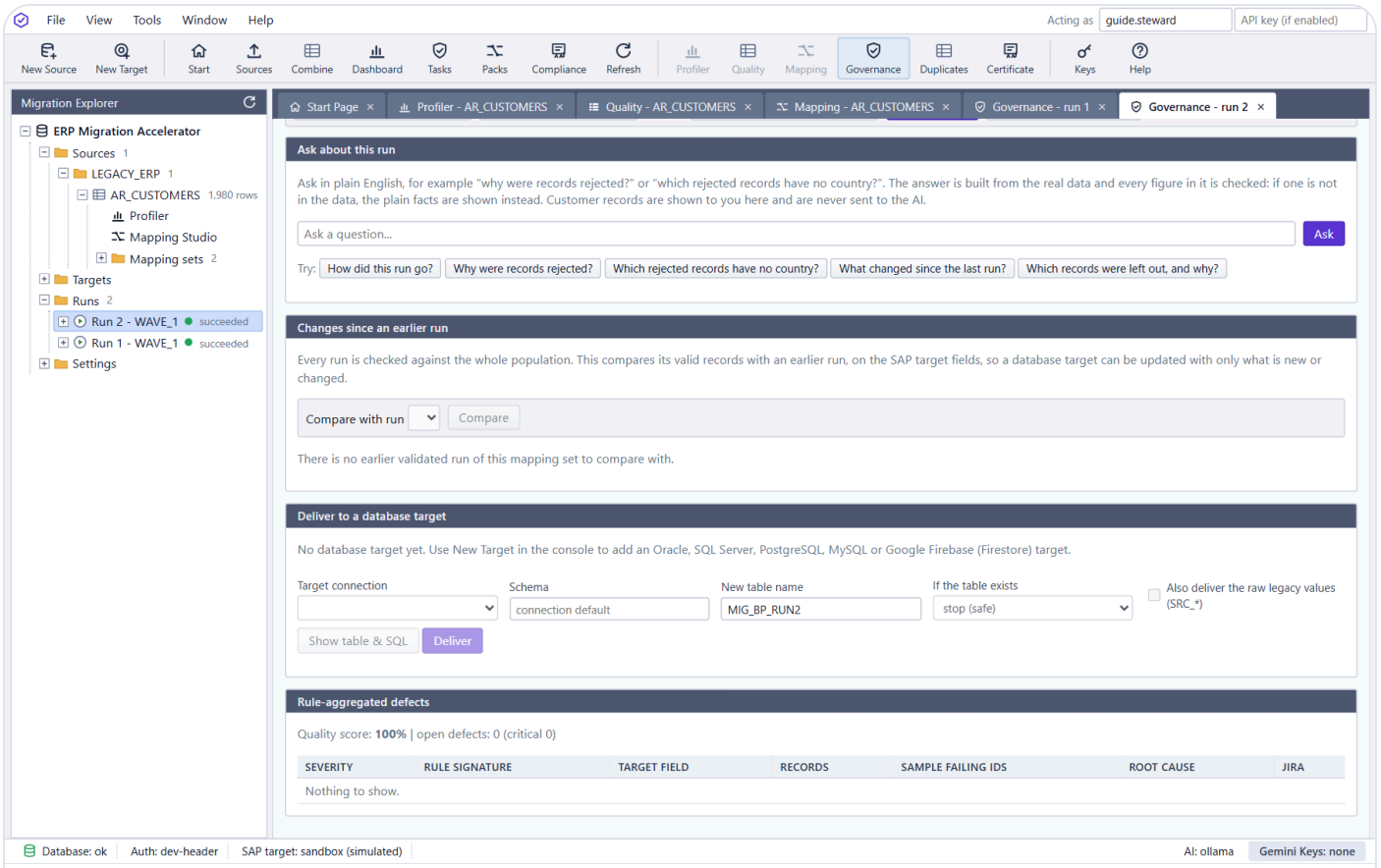


Figure 51. The defects table is empty; quality score 100%.

CAUTION

Leaving out 147 customers (7%) is a business decision. The application makes it visible, recorded and reversible; it does not take it for you. Cutting a name at 40 characters can cut a word in half: it is the quickest fix, not the neatest.

CONTROL

Excluded rows are not deleted. They are kept with the reason and listed on the certificate under `EXCLUDED_BY_RULE`.

6.11 Review duplicates

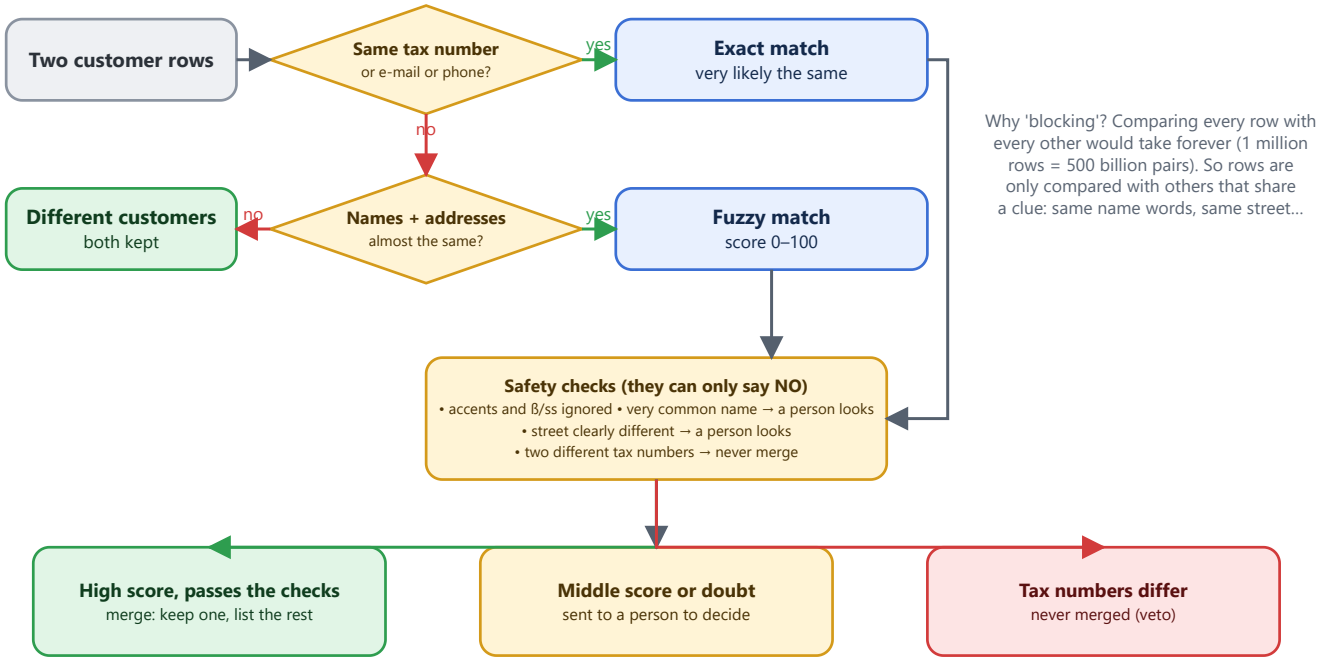


Figure 52. How two rows are judged: safety checks can only veto a merge, never create one.

PR-11 Review duplicate customers

PURPOSE

Make sure one company arrives once, and that two different companies are never merged.

PERFORMED BY

Author or steward

BEFORE YOU START

A finished run.

PROCEDURE

1. Select the run and press **Duplicates**.
2. Read the summary and the lists: merges made on a name alone, **possible duplicates** awaiting a decision, and decisions already taken.
3. For each uncertain pair choose **same customer** or **different** and give a reason.

EXPECTED RESULT

Decisions are remembered and applied in every later run, with your name and reason.

MATCH	EXAMPLE	OUTCOME
Exact: same tax number, e-mail or phone	Two rows with tax number `AA999999999`	Merged; one survivor, the others listed as not loaded.
Fuzzy: names and addresses nearly equal	Müller GmbH, Hauptstr. 5 / Mueller GmbH, Hauptstrasse 5	Score 0 to 100: high and safe is merged; in between goes to a person.
Vetoed	Same name, different tax numbers	Never merged.

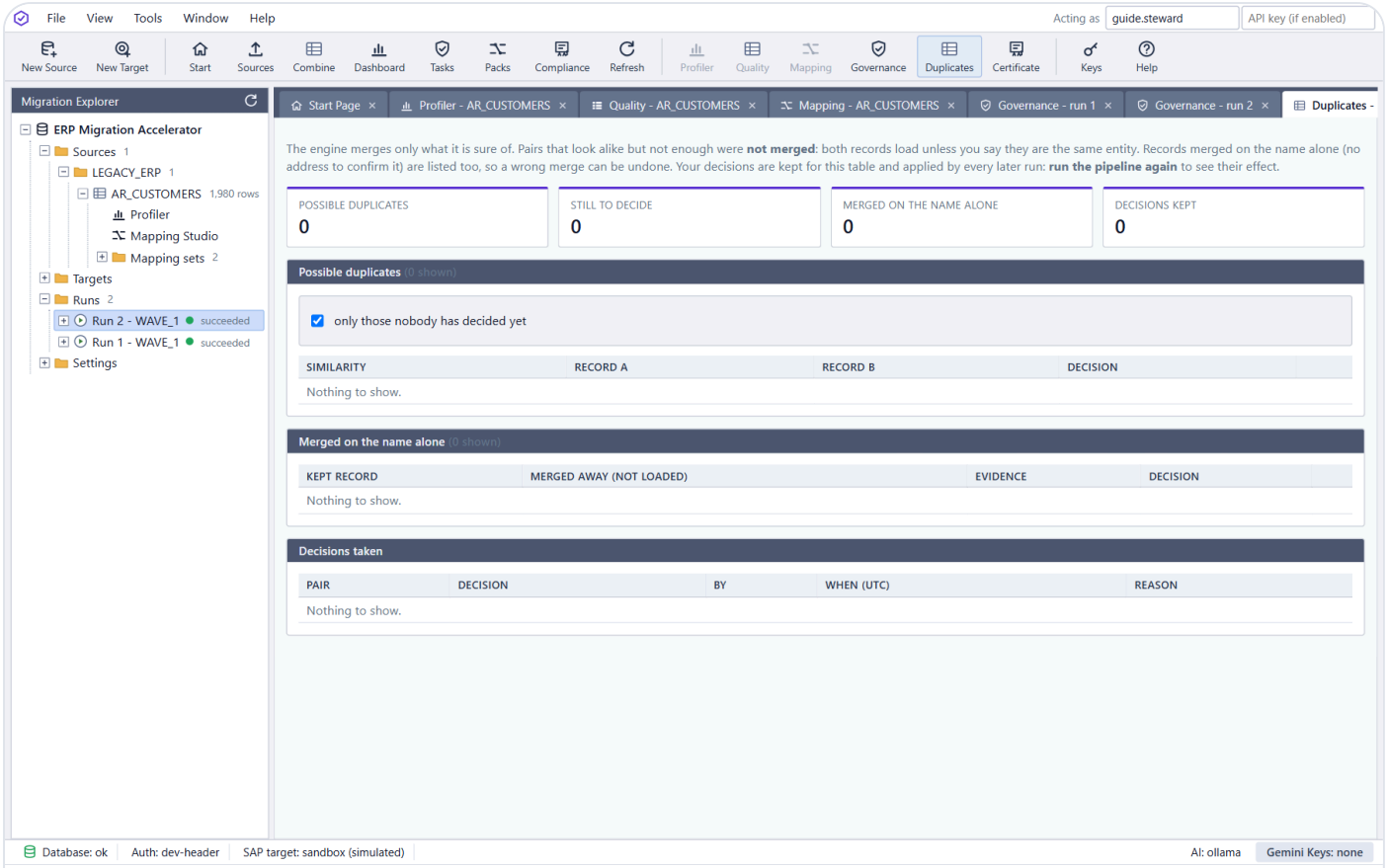


Figure 53. The Duplicates page; in this run every match was certain, so the review lists are empty.

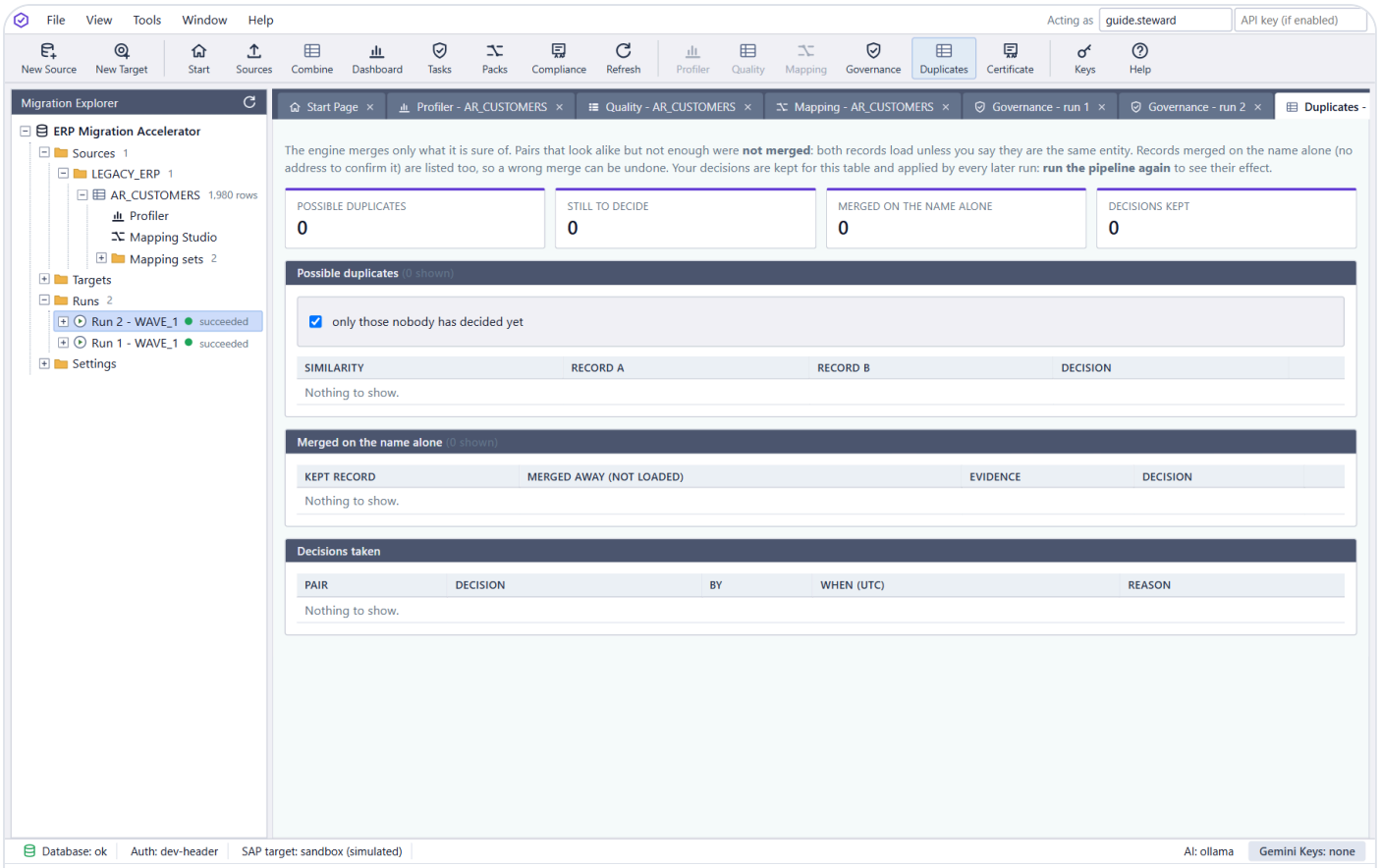


Figure 54. Merges made on a name alone.

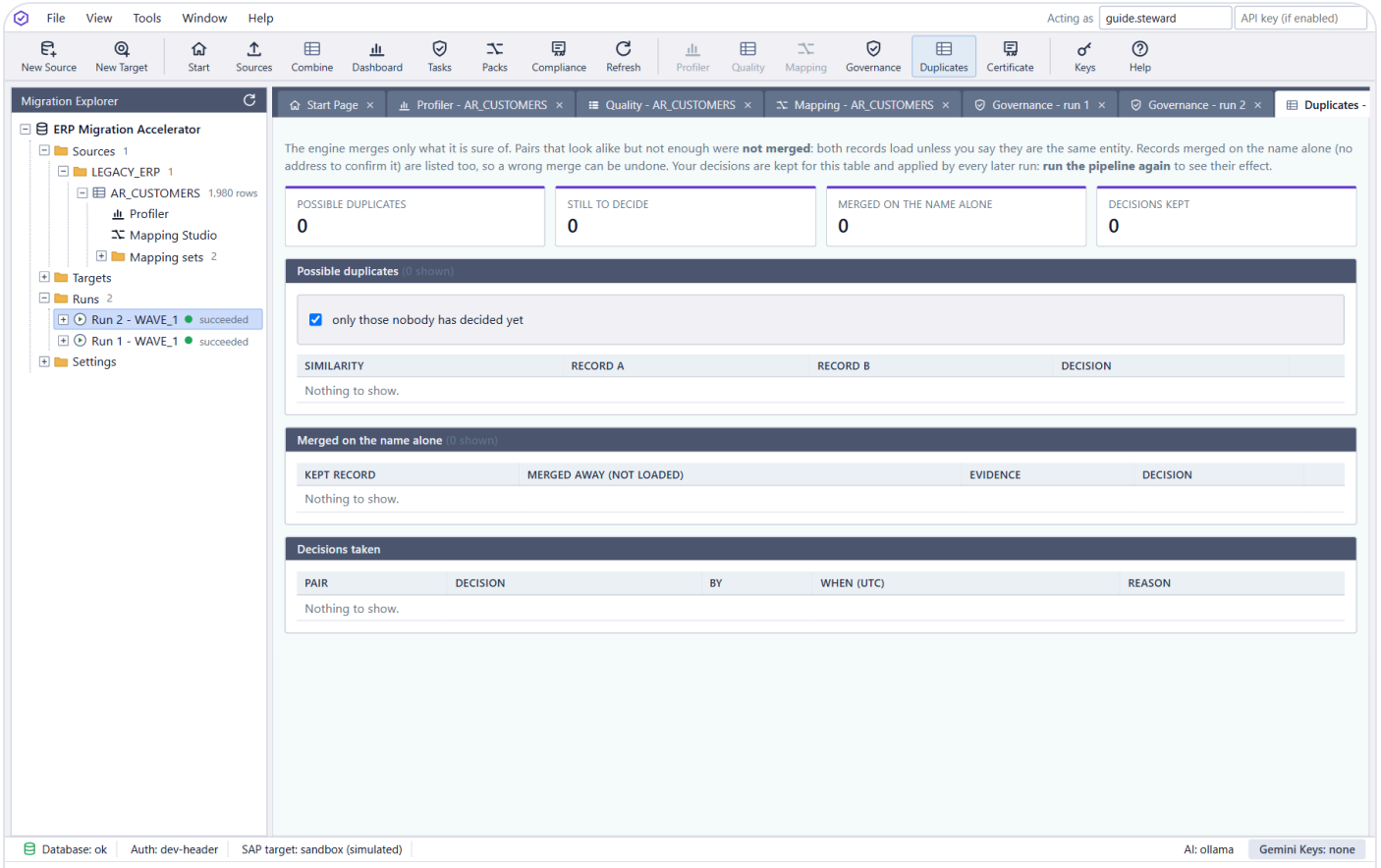


Figure 55. Possible duplicates awaiting a human decision.

NOTE

Accents and German spellings (ü/ue, ß/ss) are ignored when comparing; a very common name such as Müller is never merged on the name alone; a clearly different address sends the pair to a person.

6.12 Sign off

PR-12 Steward sign-off

PURPOSE

Record that a named person has read the evidence and accepts the run for loading.

PERFORMED BY

Steward

BEFORE YOU START

Gate READY FOR LOAD. You must be an allowed approver and not the author of the mapping.

PROCEDURE

1. Open the Governance page of the run and set **Acting as** to the steward.
2. Press **Steward sign-off for load**.

EXPECTED RESULT

A message names the person; the page and the certificate record who and when.

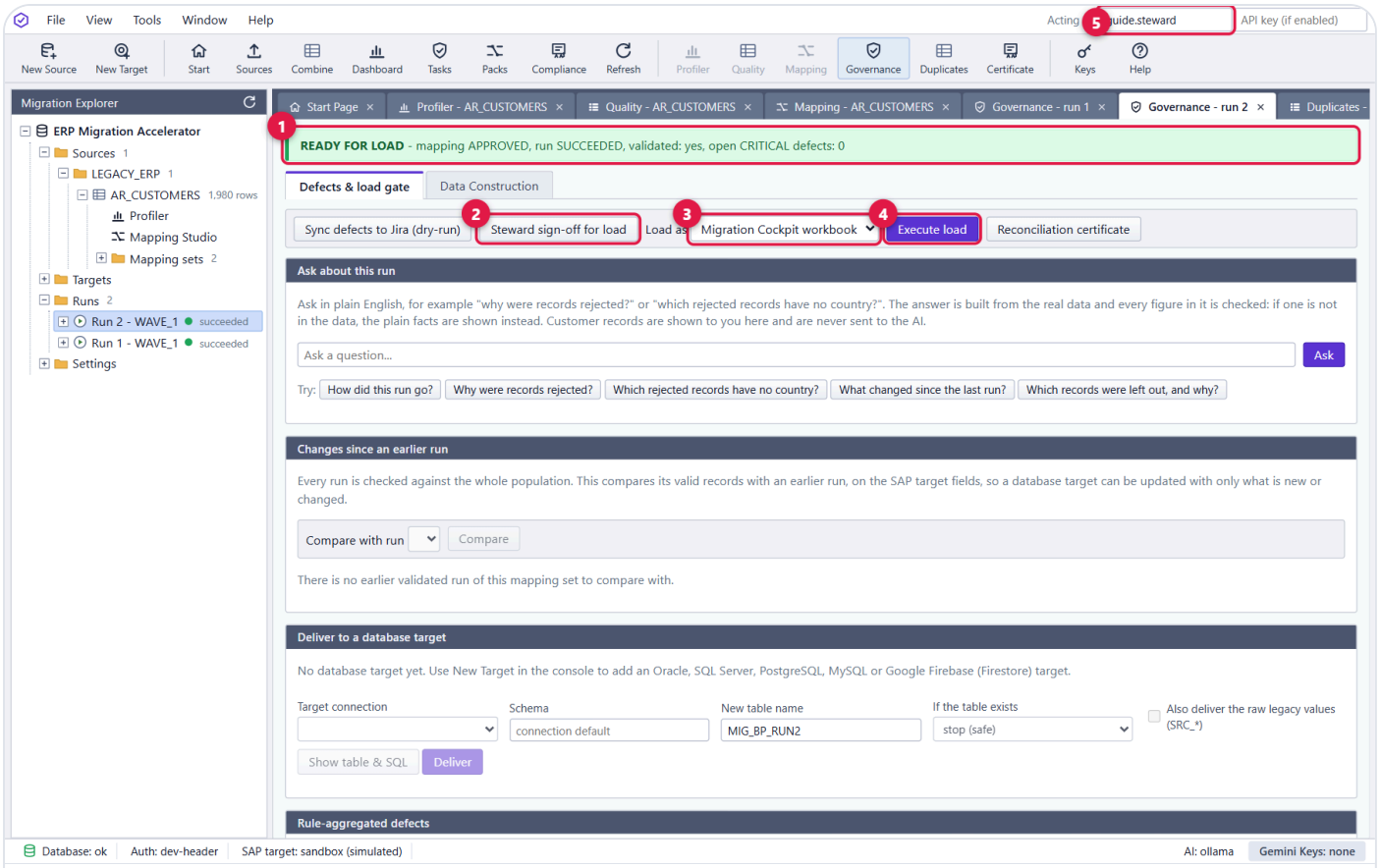


Figure 56. The final controls. 1 Gate · 2 Steward sign-off · 3 Load as · 4 Execute load · 5 Acting as.

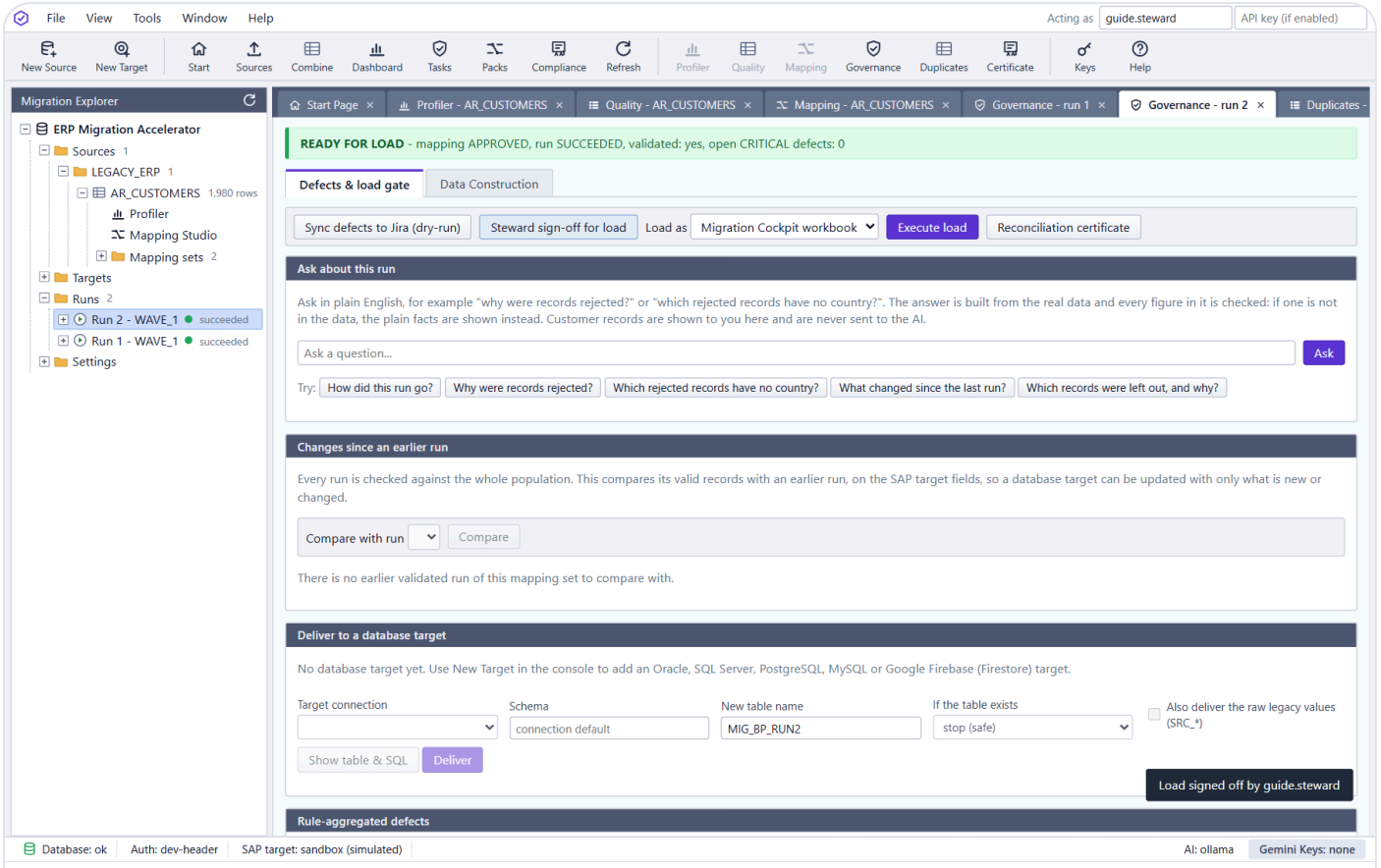


Figure 57. Signed off.

CONTROL

Approval answers “are the rules acceptable?”; sign-off answers “is the result acceptable and do I take responsibility for loading it?” Two questions, two moments, two records.

6.13 Deliver and verify

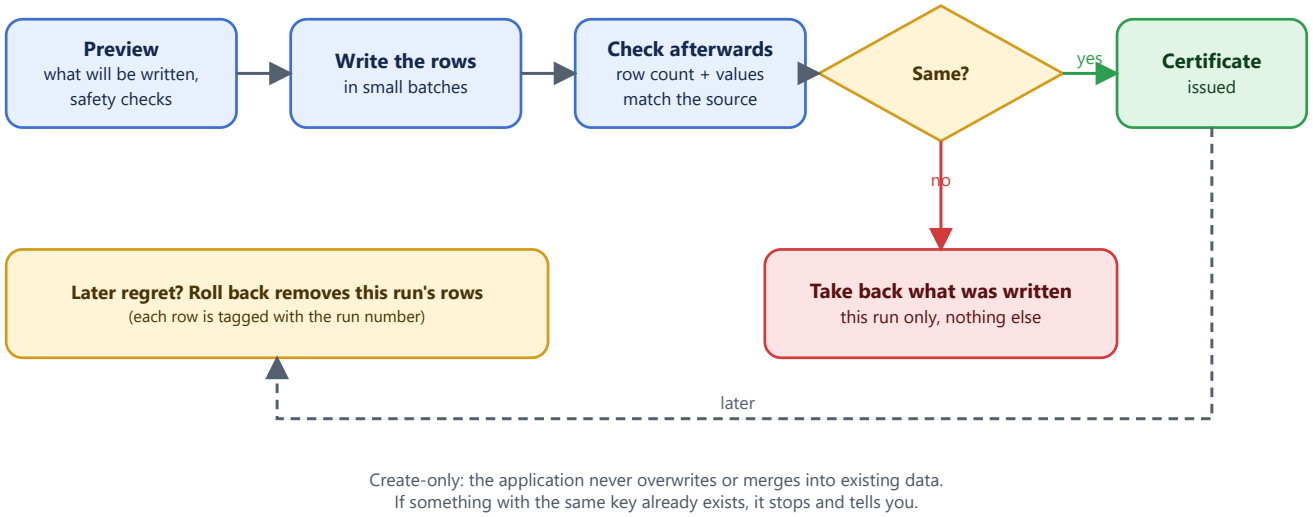


Figure 58. Delivery: preview, write in batches, count and compare, undo if wrong.

ROUTE	WHAT HAPPENS	USE WHEN
Migration Cockpit workbook	An Excel file in the format of SAP's Migration Cockpit, uploaded into SAP by a person.	The SAP team prefers to load itself.
SAP sandbox (simulated)	Each customer is sent to the stand-in gateway; accepted and failed counts are reported.	Rehearsals.
SAP live	Each customer is sent to a real SAP over OData.	Go-live. Needs the SAP address and a technical user; **not** exercised in this manual.
Database target or **Firestore**	The finished rows are written to a new table (or collection).	The new home is not SAP, or to inspect the result with ordinary tools.

PR-13 Define a database target (once)

PURPOSE

Describe where data will be written, and prove the account may write there.

PERFORMED BY

Administrator or steward

BEFORE YOU START

A reachable database and an account with permission to create and fill a table.

PROCEDURE

1. Press **New Target** and choose **Database**; name it.
2. Enter server, port, database, user and password; press **Test**.
3. Choose the schema and run the write test: the wizard writes one test row, reads it back and deletes it.
4. Review the summary and press **Finish**.

EXPECTED RESULT

The target appears under **Targets** in the Explorer.

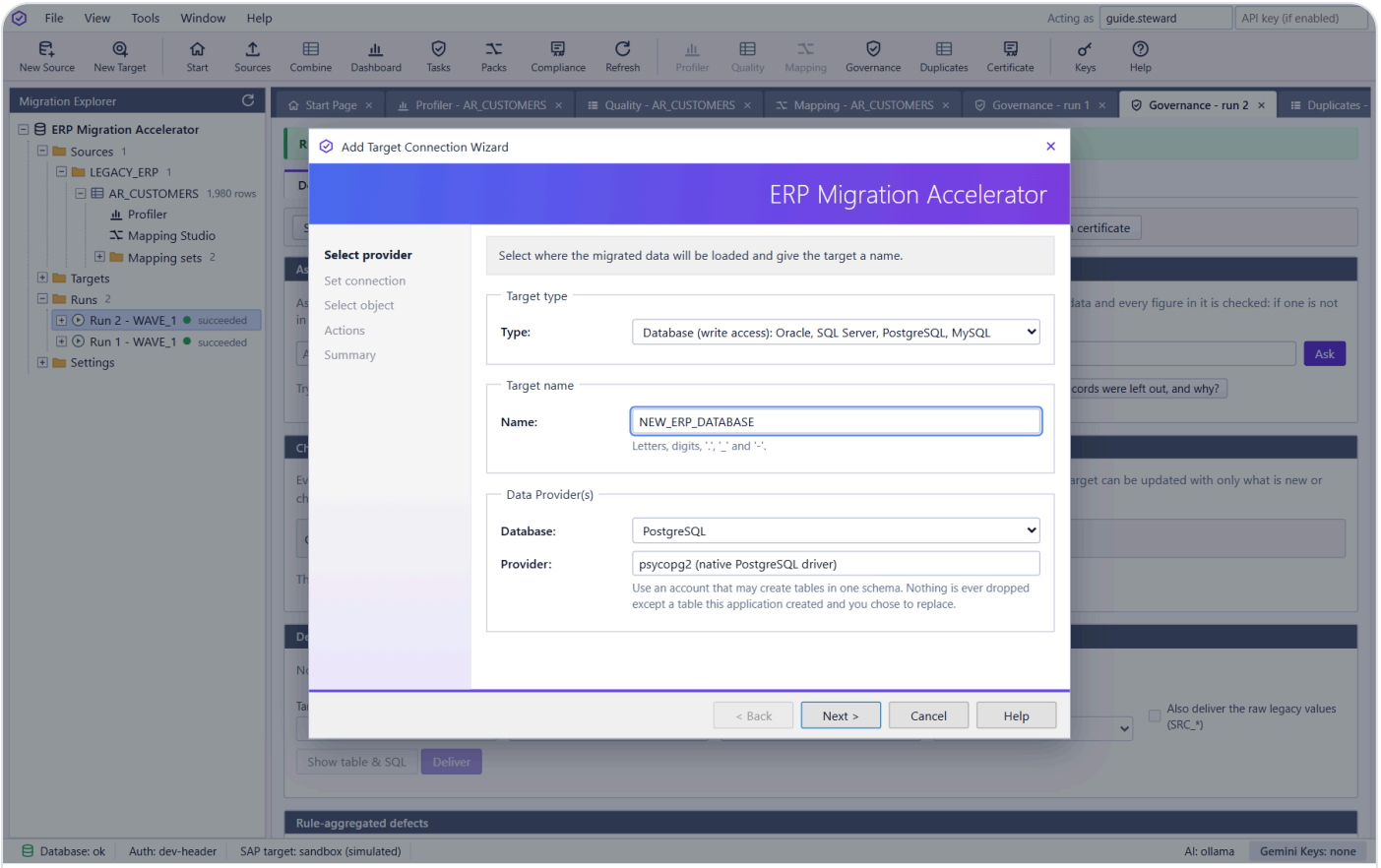


Figure 59. New Target, page 1: kind and name.

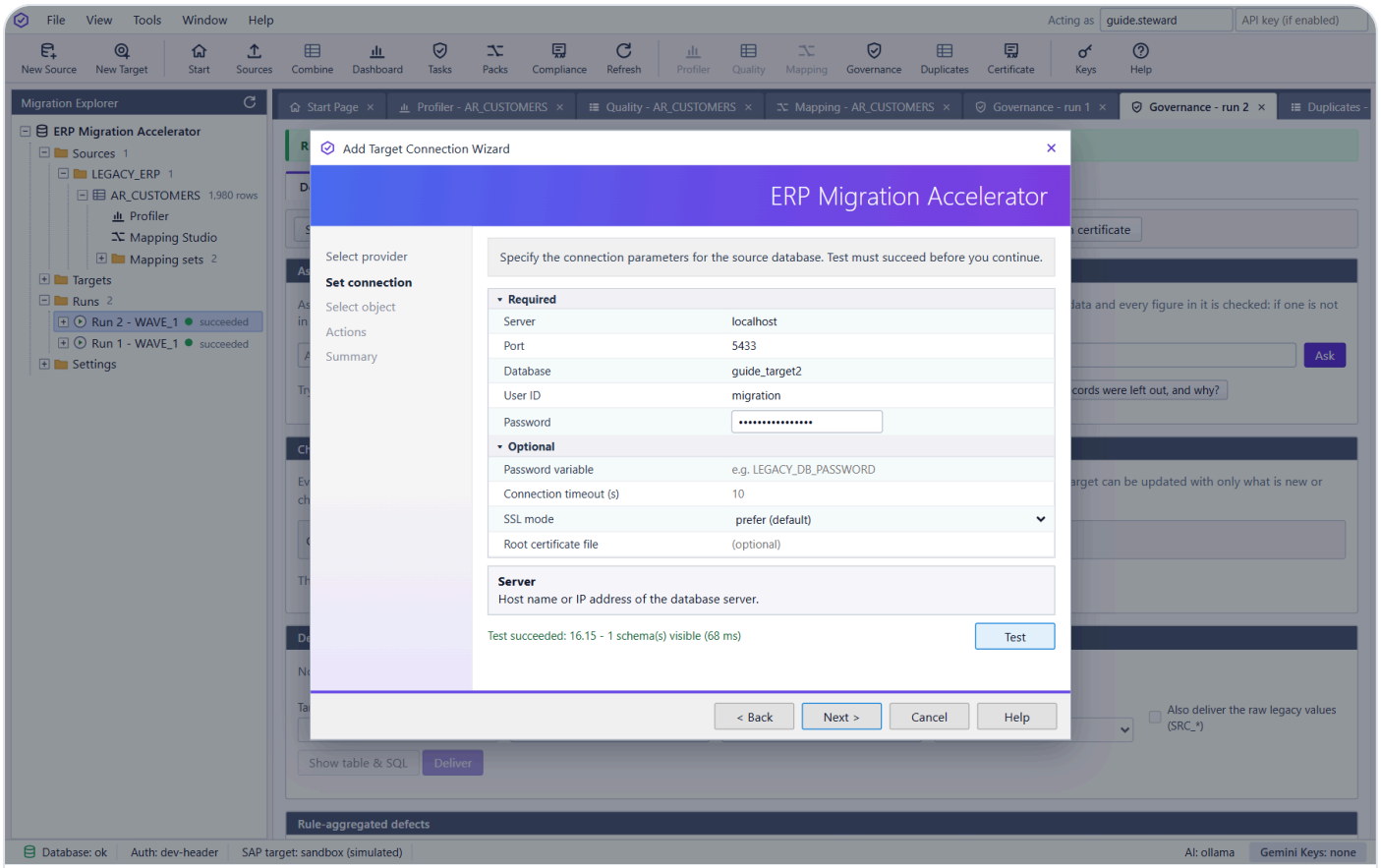


Figure 60. Page 2: connection; the password is kept only in memory.

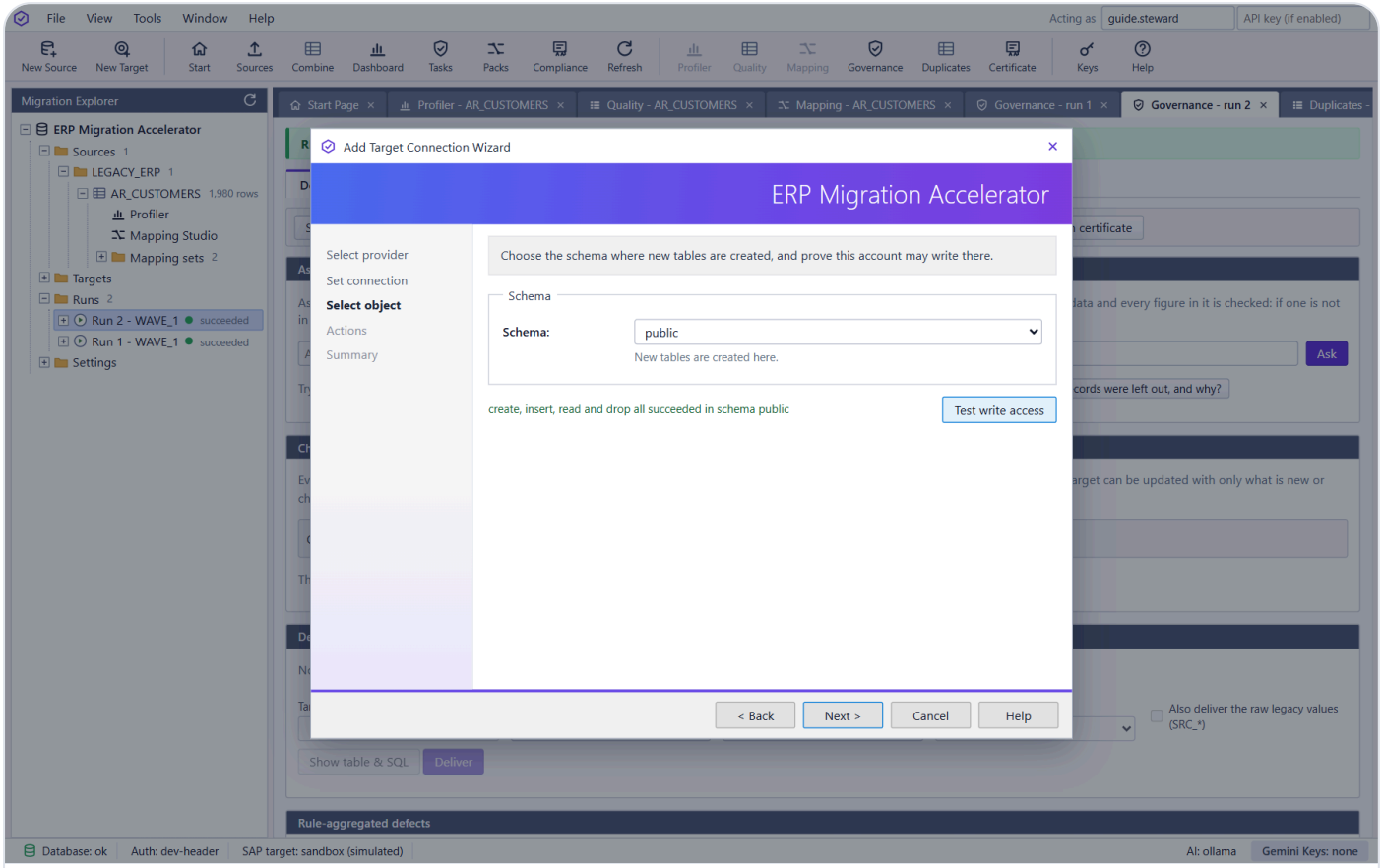


Figure 61. Page 3: the write test proves the account may create and fill tables.

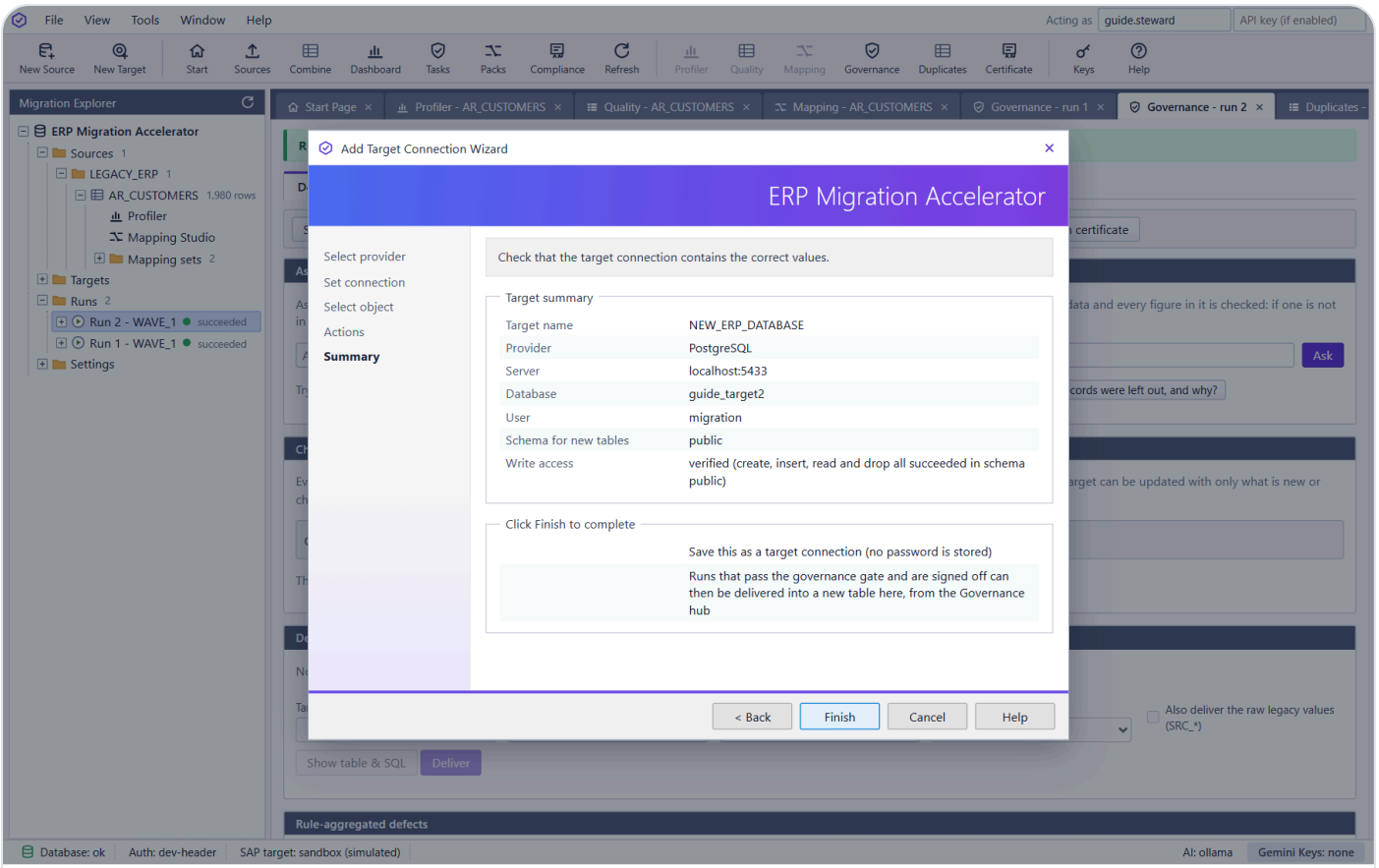


Figure 62. Summary.

PR-14 Deliver to a database target and verify

PURPOSE

Write the finished rows, then prove they arrived unchanged.

PERFORMED BY

Steward

BEFORE YOU START

A signed-off run and a defined target.

PROCEDURE

1. In **Deliver to a database target** choose the target and enter a **new table name** (`BP_CUSTOMERS`).
2. Press **Show table & SQL** to preview the table that will be created.
3. Press **Deliver**.
4. Press **Verify now** whenever you want to re-check the table against the approved result.

EXPECTED RESULT

Rows sent equal rows counted in the target, and every cell is read back and compared (25,995 cells in the practice run).

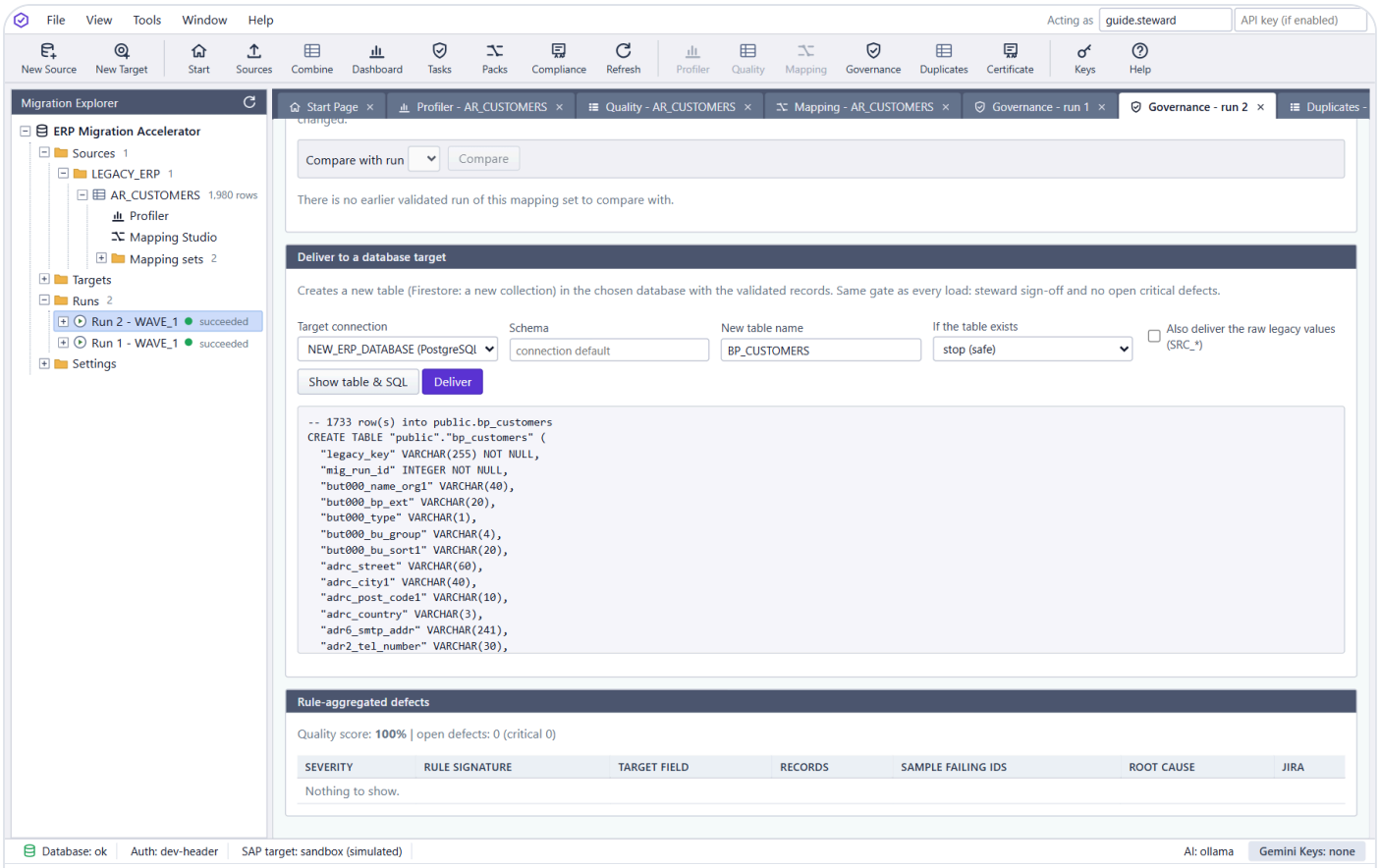


Figure 63. Preview: 1,733 rows into `public.bp_customers`, with exact column lengths and two bookkeeping columns, `legacy_key` and `mig_run_id`.

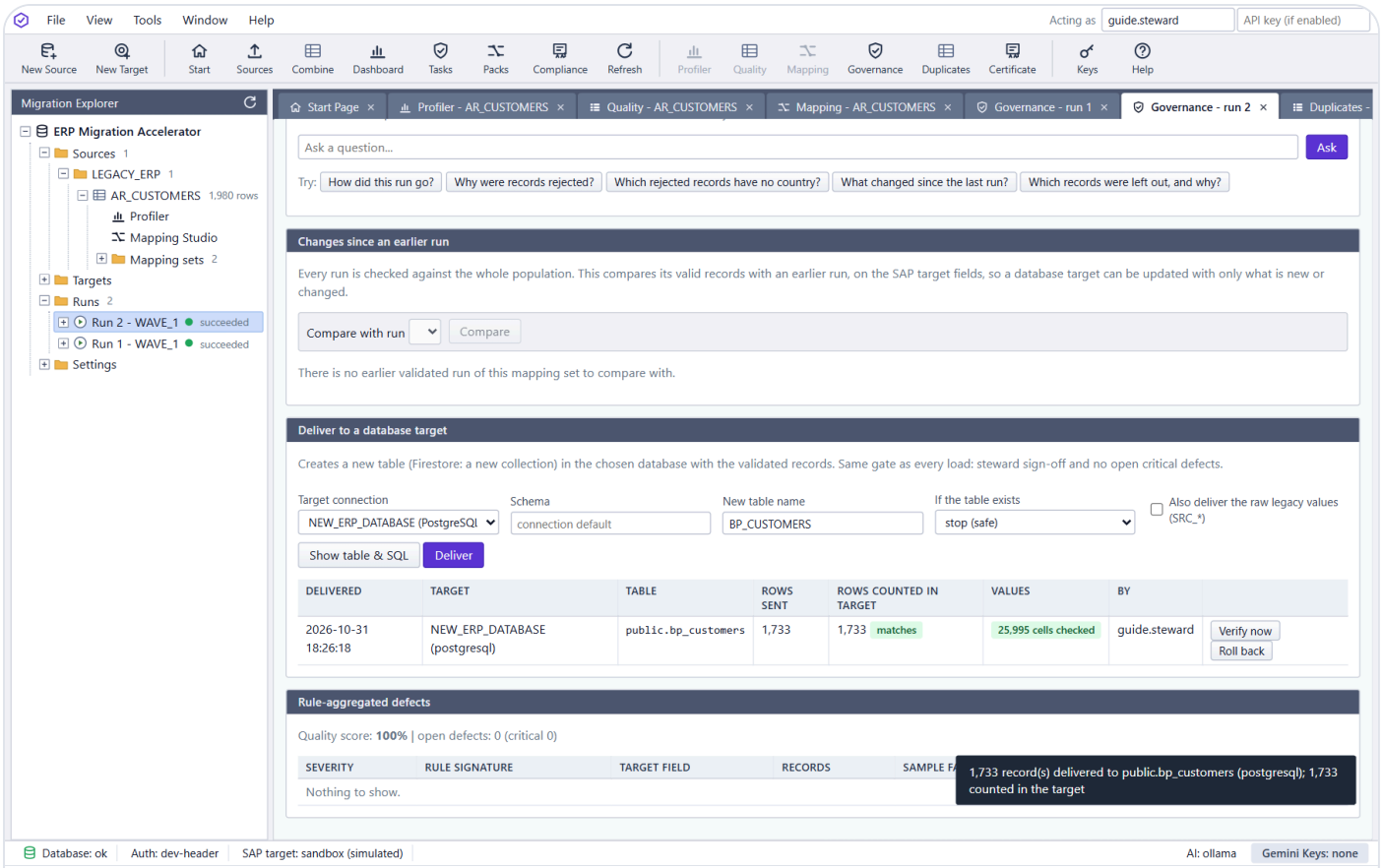


Figure 64. Delivered: 1,733 sent, 1,733 counted (matches), 25,995 cells checked.

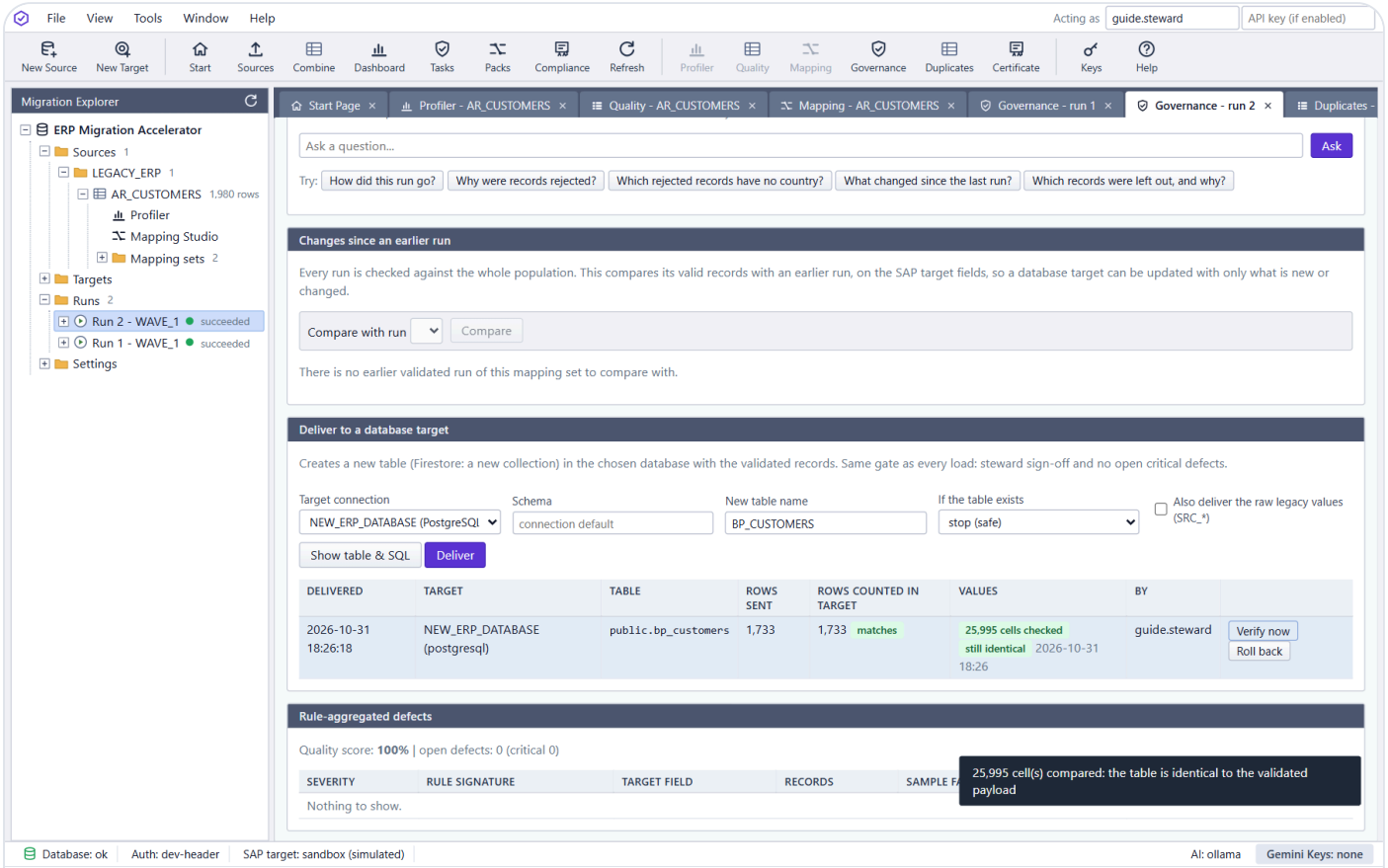


Figure 65. Verified again: *still identical*. A value changed afterwards would turn this red.

PR-15 Load into SAP (sandbox or live)

PURPOSE

Create business partners in SAP, or rehearse it.

PERFORMED BY

Steward

BEFORE YOU START

A signed-off run. For live: SAP address and technical user configured.

PROCEDURE

1. On the Governance page choose **Load as** (Migration Cockpit workbook, SAP sandbox, SAP live).
2. Press **Execute load**.

EXPECTED RESULT

For the sandbox: *1733 loaded, 0 failed*, each customer with a business-partner number from the stand-in.

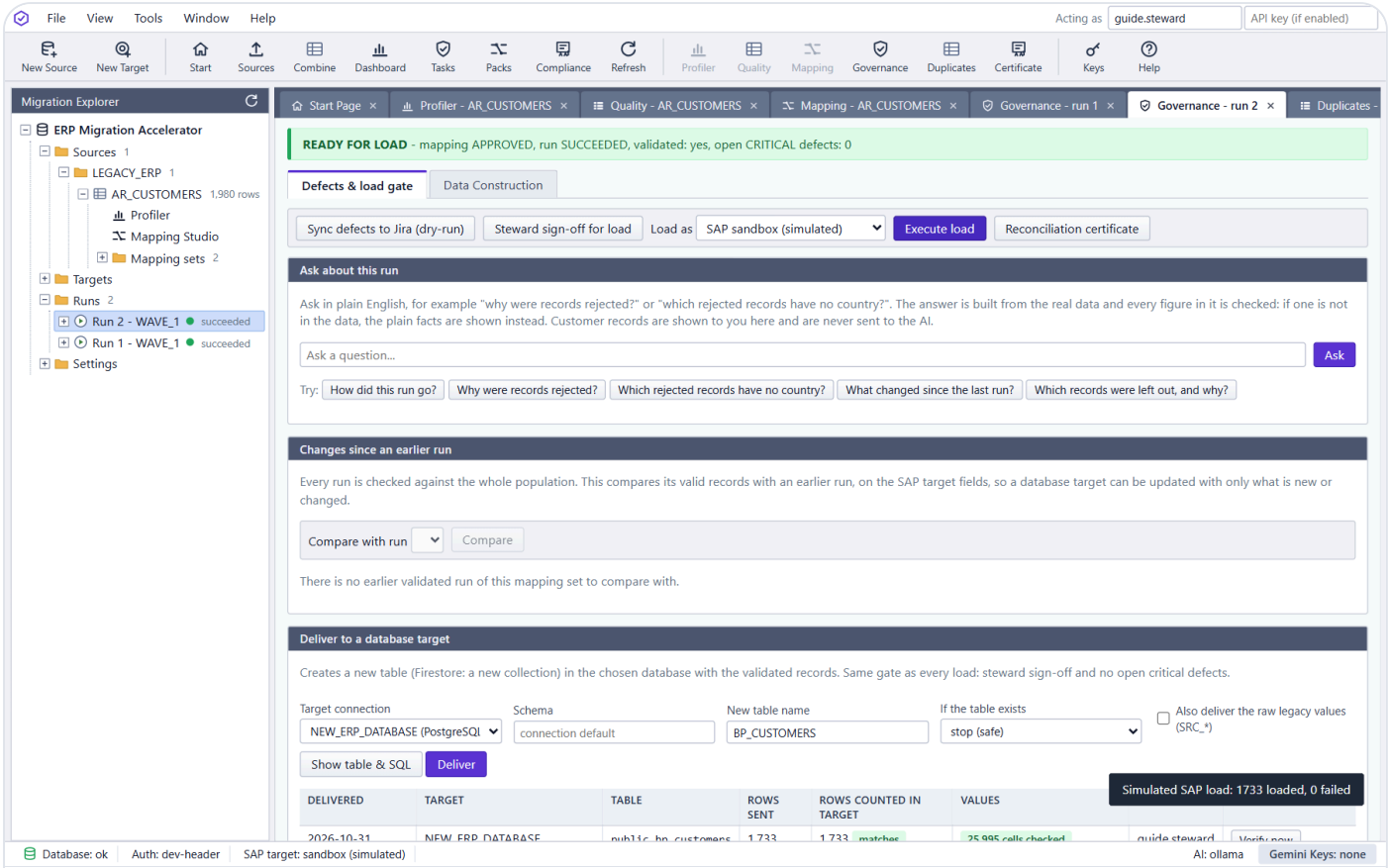


Figure 66. A rehearsal load into the SAP stand-in.

CONTROL

Create-only. The application never overwrites or merges into existing data. If a table of that name exists it stops and says so.

CAUTION

Before a real go-live repeat PR-15 against the company's own SAP test client. The stand-in does not prove that real customising, number ranges and authorisations accept the data.

6.14 Reconciliation certificate

PR-16 Produce and read the certificate

PURPOSE

Obtain the proof that every source row is accounted for.

PERFORMED BY

Steward, auditor

BEFORE YOU START

A run that has been loaded.

PROCEDURE

1. Select the run and press **Certificate**.
2. Read the banner, the drop-off accounting and the control checks.
3. Download the page as HTML or Markdown if it must be kept outside the application.

EXPECTED RESULT

RECONCILIATION PASSED, variance 0 rows.

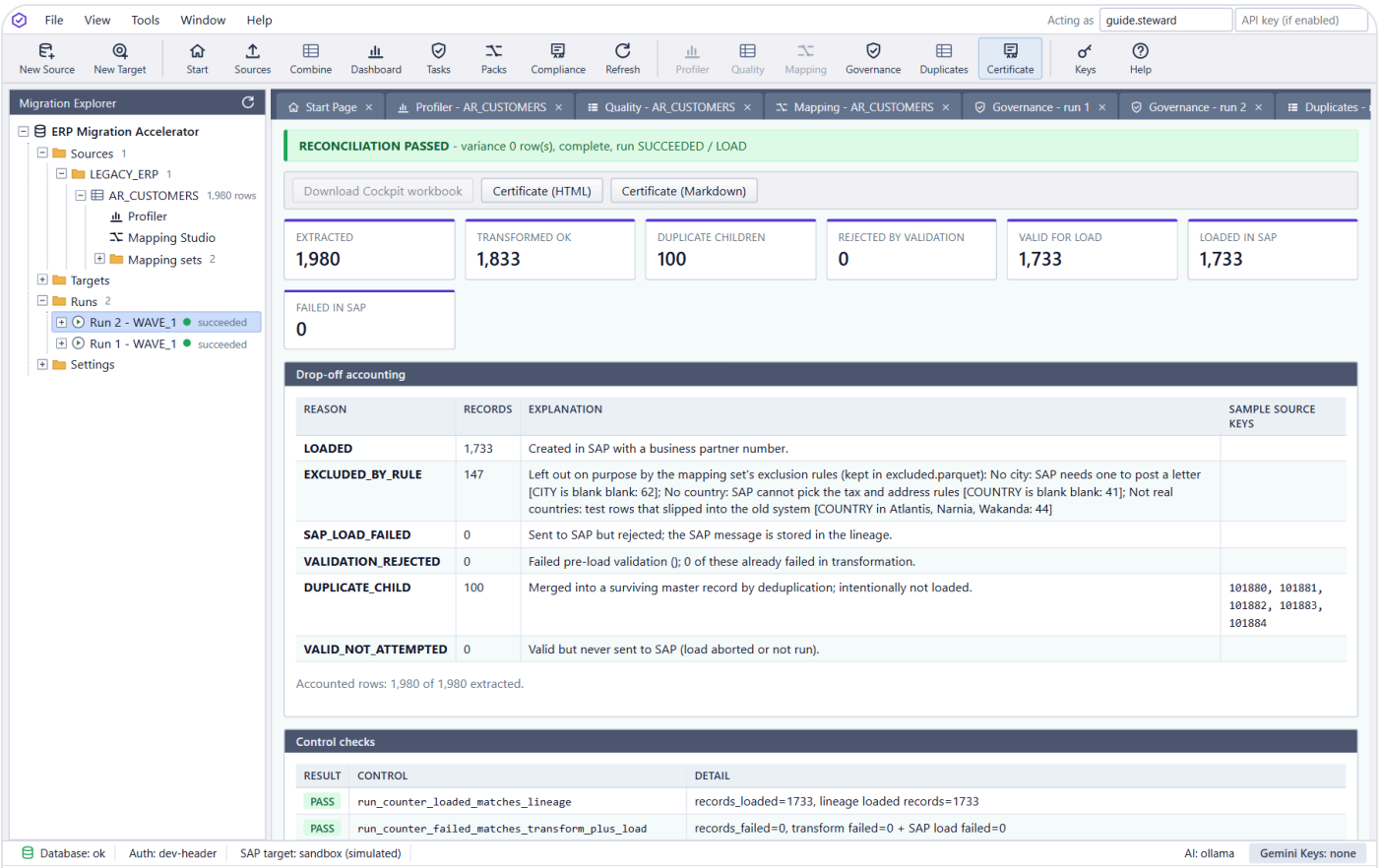


Figure 67. The certificate: extracted 1,980 · transformed 1,833 · duplicate children 100 · valid 1,733 · loaded 1,733 · failed 0.

REASON	ROWS	MEANING
LOADED	1,733	Created in the target.
EXCLUDED_BY_RULE	147	Left out on purpose, with reasons: no city 62, no country 41, not real countries 44.
DUPLICATE_CHILD	100	The same company as another row; merged into the survivor.
VALIDATION_REJECTED	0	Failed SAP's constraints (none: the gate made sure).
SAP_LOAD_FAILED	0	Sent but refused by the system.
Total	**1,980**	**1,733 + 147 + 100 = 1,980**

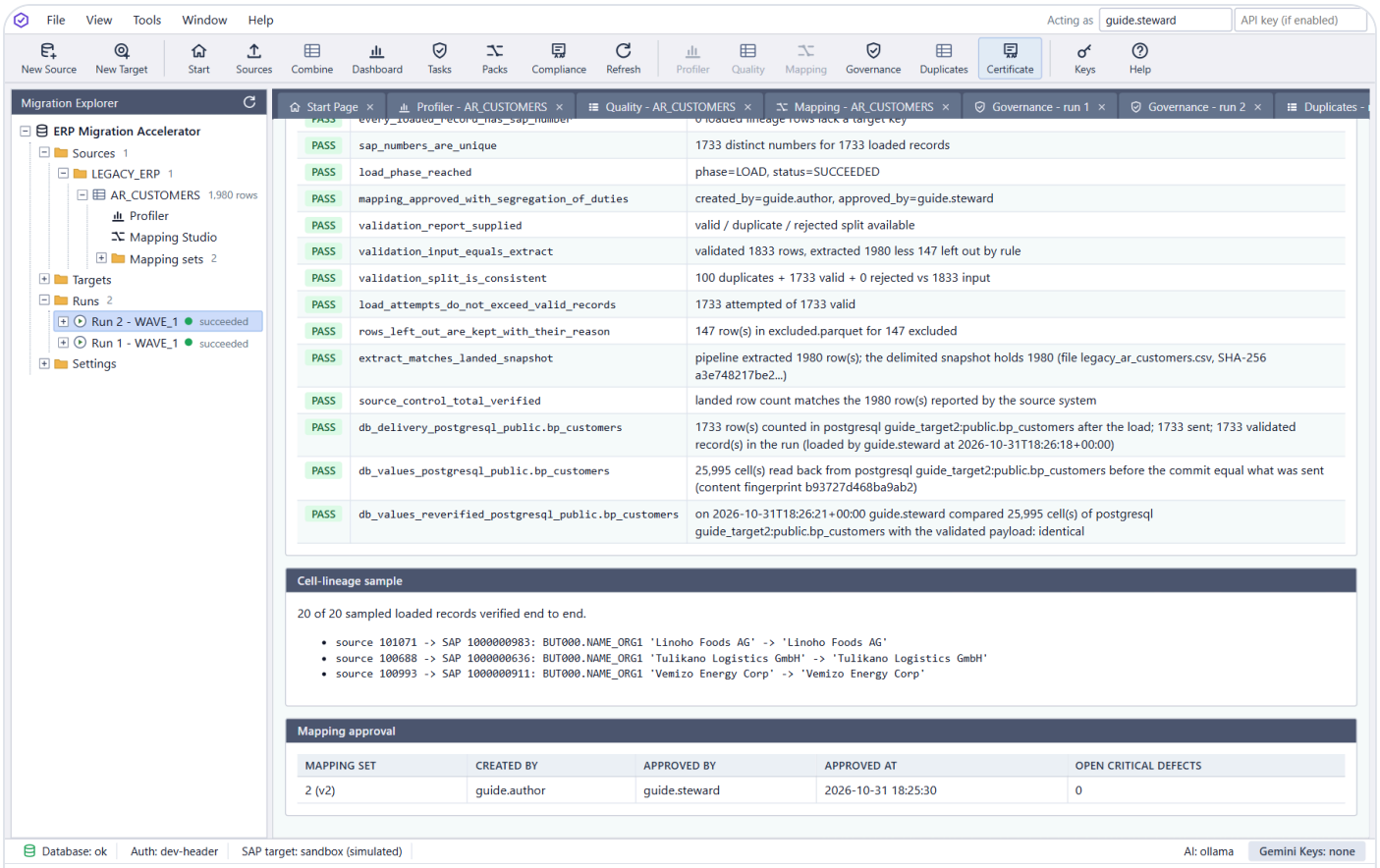


Figure 68. Control checks, each PASS: counts agree with the audit trail; author and approver differ; extract equals snapshot; control total verified; rows counted after load; cells read back and equal, with the fingerprint.

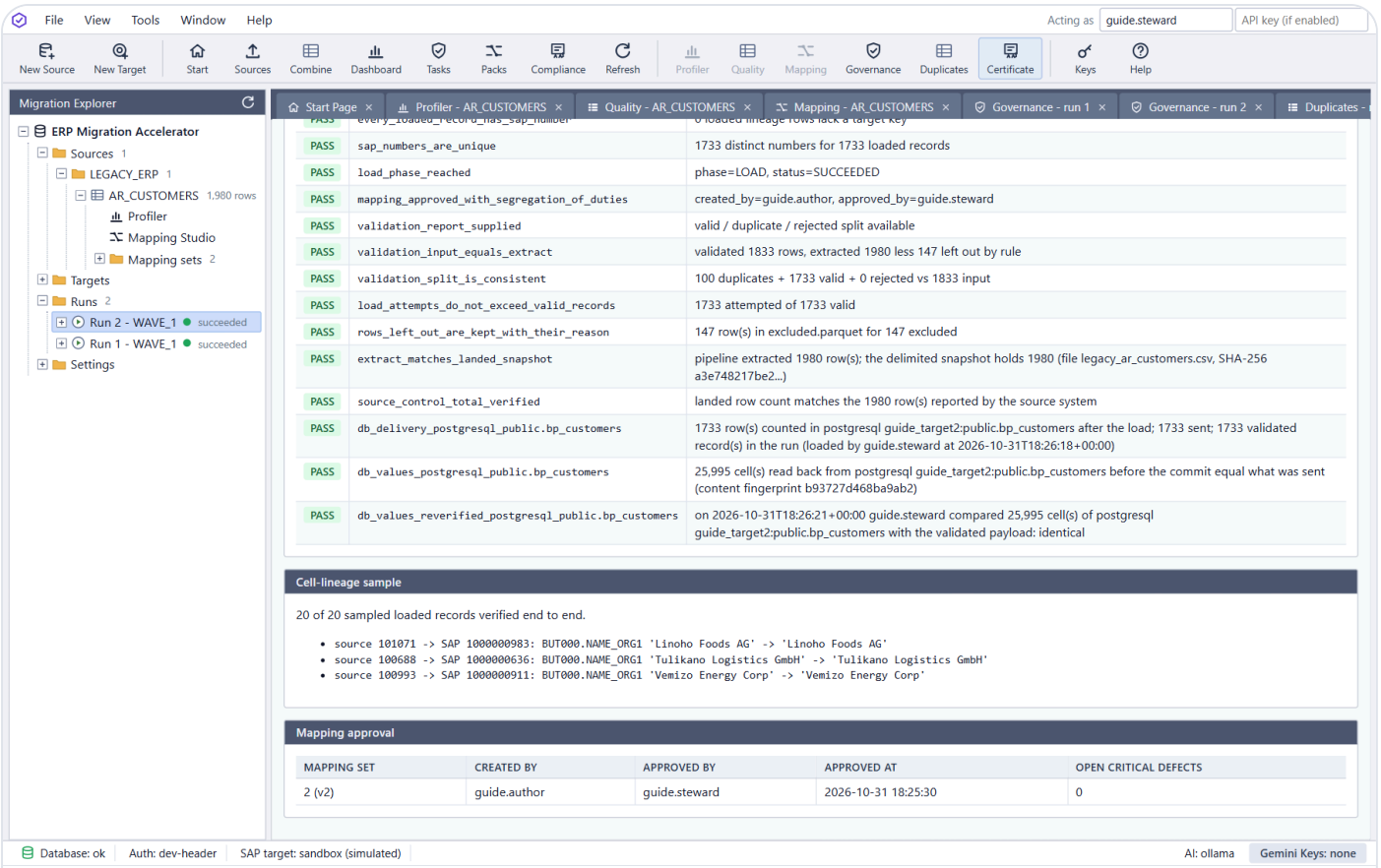


Figure 69. Samples traced from source to SAP number, and who approved the mapping and when.

CAUTION

The banner reads PASSED only when a load really happened. A run that was never loaded (neither into SAP nor the stand-in) shows *not passed* for that single reason (`load_phase_reached`).

6.15 Roll back a delivery

PR-17 Roll back one delivery

PURPOSE

Remove exactly the rows one delivery wrote.

PERFORMED BY

Steward

BEFORE YOU START

A delivery listed on the Governance page.

PROCEDURE

1. Find the delivery in the table and press **Roll back**.
2. Enter the reason (it is kept) and confirm.

EXPECTED RESULT

A message such as *1,733 row(s) removed from public.bp_customers; the table was empty, so it was dropped*. The history row reads *rolled back by ….*

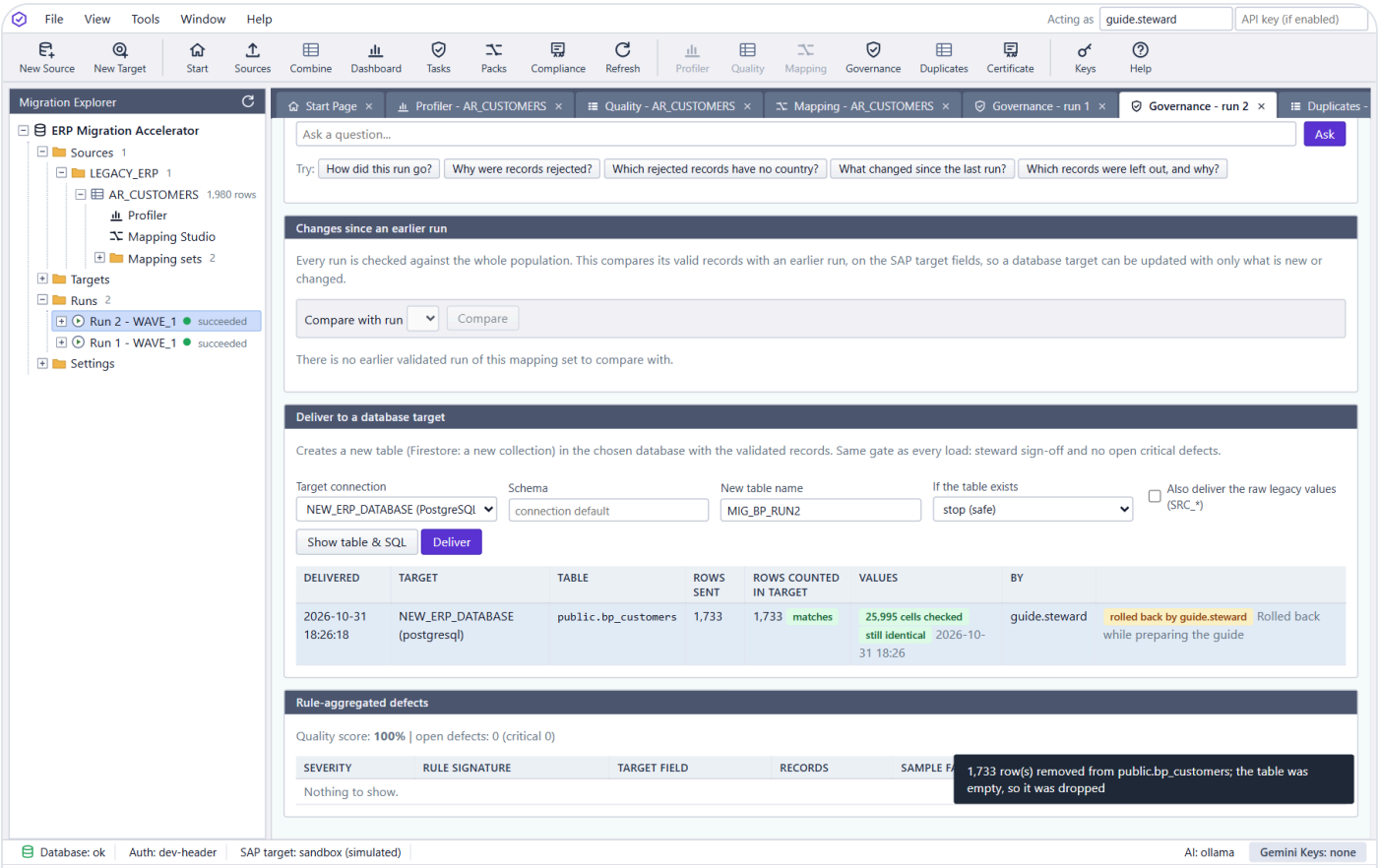


Figure 70. Rolled back, with the reason recorded.

CONTROL

Every written row carries `mig_run_id`, so rows of other runs, or rows added by someone else, are never touched.

6.16 Reuse an approved mapping (content packs)

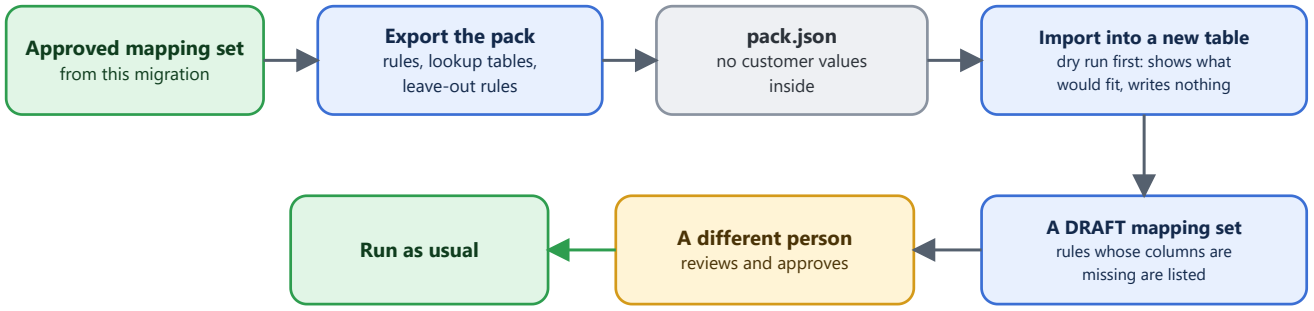


Figure 71. A content pack carries approved rules to the next migration; an import is a draft that a different person must approve.

PR-18 Export and import a content pack

PURPOSE

Reuse rules, lookup dictionaries and leave-out rules from one migration in another, without copying any customer values.

PERFORMED BY

Author; a different person approves the import

BEFORE YOU START

An approved mapping set (export). A registered table (import).

PROCEDURE

1. Press **Packs**. Under export enter the **Mapping set** number and an optional **Pack name**; press **Download the pack (.json)**.
2. To import, enter the table number (**Into table**), choose the pack file and leave **dry run** ticked; press **Check / import**.
3. Read what would happen: rules that fit, rules whose columns are missing. Untick **dry run** and import.
4. Open the draft in the Mapping Studio, check it (PR-6.4) and have it approved (PR-7).

EXPECTED RESULT

A DRAFT mapping set in the new table. It is not approved.

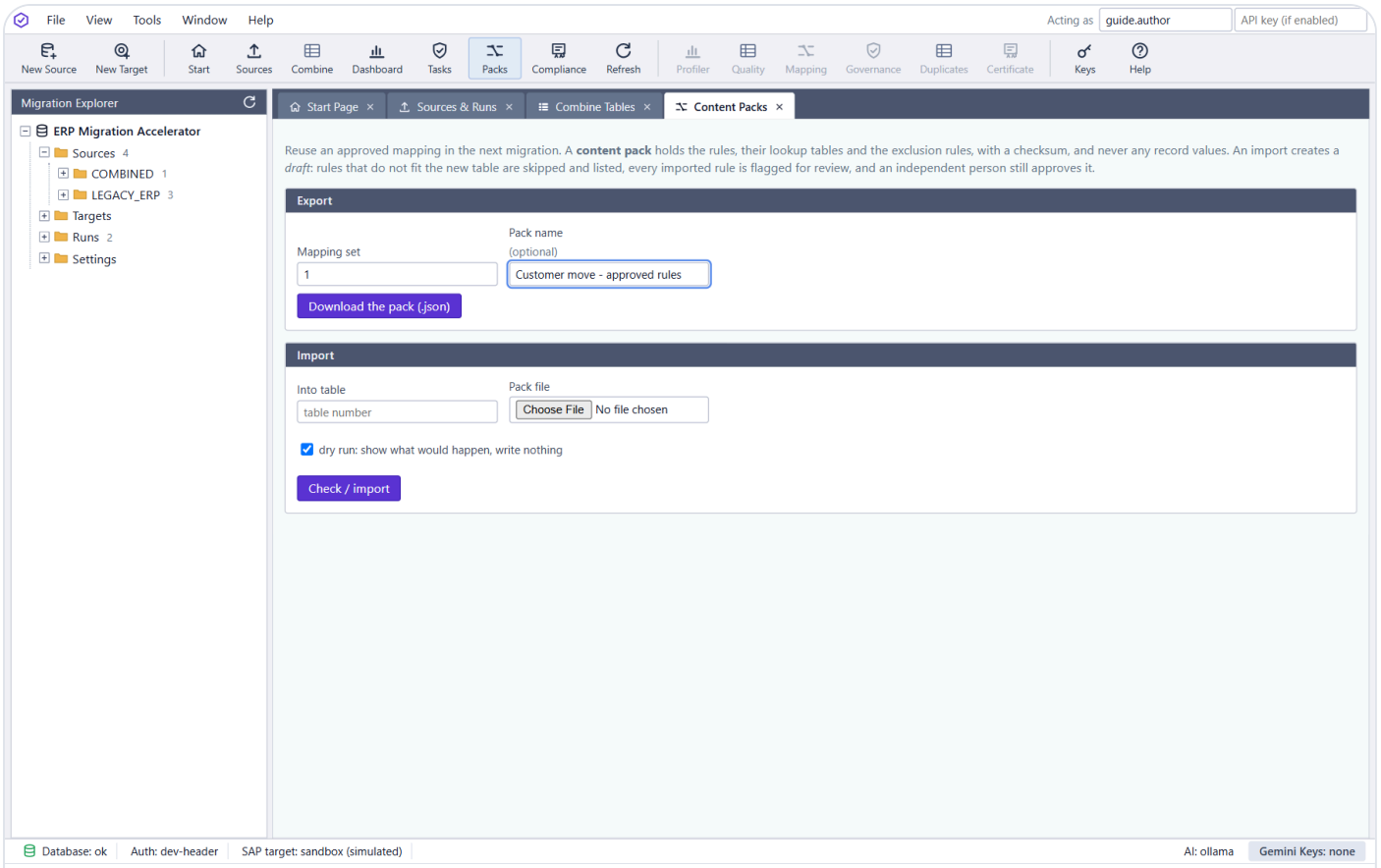


Figure 72. Export: choose the mapping set and give the pack a name.

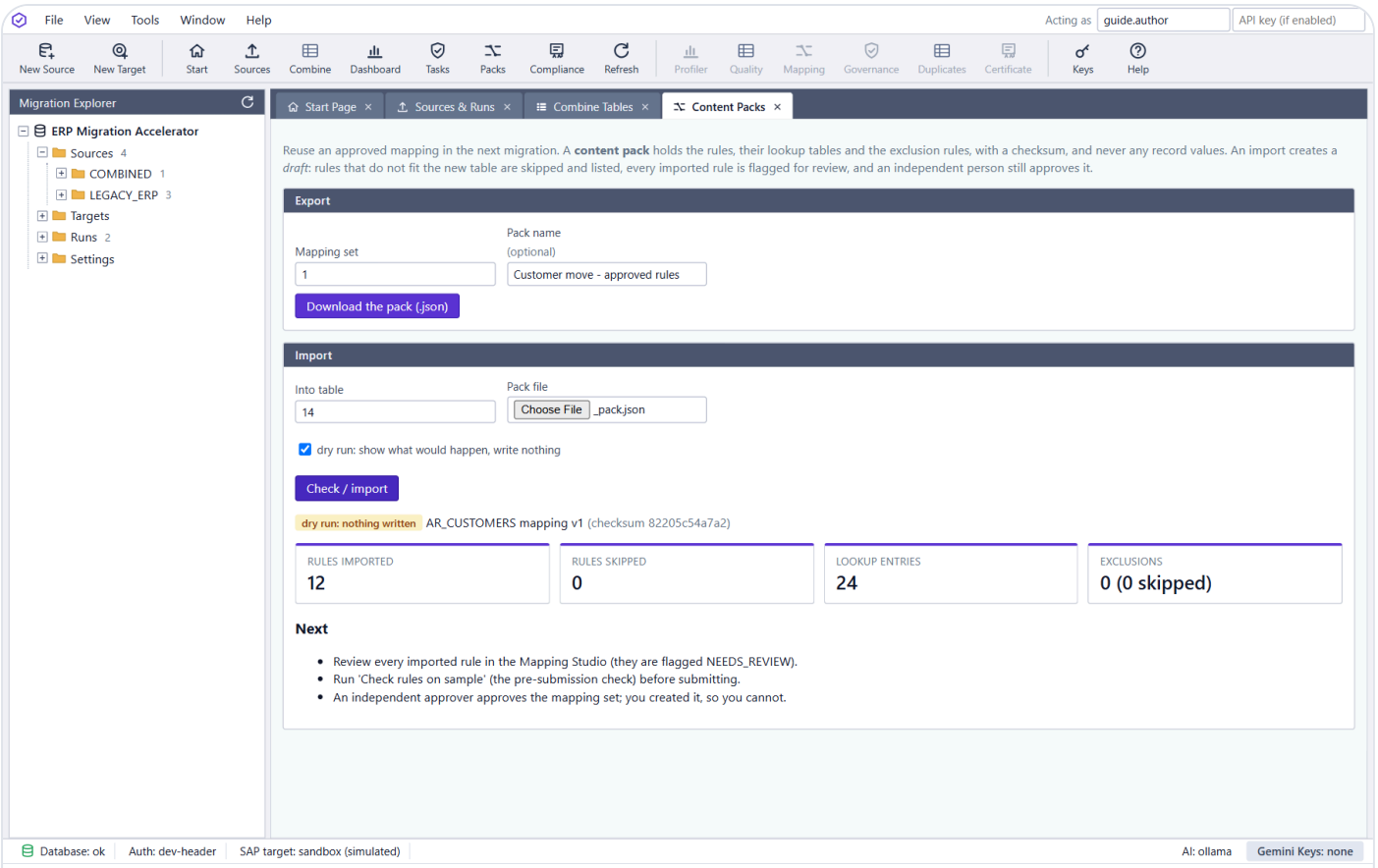


Figure 73. A dry-run import of mapping set 1 into table 14: 12 rules imported, 0 skipped, 24 lookup entries, 0 exclusions, and the next steps. Imported rules are flagged NEEDS_REVIEW; nothing is written.

CONTROL

A pack contains rules and dictionaries only. An import never approves itself.

6.17 Dashboard, tasks, compliance and the assistant

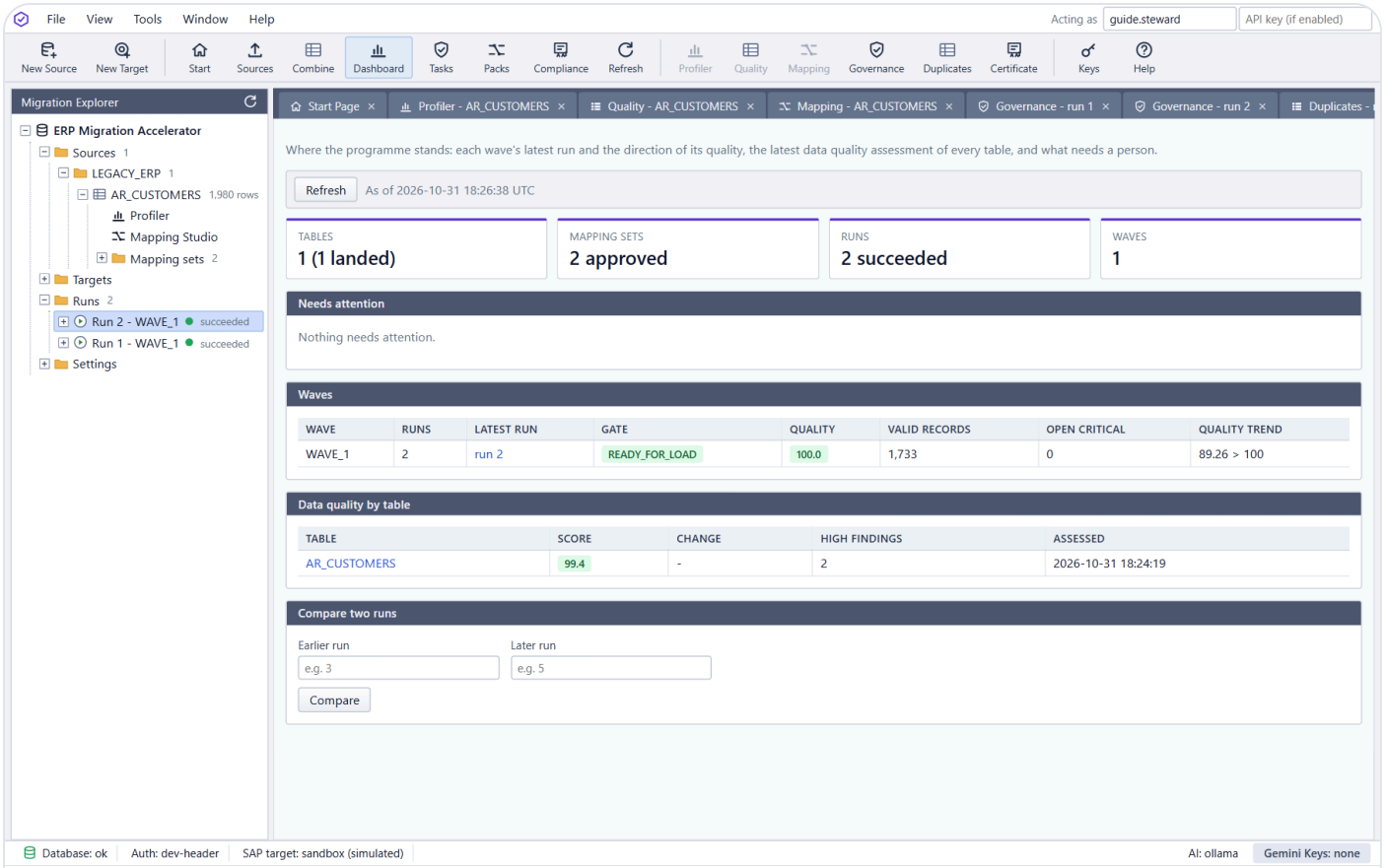


Figure 74. Dashboard. Tables, mapping sets and runs; per wave the latest run, gate, quality score and valid records, and the trend (89.26 → 100). **Compare two runs** shows what changed.

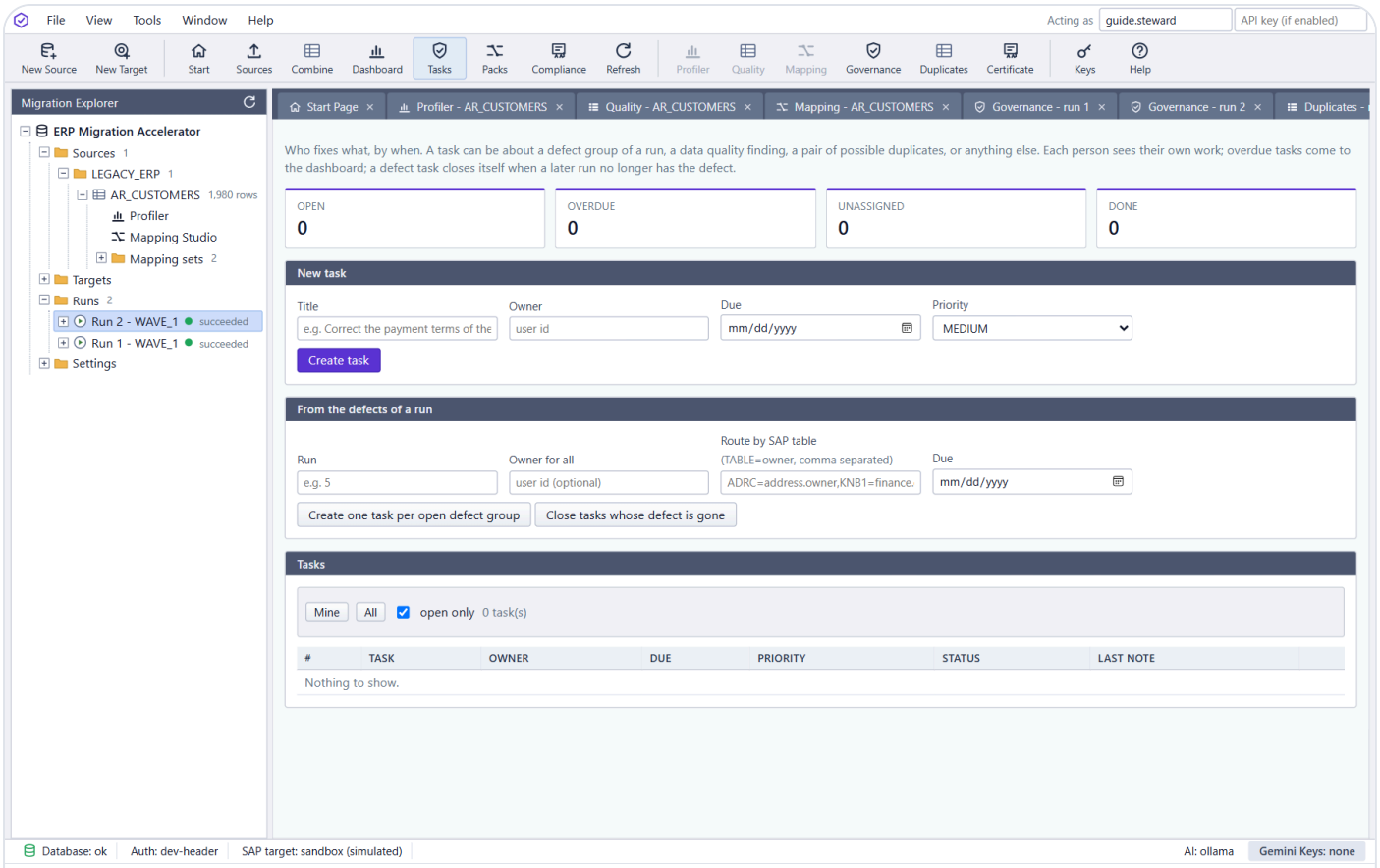


Figure 75. Tasks. Who fixes what, by when. One task can be created for every open defect group, with an owner chosen by SAP table (for example `ADRC=address.owner`); a task closes itself when a later run no longer has the defect.

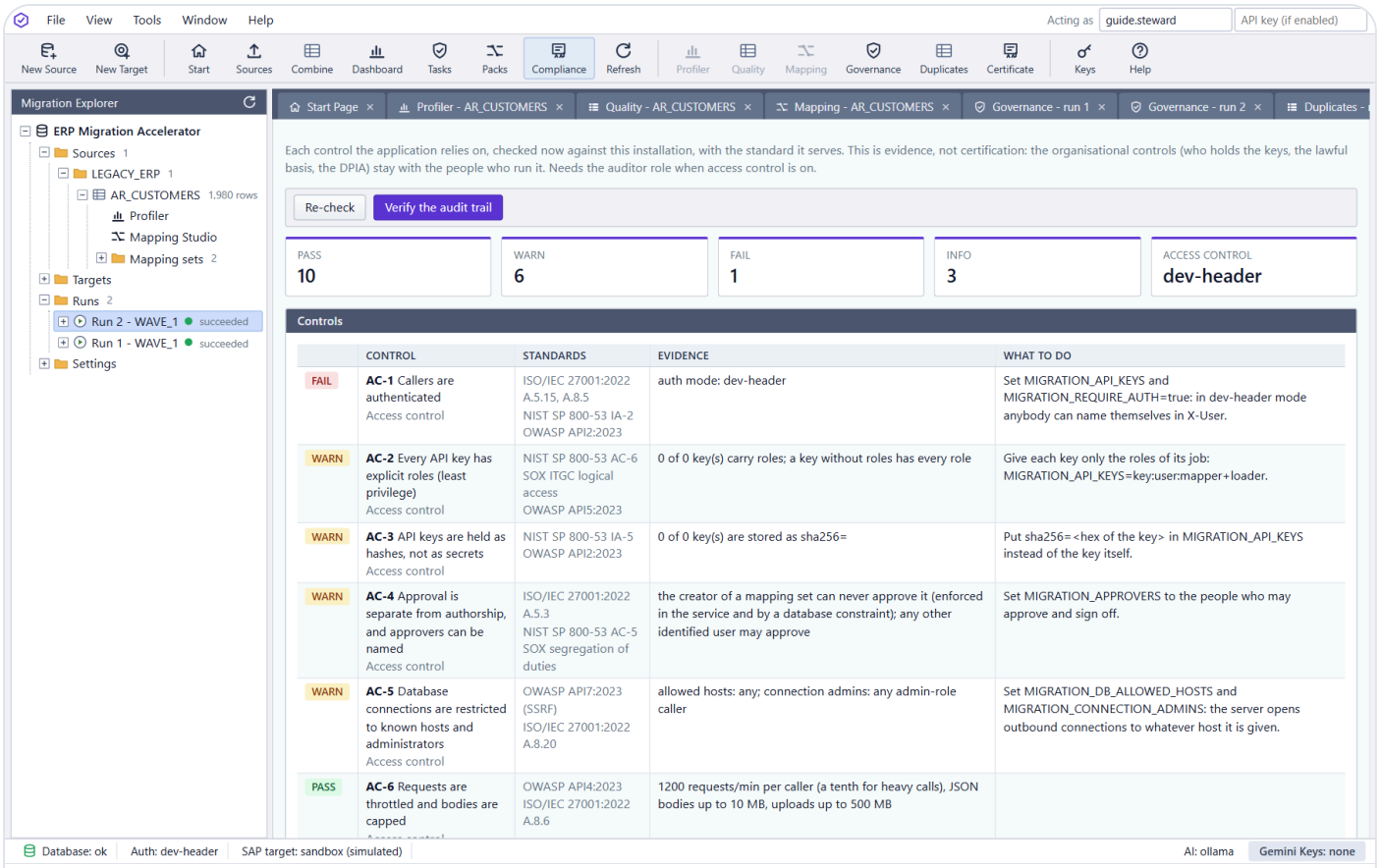


Figure 76. Compliance. Each control checked now against this installation, with the standard it serves. It is evidence, not a certificate: human controls (key custody, legal basis) remain with the people who run the system.

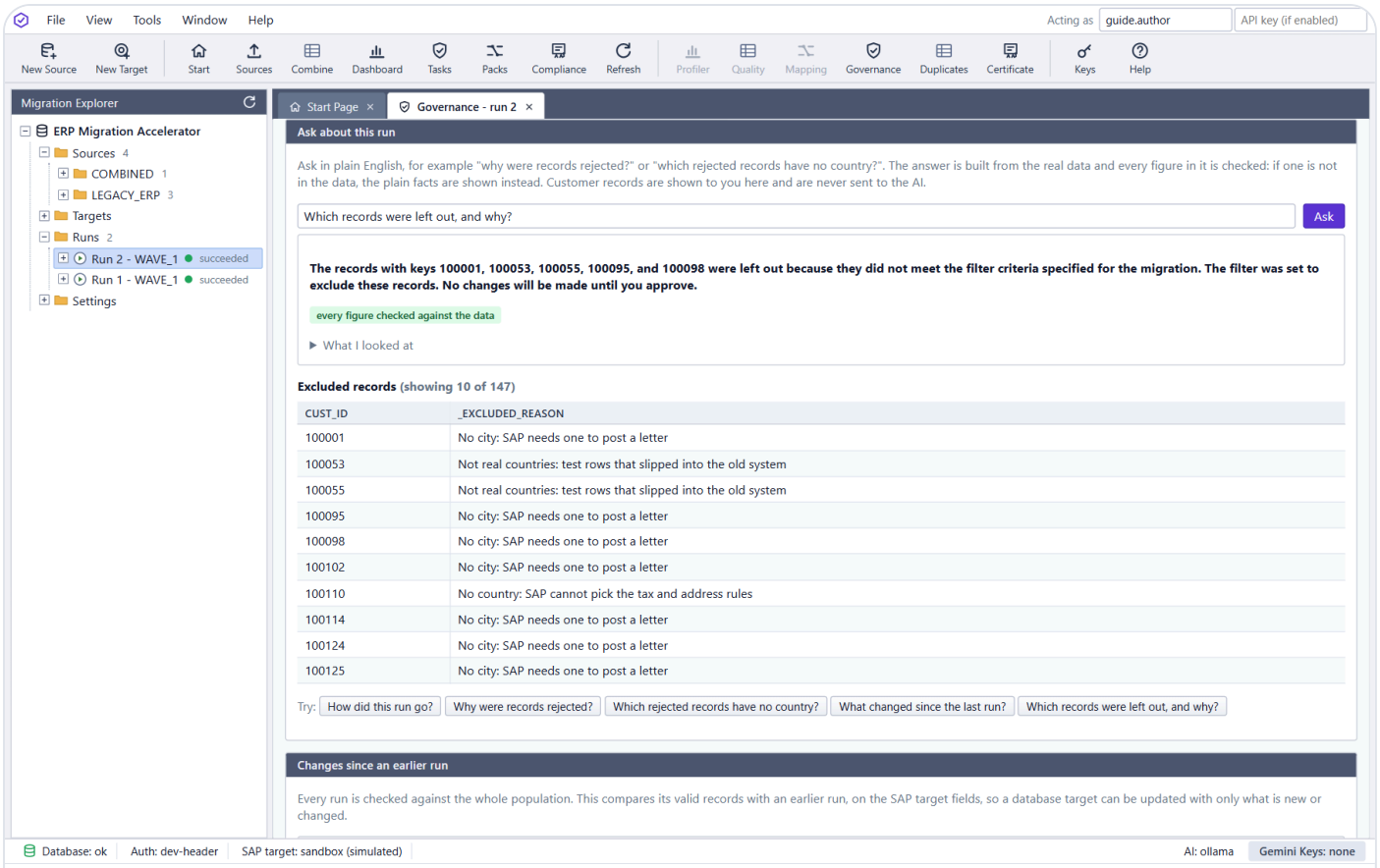


Figure 77. Ask about this run. The question is answered from the run's own records and the figures are checked against the data (green badge); the table below the answer lists the 147 excluded customers with their reasons. Customer records are not sent to the AI. Note the wording of the sentence: a small local model writes it and it is generic ("did not meet the filter criteria"). Rely on the checked table, not on the prose.

6.18 AI keys and the menus

PR-19 Add or remove an online AI key

PURPOSE

Allow AI mapping through Gemini.

PERFORMED BY

Administrator

BEFORE YOU START

A Gemini API key and a policy that allows sending column names and a few sample values to an online service.

PROCEDURE

1. Press **Keys** (or the key indicator in the status bar).
2. Paste the key (at least 16 characters), optionally label it, press **Test key**, then add it.
3. Remove a key from the same dialog.

EXPECTED RESULT

The indicator turns active. Keys live in memory only and are lost when the server restarts.

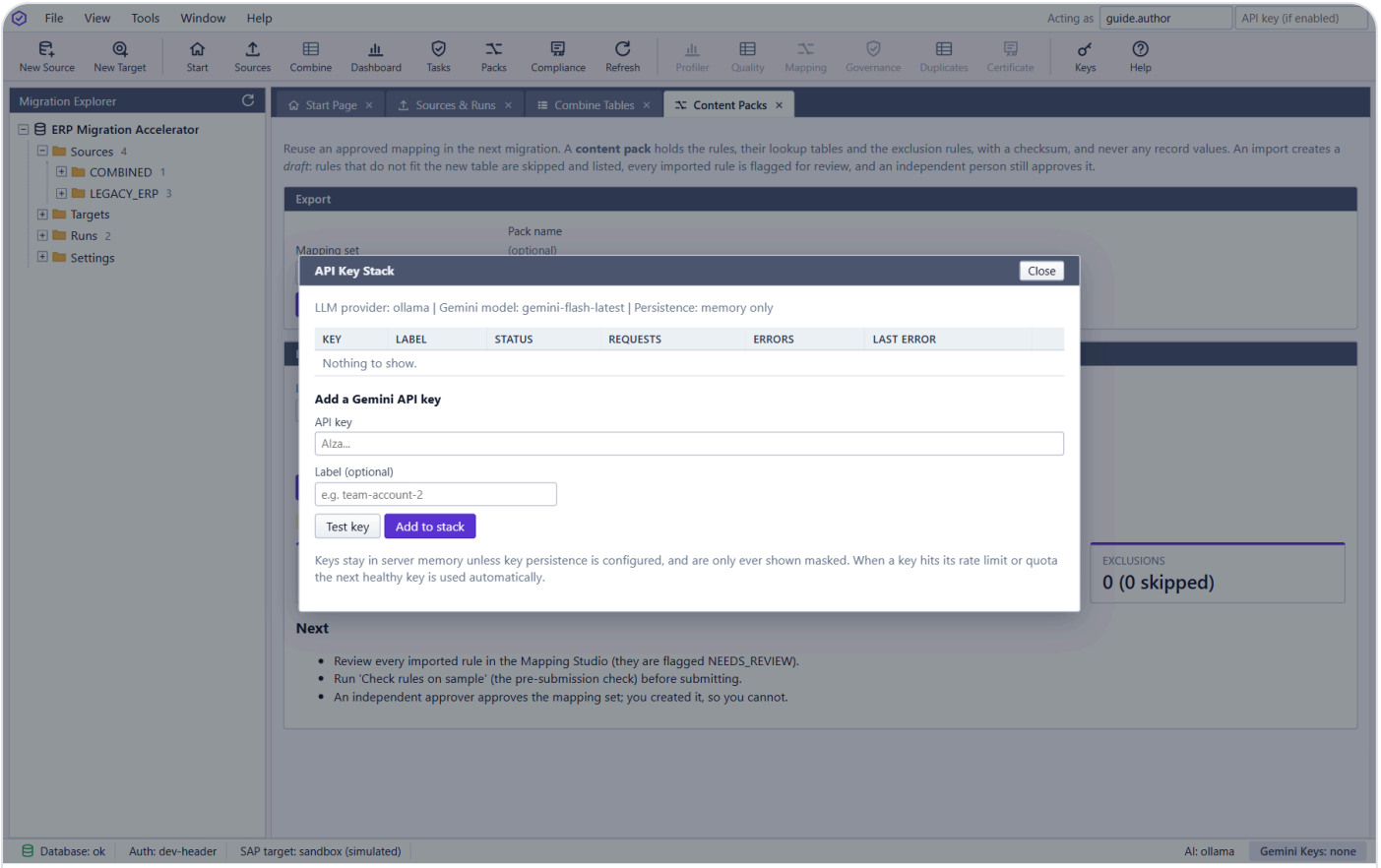


Figure 78. The API key dialog.

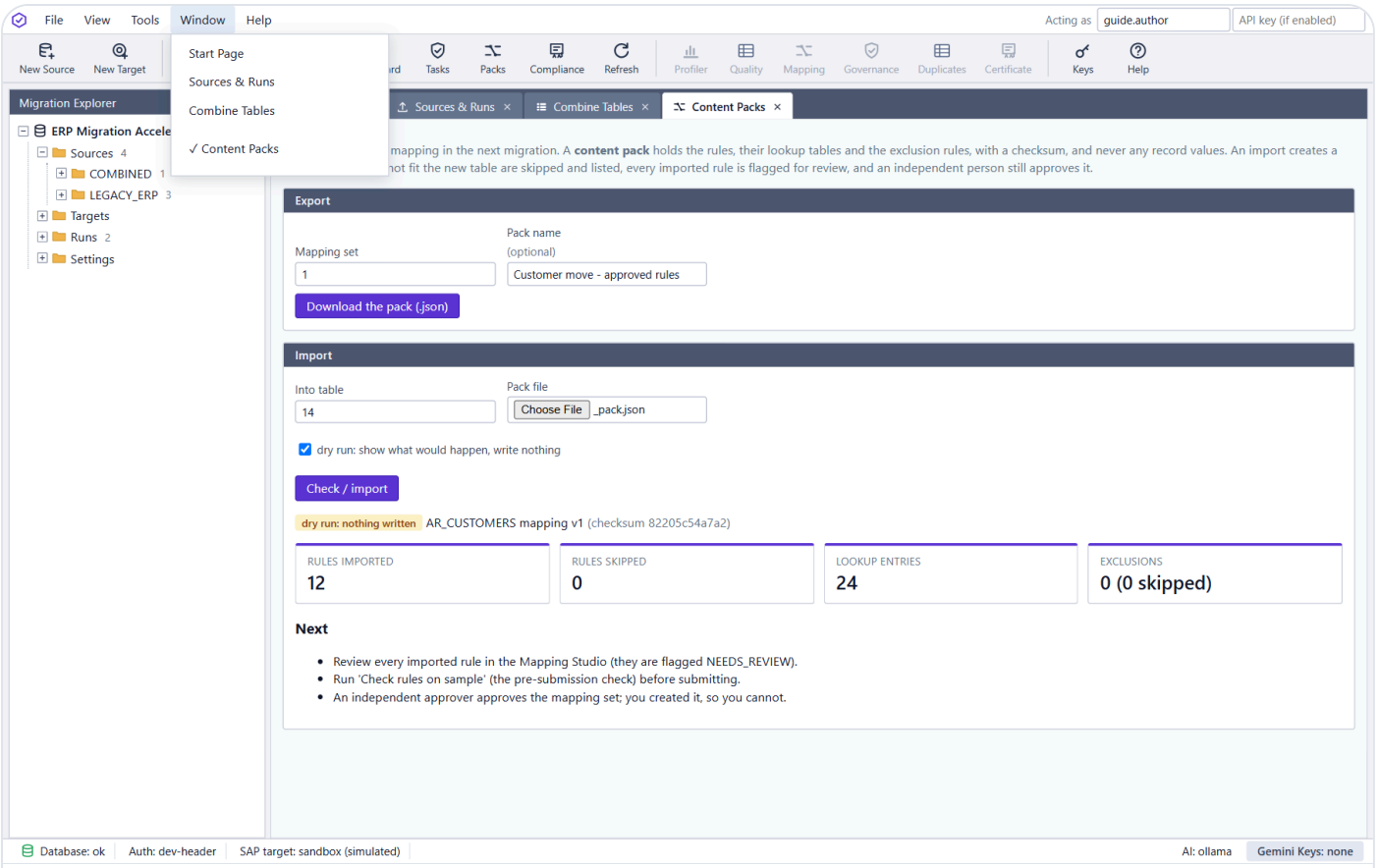


Figure 79. The Window menu lists the open pages (the current one is ticked); the other menus offer the same actions as the toolbar.

CONTROL

Keys and passwords are never written to a file or returned by the API. Rotate a key that has been pasted anywhere else.

7 Controls and audit evidence

What the application enforces and what it leaves behind as proof.

7.1 Control framework

The table lists what the application **enforces** (it cannot be bypassed from the user interface) and the evidence it leaves. Further detail is in `docs/controls.md` and `docs/compliance.md`; the **Compliance** page re-checks the technical controls against the running installation.

CONTROL	ENFORCED BY	EVIDENCE
The legacy data is never changed	Sources are read only; work uses a snapshot	Snapshot row count and fingerprint on the certificate
The extract is complete	Control total checked at registration	Certificate: control total verified
Independent review of the rules	Approval refused for the author	Approver and time on the certificate; audit log
Approved rules cannot drift	Approved sets are locked; a change makes a new version	Version history in the Explorer
Nothing is loaded with critical defects	The gate blocks sign-off and load	Gate state and defect table of each run
Every row is accounted for	Run accounting identity	Certificate drop-off table (sum equals extracted)
Exclusions are deliberate	Reason required; rows kept	Exclusion list with reasons
A responsible person accepts the result	Steward sign-off before load	Name and time recorded
Delivery cannot damage existing data	Create-only; stops if the table exists	Delivery history
Data arrived unchanged	Row count and every cell read back	Certificate: cells checked and equal; fingerprint
Undo is exact	Rows tagged with the run number	Rollback record with reason

CONTROL	ENFORCED BY	EVIDENCE
Secrets are not stored	Keys and passwords are memory-only	Compliance page
AI cannot act alone	Candidate-only choice, validator, review, second approval	AI-suggested tag and model name on each rule

7.2 What the controls do not cover

- **Who the user is.** Without `MIGRATION_API_KEYS` a typed name is trusted. There is no company single sign-on.
- **Whether a business decision is right.** The application records excluding 147 customers; it does not judge whether that is acceptable.
- **SAP acceptance.** Only the stand-in has been exercised (appendix D).
- **Operating practice.** Backup of the workspace, key custody and the legal basis for processing personal data remain with the organisation.

8 Troubleshooting and FAQ

Look up the symptom; each row gives a cause and an action.

8.1 Troubleshooting

SYMPTOM	LIKELY CAUSE	ACTION
A toolbar button is grey	Nothing is selected in the Explorer.	Click the table or run first.
The explorer is empty or the page says *Loading…*	The page was opened while the server was down, or the catalogue database is unreachable.	Check the status bar; start the database; reload with Ctrl+F5.
After **Test** everything is in one column, or accents are damaged	Wrong delimiter or encoding was detected.	Change **Delimiter** or **Encoding** and test again.
Row count differs from the control total	Incomplete extract, or wrong number.	Obtain a new extract or correct the total.
Approve or **Sign off** is refused	You authored the mapping, or you are not an allowed approver.	Act as a different, authorised person.
Edit is refused	The set is approved and locked.	Use a Fix-it card (makes a copy) or create a new set.
Gate BLOCKED	A critical defect is open.	See 6.9 and 6.10.
A **Combine** preview is refused	The join would repeat records.	Change the join key, or allow repeats on purpose.
AI mapping is very slow or returns nothing	A local model on a small computer is slow; or no AI is configured.	Write rules by hand, or add an online key under **Keys**.
An AI rule is marked NEEDS_REVIEW	It rewrites values (padding, replacing, translating) or its samples do not fit.	Read it; edit or delete it.

SYMPTOM	LIKELY CAUSE	ACTION
Deliver says the table exists	Delivery is create-only.	Choose another name, or roll back the earlier delivery.
Deliver is grey	No sign-off, or no target defined.	Sign off; define a target (PR-13).
Firestore: quota exceeded	The free plan allows about 20,000 writes a day.	Wait, deliver fewer rows, or change plan.
Certificate says *not passed*	No load has happened.	Execute a load (sandbox or live).
A connection test fails for a server you typed	The host is not in `MIGRATION_DB_ALLOWED_HOSTS`, or a firewall blocks it.	Ask the administrator.
Port busy at start	Another copy is running.	Use another port or stop the other copy.
Docker/PostgreSQL stops answering though the port is open	The container engine hung.	Restart Docker (`docker desktop restart`), then reload the page.

8.2 Frequently asked questions

Can the original data be damaged?

No. It is only read; all work is on a snapshot.

Can I use my own file?

Yes. CSV, Excel, JSON, fixed-width and Parquet files, databases (PostgreSQL, SQL Server, Oracle, MySQL), MongoDB, Firestore and cloud storage are supported.

Is the AI required?

No. Everything can be done by hand; the AI proposes and a person decides.

Is customer data sent to the internet?

Not with the local model. With Gemini, column names and a few sample values are sent for mapping; ask the administrator what policy applies. The assistant never sends customer records.

What if I change my mind after delivery?

Roll back (PR-17), correct, and deliver again.

What does “sandbox, simulated” in the status bar mean?

The application is connected to the SAP stand-in, not to a real SAP.

9 Appendices

Quick reference, defect codes, files, test status and provenance.

9.1 Appendix A: quick reference

#	TASK	WHERE	YOU SHOULD SEE
1	Register a source	New Source → wizard → Finish	Profiler tab with the right row count
2	Profile / assess	Profiler; Quality → Assess	Health numbers; findings by severity
3	Combine (optional)	Combine → Preview → Create table	A new source table
4	Write rules	Mapping → Generate AI mapping or Add manual rule	A card per SAP field with a pass rate
5	Check	Check rules on sample	"can be submitted"
6	Submit / approve (someone else)	Submit for approval; Approve	SUBMITTED → APPROVED
7	Run	Run pipeline	Governance tab opens
8	Read the gate	Governance	BLOCKED + defects, or READY FOR LOAD
9	Fix	Fix-it card; Rows to leave out; Edit rule	A new DRAFT; repeat 5 to 8
10	Sign off (steward)	Steward sign-off for load	"signed off by …"
11	Deliver	New Target (once) → Deliver; or Load as → Execute load	Rows sent = rows counted; cells checked
12	Prove	Certificate	RECONCILIATION PASSED, variance 0
–	Undo	Roll back	"rolled back by …"

9.2 Appendix B: defect codes

PATTERN	MEANING	USUAL REMEDY
`ERR_MANDATORY_<FIELD>`	A field SAP requires is empty after mapping.	Add or correct a rule; supply a value; exclude the rows.
`ERR_LOOKUP_FAILED_<FIELD>`	A value is not in the rule's lookup dictionary.	Add the value to the lookup, or exclude the rows.
`ERR_LENGTH_<FIELD>`	A value is longer than the SAP field.	Shorten in the rule, or correct the source.

Severity: CRITICAL blocks the gate; HIGH, MEDIUM and LOW are reported without blocking.

9.3 Appendix C: files and scripts

PATH	PURPOSE
`mock_data/legacy_ar_customers.csv`	The 1,980-row practice file.
`docs/user-guide/`	The illustrated, story-style guide for first-time readers (HTML and PDF).
`docs/user-manual/`	This manual (HTML and PDF).
`docs/controls.md`, `docs/compliance.md`	Control detail and the standards each control serves.
`docs/ai-evaluation.md`	How the AI mapping was measured and what it gets wrong.
`docs/scale-test.md`	The million-row test.
`scripts/make_guide_screenshots.py`	Takes the real screenshots by operating the application.
`scripts/guide_diagrams.py`	Draws the flow charts.
`scripts/build_user_manual.py`	Assembles this manual (`--pdf` also writes the PDF).

9.4 Appendix D: what was tested

AREA	STATUS
Files, profiling, quality rules	automated tests; used in this manual
AI mapping with a local model, and its safeguards	tested; photographed once (the step takes about 40 minutes on a laptop). The Gemini path is tested with a fake service.
Mapping, approval, run, de-duplication, gate, sign-off, certificate, rollback	automated tests; used in this manual
Combine tables, content packs, assistant, tasks, dashboard, compliance	automated tests and browser tests; used in this manual
Delivery to PostgreSQL, SQL Server, Oracle, MySQL, Firestore	tested against real servers (containers) and a real Firebase project; PostgreSQL used in this manual
MongoDB and Firestore as sources	tested against a MongoDB container and the Firestore emulator; not connected in this manual (the screenshots show the questions the wizard asks)
Million-row runs	tested at 1,000,000 rows
SAP: Migration Cockpit workbook, sandbox, live OData	against a stand-in only; never a real SAP system
Single sign-on	not built in; sign-in by API key is supported

9.5 Appendix E: how this manual was made

The screenshots are unretouched, taken by `scripts/make_guide_screenshots.py` driving Microsoft Edge through the whole practice migration on a fresh copy of the application and a made-up file. Five pictures of the AI step are the ones taken for the illustrated guide. The flow charts are drawn by `scripts/guide_diagrams.py`. Rebuild with `python scripts/build_user_manual.py --pdf`.

ERP Migration Accelerator User Manual v1.0 · built by scripts/build_user_manual.py · screenshots are from a practice run on made-up data.